Calculation of Heater-Tube Thickness in Petroleum Refineries

API STANDARD 530
FOURTH EDITION, OCTOBER 1996



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Calculation of Heater-Tube Thickness in Petroleum Refineries

Manufacturing, Distribution and Marketing

API STANDARD 530
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FOREWORD

This standard is based on the accumulated knowledge and experience of petroleum refiners, fired-heater manufacturers, and engineering contractors. The objective of this publication is to provide a calculation procedure that will facilitate the design and procurement of fired-heater tubes used in general refinery service and related process facilities.

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Calculation of Heater-Tube Thickness in Petroleum Refineries

SECTION 1—GENERAL

1.1 Scope

This standard provides procedures and design criteria for calculating the required wall thickness of new tubes for petroleum refinery heaters. These procedures are appropriate for designing tubes that will be used in both corrosive and noncorrosive services. These procedures have been developed specifically for the design of refinery and related process-fired heater tubes (direct-fired, heat-absorbing tubes within enclosures). These procedures are not intended to be used for the design of external piping.

This standard represents accepted engineering approaches based on up-to-date knowledge of the subject. The bases and sources of the procedures, particularly the design equations and stresses, are described in Sections 2 and 3.

Appendix A describes in detail the sources of the data used to develop the allowable stresses. Appendix B presents the derivations of the equations for corrosion fraction and temperature fraction. Appendix C describes a procedure for calculating the skin temperature of a heater tube. Appendix D describes limits for thermal stresses in heater tubes.

This standard does not include recommendations concerning tube retirement thickness, but Appendix E describes a technique for estimating the life that remains in a heater tube.

1.2 Information Required

When the use of this standard is specified, the usual design parameters—design pressures, design fluid temperature, corrosion allowance, and tube material—must be defined. In addition, the following information must be furnished:

a. The design life of the heater tube.
b. Whether the equivalent-temperature concept is to be applied. If so, the operating conditions at the start and the end of the run must be furnished.
c. The temperature allowance, if any.
d. The corrosion fraction (if different from that shown in Figure 1).
e. Whether elastic-range thermal-stress limits are to be applied.

If any of Items a–e are not furnished, the following applicable parameters should be used:

a. A design life equal to 100,000 hours.
b. A design metal temperature based on the maximum metal temperature. (The equivalent-temperature concept shall not apply.)
c. A temperature-allowance equal to 25°F (15°C).
d. The corrosion fraction given in Figure 1.
e. The elastic-range thermal-stress limits.

1.3 Limitations

The design procedures described in this standard are subject to the limitations of 1.3.1 through 1.3.7.

1.3.1 The allowable stresses are based on a consideration of yield strength and rupture strength only; plastic or creep strain has not been considered. Using these allowable stresses may result in small permanent strains in some applications; however, these small strains will not affect the safety or operability of heater tubes.

1.3.2 No considerations are included for adverse environmental effects such as graphitization, carburization, or hydrogen attack. Limitations imposed by hydrogen attack can be developed from the Nelson curves [Reference 1].[1]

1.3.3 These design procedures have been developed for seamless tubes. When they are applied to tubes that have a longitudinal weld, the allowable stress values should be multiplied by the appropriate joint efficiency factor. Joint efficiency factors shall not be applied to circumferential welds.

1.3.4 These design procedures have been developed for thin tubes (tubes with a thickness/outside diameter t_m/D_o ratio of less than 0.15). Additional considerations may apply to the design of thicker tubes.

1.3.5 No considerations are included for the effects of cyclic pressure or cyclic thermal loading.

1.3.6 The design loading includes only internal pressure. Limits for thermal stresses are provided in Appendix D. Limits for stresses developed by weight, supports, end connections, and so forth are not discussed in this standard.

1.3.7 Most of the Larson-Miller parameter curves in 3.8 are not Larson-Miller curves in the traditional sense but are derived from the 100,000-hour rupture strength as explained in A.3. Consequently, the curves may not provide a reliable estimate of the rupture strength for a design life that is less than 20,000 hours or more than 200,000 hours.

[1]References in brackets are listed in Appendix G.

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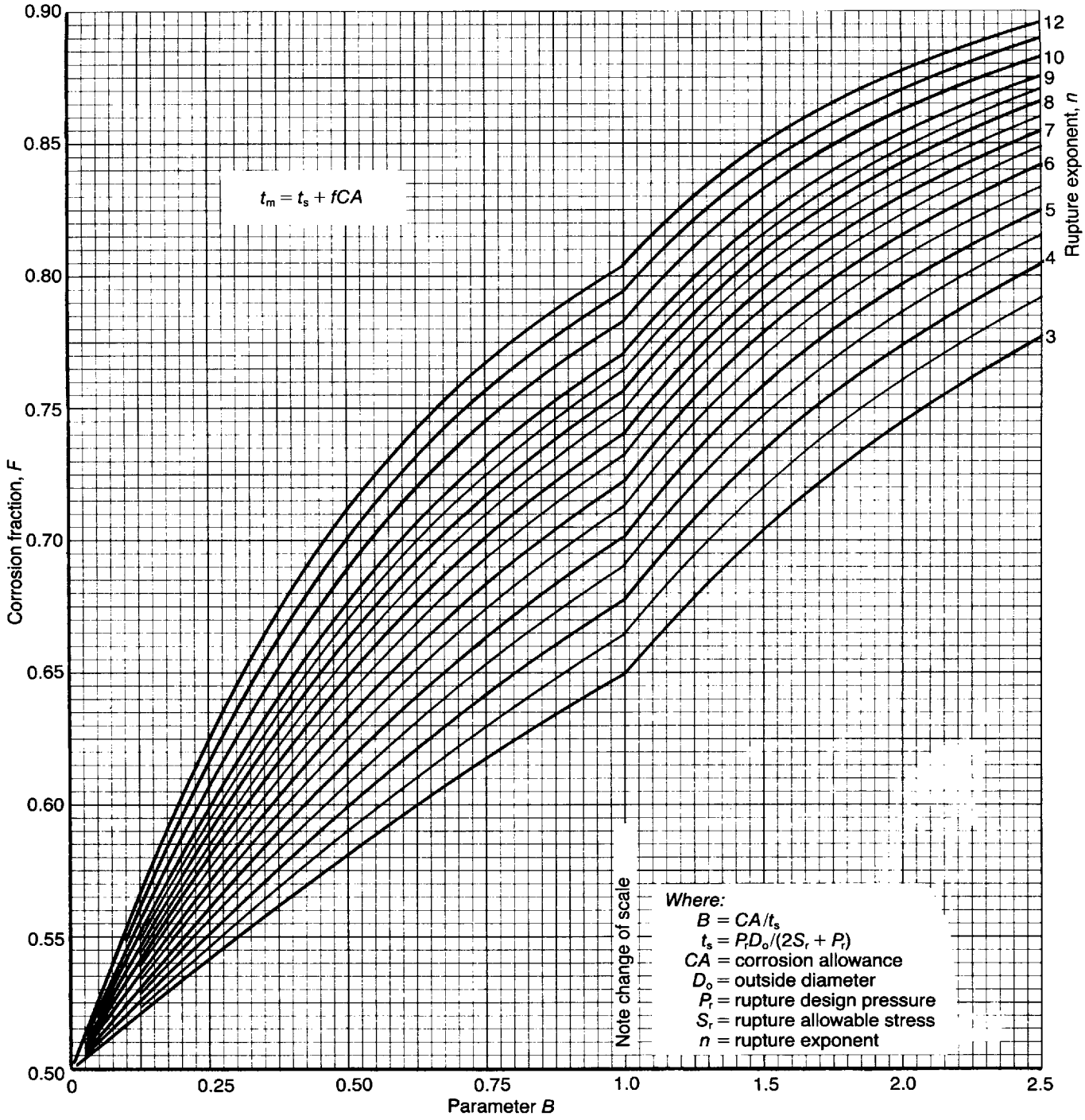


Figure 1—Corrosion Fraction

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1.4 Definitions of Terms

Terms used in this standard are defined in 1.4.1 through 1.4.15.

1.4.1 The *design metal temperature* (T_d), in degrees Fahrenheit (degrees Celsius), is the tube metal, or skin, temperature used for design. It shall be determined by calculating the maximum tube metal temperature (T_m in Appendix C) or the equivalent tube metal temperature (T_e in 1.4.2) and adding an appropriate temperature allowance (see 1.4.6). A procedure for calculating the maximum tube metal temperature from the heat flux is included in Appendix C. When the equivalent tube metal temperature is used, the maximum operating temperature can be higher than the design metal temperature.

1.4.2 The *equivalent tube metal temperature* (T_e), in degrees Fahrenheit (degrees Celsius), is a calculated constant metal temperature that in a specified period of time produces the same creep damage as does a linearly changing metal temperature (see 2.8).

1.4.3 The *elastic design pressure* (P_e), in pounds per square inch gauge (megapascals gauge), is the maximum pressure that the heater coil will sustain for short periods of time. This pressure is usually related to relief valve settings, pump shut-in pressures, and so forth.

1.4.4 The *rupture design pressure* (P_r), in pounds per square inch gauge (megapascals gauge), is the maximum operating pressure that the coil section will sustain during normal operation. The tube must withstand this pressure during long periods of operation. If the pressure changes during an operating run, the highest operating pressure should be used.

Note: The rupture design pressure is usually less than the elastic design pressure. The characteristic that differentiates these two pressures is the relative length of time over which they are sustained. The rupture design pressure is a long-term loading condition that remains relatively uniform over a period of years. The elastic design pressure is usually a short-term loading condition that typically lasts only hours or days. The rupture design pressure is used in the rupture design equation, since creep damage accumulates as a result of the action of the operating, or long-term, stress. The elastic design pressure is used in the elastic design equation to prevent excessive stresses in the tube during periods of operation at the maximum pressure.

1.4.5 The *corrosion allowance* (*CA*), in inches (millimeters), is the part of the tube thickness that is included for corrosion.

1.4.6 The *temperature allowance* (*TA*), in degrees Fahrenheit (degrees Celsius), is the part of the design metal temperature that is included for process- or flue-gas maldistribution, operating unknowns, and design inaccuracies. The temperature allowance is added to the calculated maximum tube metal temperature or to the equivalent tube metal temperature to obtain the design metal temperature (see 1.4.1).

1.4.7 The *minimum thickness* (t_m), in inches (millimeters), is the minimum required thickness of a new tube, taking into account all appropriate allowances. (See Equation 5).

1.4.8 The *stress thickness* (t_s), in inches (millimeters), excludes all thickness allowances and is calculated from Equations 2 and 4 that use the allowable stress.

1.4.9 The *outside diameter* (D_o), in inches (millimeters), is the outside diameter of a new tube.

1.4.10 The *actual inside diameter* (D_i), in inches (millimeters), is the inside diameter of a new tube. The actual inside diameter shall be used to calculate the tube skin temperature in Appendix C and the thermal stress in Appendix D.

1.4.11 The *inside diameter* (D'_i), in inches (millimeters), is the inside diameter of a tube with the corrosion allowance removed. The inside diameter of a cast tube is the inside diameter of the tube with the porosity and corrosion allowances removed. This inside diameter shall be used in the design calculations.

1.4.12 The *design life* (L_d), in hours, is the operating time used as a basis for tube design. The design life is not necessarily the same as the retirement or replacement life.

1.4.13 The *elastic allowable stress* (S_e), in pounds per square inch (megapascals), is the allowable stress for the elastic range (see 3.2). (See 1.3.3 for information about tubes that have longitudinal welds.)

1.4.14 The *rupture allowable stress* (S_r), in pounds per square inch (megapascals), is the allowable stress for the creep-rupture range (see 3.3). (See 1.3.3 for information about tubes that have longitudinal welds.)

1.4.15 The *rupture exponent* (n) is a parameter used for design in the creep-rupture range. (See A.4).

1.5 Referenced Material Specifications

The current editions of the following ASTM[2] specifications are cited in 3.8:

A 53	*Zinc-Coated Welded and Seamless Black and Hot-Dipped Steel Pipe*
A 106	*Seamless Carbon Steel Pipe for High-Temperature Service*
A 161	*Seamless Low-Carbon and Carbon-Molybdenum Steel Still Tubes for Refinery Service*
A 192/A 192M	*Seamless Carbon Steel Boiler Tubes for High-Pressure Service*

[2]American Society for Testing and Materials, 1916 Race Street. Philadelphia. Pennsylvania 19103.

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A 200	*Seamless Intermediate Alloy-Steel Still Tubes for Refinery Service*
A 209/A 209M	*Seamless Carbon-Molybdenum Alloy-Steel Boiler and Superheater Tubes*
A 210/A 210M	*Seamless Medium-Carbon Steel Boiler and Superheater Tubes*
A213/A 213M	*Seamless Ferritic and Austenitic Alloy Steel Boiler, Superheater and Heat Exchange Tubes*
A 271	*Seamless Austenitic Chromium-Nickel Steel Still Tubes for Refinery Service*
A 312/A 312M	*Seamless and Welded Austenitic Stainless Steel Pipe*
A 335/A 335M	*Seamless Ferritic Alloy Steel Pipe for High-Temperature Service*
A 376/A 376M	*Seamless Austenitic Steel Pipe for High-Temperature Central-Station Service*
A 608	*Centrifugally Cast Iron-Chromium-Nickel High-Alloy Tubing for Pressure Application at High Temperatures*
B407	*Nickel-Iron-Chromium Alloy Seamless Pipe and Tube*

SECTION 2—DESIGN

2.1 General

There is a fundamental difference between the behavior of carbon steel in a hot-oil heater tube operating at 575°F (300°C) and that of chromium-molybdenum steel in a catalytic-reformer heater tube operating at 1110°F (600°C). The steel operating at the higher temperature will creep, or deform permanently, even at stress levels well below the yield strength. When the tube metal temperature is high enough for the effects of creep to be significant, the tube will eventually fail from creep rupture, even when a corrosion or oxidation mechanism is not active. For the steel operating at the lower temperature, the effects of creep will be nonexistent or negligible. Experience indicates that in this case the tube will last indefinitely unless a corrosion or oxidation mechanism is active.

Since there is a fundamental difference between the behavior of the materials at these two temperatures, there are two different design considerations for heater tubes—elastic design and creep-rupture design. Elastic design is design in the elastic range, at lower temperatures, in which allowable stresses are based on the yield strength (see 2.3). Creep-rupture design (which is referred to below as rupture design) is design in the creep-rupture range, at higher temperatures, in which allowable stresses are based on the rupture strength (see 2.4).

The temperature that separates the elastic and creep rupture ranges of a heater tube is not a single value; it is a range of temperatures that depends on the alloy. For carbon steel, the lower end of this temperature range is about 800°F (425°C); for Type 347 stainless steel, the lower end of this temperature range is about 1100°F (590°C). The considerations that govern the design range also include the elastic and rupture design pressures, the design life, and the corrosion allowance.

In the temperature range near or above the point where the elastic and rupture allowable stress curves cross, both elastic and rupture design equations must be used. The larger value of t_m should govern the design (see 2.5). A sample calculation that uses these methods is included in Section 4. Calculation sheets (see Appendix F) are available for summarizing the calculations of minimum thickness and equivalent tube metal temperature.

The minimum allowable thickness of a new tube is given in Table 1.

All of the design equations described in this section are summarized in Table 2.

2.2 Equation for Stress

In both the elastic range and the creep-rupture range, the design equation is based on the mean-diameter equation for stress in a tube. In the elastic range, the elastic design pressure (P_e) and the elastic allowable stress (S_e) are used. In the creep-rupture range, the rupture design pressure (P_r) and the rupture allowable stress (S_r) are used.

The mean-diameter equation gives a good estimate of the pressure that will produce yielding through the entire tube wall in thin tubes (see 1.3.4 for a definition of thin tubes).The mean-diameter equation also provides a good correlation between the creep rupture of a pressurized tube and a uniaxial test specimen. It is therefore a good equation to use in both the elastic range and the creep-rupture range [References 2, 3, 4, 5]. The mean diameter equation for stress is as follows:

$$S = \frac{P}{2}\left(\frac{D_o}{t} - 1\right) = \frac{P}{2}\left(\frac{D_i'}{t} + 1\right) \quad (1)$$

Where:

S = stress, in pounds per square inch (megapascals).
P = pressure, in pounds per square inch (megapascals).
D_o = outside diameter, in inches (millimeters).
D'_i = inside diameter, in inches (millimeters).
t = thickness, in inches (millimeters).

The equations for the stress thickness (t_s) in 2.3 and 2.4 were derived from Equation 1.

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2.3 Elastic Design (Lower Temperatures)

The elastic design is based on preventing failure by bursting when the pressure is at its maximum (that is, when a pressure excursion has reached P_e) near the end of the design life after the corrosion allowance has been used up. With the elastic design, t_s and t_m (see 2.6) are calculated as follows:

$$t_s = \frac{P_e D_o}{2S_e + P_e} \quad \text{or} \quad t_s = \frac{P_e D_i'}{2S_e - P_e} \tag{2}$$

$$t_m = t_s + CA \tag{3}$$

Where:

S_e = the elastic allowable stress at the design metal temperature, in pounds per square inch (megapascals).

2.4 Rupture Design (Higher Temperatures)

The rupture design is based on preventing failure by creep rupture during the design life. With the rupture design, t_s and t_m (see 2.6) are calculated as follows:

$$t_s = \frac{P_r D_o}{2S_r + P_r} \quad \text{or} \quad t_s = \frac{P_r D_i'}{2S_r - P_r} \tag{4}$$

$$t_m = t_s + fCA \tag{5}$$

Where:

S_r = rupture allowable stress at the design metal temperature and the design life, in pounds per square inch (megapascals).

f = corrosion fraction given as a function of B and n in Figure 1.

B = CA/t_s.

n = rupture exponent at the design metal temperature (shown on the figures in 3.8).

The derivation of the corrosion fraction is described in Appendix B. It is recognized in this derivation that stress is reduced by the corrosion allowance; correspondingly, the rupture life is increased.

This design equation is suitable for heater tubes; however, if special circumstances require that the user choose a more conservative design, a corrosion fraction of unity (f = 1) may be specified.

2.5 Intermediate Temperature Range

At temperatures near or above the point where the curves of S_e and S_r intersect on the figures in 3.8, either elastic or rupture considerations will govern the design. In this temperature range, both the elastic and rupture designs should be applied. The larger value of t_m shall govern the design.

2.6 Minimum Allowable Thickness

The minimum thickness (t_m) of a new tube (including the corrosion allowance) shall not be less than that shown in Table 1. For ferritic steels, the values shown are the minimum thicknesses of Schedule 40 average wall pipe. For austenitic steels, the values are the minimum thicknesses of Schedule 10S average wall pipe. (Table 5 shows which alloys are ferritic and which are austenitic.) The minimum thicknesses are 0.875 times the average thicknesses. These minimums are based on industry practice. The minimum thickness is not the retirement or replacement thickness of a used tube.

2.7 Minimum and Average Thicknesses

The minimum thickness (t_m) is calculated as described in 2.3 and 2.4. Tubes that are purchased to this minimum thickness will have an average thickness that is greater. A thickness tolerance is specified in each ASTM specification. For most of the ASTM specifications shown on the figures in 3.8, the tolerance on the minimum thickness is –0, +28 percent for hot-finished tubes and –0, +22 percent for cold-drawn tubes. This is equivalent to tolerances on the average thickness of ±12.3 percent and ±9.9 percent, respectively. The remaining ASTM specifications require that the minimum thickness be greater than 0.875 times the average thickness, which is equivalent to a tolerance on the average thickness of ±12.5 percent.

Table 1—Minimum Allowable Thickness of New Tubes

Tube Outside Diameter		Minimum Thickness			
		Ferritic Steel Tubes		Austenitic Steel Tubes	
Inches	Millimeters	Inches	Millimeters	Inches	Millimeters
2.375	60.3	0.135	3.4	0.095	2.4
2.875	73.0	0.178	4.5	0.105	2.7
3.50	88.9	0.189	4.8	0.105	2.7
4.00	101.6	0.198	5.0	0.105	2.7
4.50	114.3	0.207	5.3	0.105	2.7
5.563	141.3	0.226	5.7	0.117	3.0
6.625	168.3	0.245	6.2	0.117	3.0
8.625	219.1	0.282	7.2	0.130	3.3
10.75	273.1	0.319	8.1	0.144	3.7

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Table 2—Summary of Working Equations

Elastic design (lower temperatures):

$$t_s = \frac{P_e D_o}{2S_e + P_e} \quad \text{or} \quad t_s = \frac{P_e D_i'}{2S_e - P_e} \tag{2}$$

$$t_m = t_s + CA \tag{3}$$

Rupture design (higher temperatures):

$$t_s = \frac{P_r D_o}{2S_r + P_r} \quad \text{or} \quad t_s = \frac{P_r D_i'}{2S_r - P_r} \tag{4}$$

$$t_m = t_s + fCA \tag{5}$$

Where:

t_s = stress thickness, in inches (millimeters).
P_e = elastic design pressure, in pounds per square inch gauge [megapascals (gauge)].
P_r = rupture design pressure, in pounds per square inch gauge [megapascals (gauge)].
D_o = outside diameter, in inches (millimeters).
D_i' = inside diameter with corrosion allowance removed, in inches (millimeters).
S_e = elastic allowable stress at the design metal temperature, in pounds per square inch (megapascals).
S_r = rupture allowable stress at the design metal temperature and design life, in pounds per square inch (megapascals).
t_m = minimum thickness, including corrosion allowance, in inches (millimeters).
CA = corrosion allowance, in inches (millimeters).
f = corrosion fraction, given in Figure 1 as a function of B and n.

$$B = \frac{CA}{t_s}$$

n = rupture exponent at the design metal temperature.

Equivalent tube metal temperature:

$$T_e = T_{sor} + f_T (T_{eor} - T_{sor}) \tag{6}$$

Where:

T_e = equivalent tube metal temperature, in degrees Fahrenheit (degrees Celsius).
T_{sor} = tube metal temperature at start of run, in degrees Fahrenheit (degrees Celsius).
T_{eor} = tube metal temperature at end of run, in degrees Fahrenheit (degrees Celsius).
f_T = temperature fraction, given in Figure 2 as a function of V and N.

$$V = n_0\left(\frac{\Delta T}{T_s}\right) \ln\left(\frac{A}{S_0}\right)$$

$$N = n_0\left(\frac{\Delta t}{t_0}\right)$$

n_0 = rupture exponent at T_{sor}.
ΔT = $T_{eor} - T_{sor}$, temperature change during operating period, in degrees Fahrenheit (degrees Celsius).
T_s = T_{sor} + 460, in degrees Rankine (T_{sor} + 273, in kelvin).
A = material constant, from Table 3, in pounds per square inch (megapascals).
S_e = initial stress at start of run using Equation 1, in pounds per square inch (megapascals).
Δt = RL_o, thickness change during operating period, in inches (millimeters).
R = corrosion rate, in inches per year (millimeters per year).
L_0 = duration of operating period, in years.
t_0 = initial thickness at start of run, in inches (millimeters).

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With a –0, +28-percent tolerance, a tube that is purchased to a 0.500-inch (12.7-millimeter) minimum-thickness specification will have the following average thickness:

(0.500)(1 + 0.28/2) = 0.570 inch (14.5 millimeters)

To obtain a minimum thickness of 0.500 inch (12.7 millimeters) in a tube purchased to a ±12.5-percent tolerance on the average thickness, the average thickness must be specified as follows:

(0.500)(1/0.875) = 0.571 inch (14.5 millimeters)

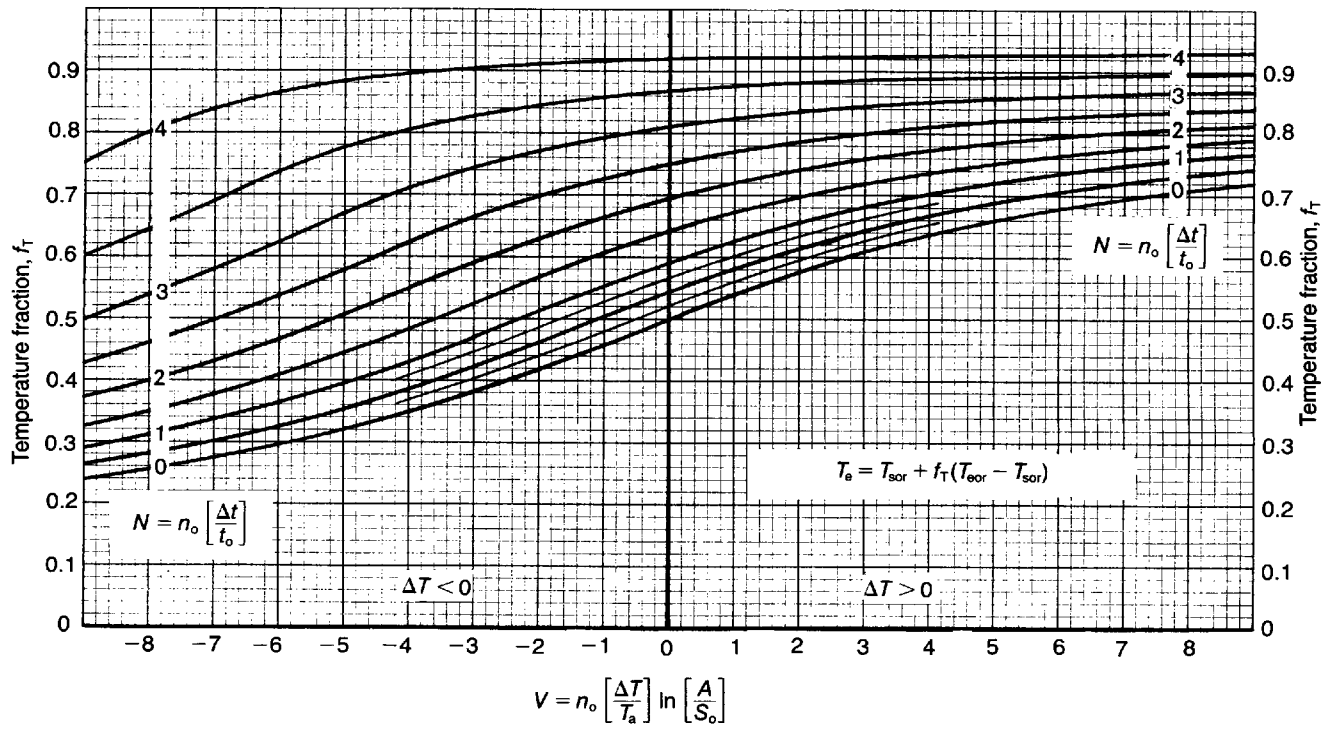
All thickness specifications shall indicate whether the specified value is a minimum or an average thickness. The tolerance used to relate the minimum and average wall thicknesses shall be the tolerance given in the ASTM specification to which the tubes will be purchased.

2.8 Equivalent Tube Metal Temperature

In the creep-rupture range, the accumulation of damage is a function of the actual operating temperature. For applications in which there is a significant difference between start-of-run and end-of-run metal temperatures, a design based on the maximum temperature may be excessive, since the actual operating temperature will usually be less than the maximum.

For a linear change in metal temperature from start of run (T_{sor}) to end of run (T_{eor}), an equivalent tube metal temperature (T_e) can be calculated as shown below. A tube operating at the equivalent tube metal temperature will sustain the same creep damage as one that operates from the start-of-run to end-of-run temperatures.

$$T_e = T_{sor} + f_T (T_{eor} - T_{sor}) \qquad (6)$$



n_0 = rupture exponent at T_{sor}.
ΔT = $T_{eor} - T_{sor}$, temperature change during operating period, in degrees Fahrenheit (degrees Celsius).
T_a = T_{sor} + 460, in degrees Rankine (T_{sor} + 273, in degrees kelvins).
ln = natural logarithm.
Δt = RL_o, thickness change during operating period, in inches (millimeters).
R = corrosion rate, in inches per year (millimeters per year).
L_o = duration of operating period, in years.
t_0 = initial thickness at start of run, in inches (millimeters).
S_0 = initial stress at start of run using Equation 1, in pounds per square inch (megapascals).
A = material constant, in pounds per square inch (megapascals). See Table 3.

Figure 2—Temperature Fraction

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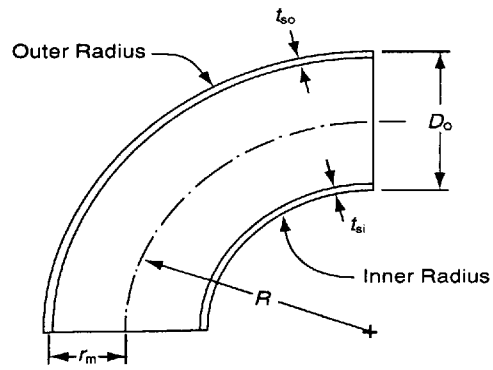


Figure 3—Return Bend and Elbow Geometry

Where:

T_e = equivalent tube metal temperature, in degrees Fahrenheit (degrees Celsius).
T_{sor} = tube metal temperature at start of run, in degrees Fahrenheit (degrees Celsius).
T_{eor} = tube metal temperature at end of run, in degrees Fahrenheit (degrees Celsius).
f_T = temperature fraction given in Figure 2.

The derivation of the temperature fraction is described in Appendix B. The temperature fraction is a function of two parameters, V and N:

$$V = n_0\left(\frac{\Delta T}{T_a}\right)\ln\left(\frac{A}{S_0}\right)$$

$$N = n_0\left(\frac{\Delta t}{t_0}\right)$$

Where:

n_0 = rupture exponent at T_{sor}.
$\Delta T = T_{eor} - T_{sor}$, temperature change during operating period, in degrees Fahrenheit (degrees Celsius).
$T_a = T_{sor} + 460$, in degrees Rankine ($T_{sor} + 273$, in degrees kelvins).
ln = natural logarithm.
A = material constant, in pounds per square inch (megapascals).
S_0 = initial stress at start of run using Equation 1, in pounds per square inch (megapascals).
$\Delta t = RL_o$, thickness change during operating period, in inches (millimeters).
R = corrosion rate, in inches per year (millimeters per year).
L_o = duration of operating period, in years.
t_0 = initial thickness at start of run in inches (millimeters).

The constant A is given in Table 3. The significance of the material constant is explained in B.5.

The temperature fraction and the equivalent temperature shall be calculated for the first operating cycle. In applications that involve very high corrosion rates, the temperature fraction for the last cycle will be greater than that for the first. In such cases, the calculation of the temperature fraction and the equivalent temperature should be based on the last cycle.

If the temperature change from start of run to end of run is other than linear, a judgment shall be made regarding the use of the value of f_T given in Figure 2. Note that the calculated thickness of a tube is a function of the equivalent temperature, which in turn is a function of the thickness (through the initial stress). A few iterations may therefore be necessary to arrive at the design (see the sample calculation in 4.4).

2.9 Return Bends and Elbows

The design procedure in this section shall be applied to austenitic stainless steel return bends and elbows (see Figure 3) located in the firebox and operating in the elastic range. In this situation, the allowable stress does not vary much with temperature. This design procedure shall also be applied in other situations, if applicable.

The stress variations in a return bend or elbow are far more complex than in a straight tube. The hoop stresses at the inner radius (or crotch) of a return bend are higher than in a straight tube of the same thickness. In the situation defined above, the minimum thickness at the crotch may need to be greater than the minimum thickness of the attached tube.

Due to the reduced exposure to radiant gases, the metal

Table 3—Material Constant for Temperature Fraction

Material	Type or Grade	Constant A	
		Pounds Per Square Inch	Megapascals
Low-carbon steel		1.08×10^8	7.46×10^5
Medium-carbon steel	B	4.17×10^7	2.88×10^5
C-$^1/_2$Mo steel	T1 or P1	2.91×10^9	2.01×10^7
1-$^1/_4$Cr-$^1/_2$Mo steel	T11 or P11	7.49×10^9	5.17×10^7
2-$^1/_4$Cr-1Mo steel	T22 or P22	1.25×10^8	8.64×10^5
3Cr-1Mo steel	T21 or P21	3.07×10^8	2.12×10^6
5Cr-$^1/_2$Mo steel	T5 or P5	7.97×10^7	5.49×10^5
5Cr-$^1/_2$Mo-Si steel	T5b or P5b	4.18×10^7	2.88×10^5
7Cr-$^1/_2$ Mo steel	T7 or P7	2.37×10^7	1.64×10^5
9Cr-1Mo steel	T9 or P9	1.09×10^9	7.54×10^6
9Cr-1Mo Va steel	T91 or P91	3.24×10^8	2.23×10^6
18Cr-8Ni steel	304 or 304H	2.25×10^8	1.55×10^6
16Cr-12Ni-2Mo steel	316 or 316H	1.79×10^8	1.24×10^6
16Cr-12Cr-2Mo steel	316L	1.99×10^8	1.37×10^6
18Cr-10Ni-Ti steel	321	1.92×10^8	1.32×10^6
18Cr-10Ni-Ti steel	321H	4.00×10^7	2.76×10^5
18Cr-10Ni-Cb steel	347 or 347H	1.79×10^8	1.23×10^6
Ni-Fe-Cr	Alloy 800H	1.50×10^7	1.03×10^5
25Cr-20Ni	HK-40	3.63×10^7	2.50×10^5

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temperature at the inner radius of a return bend in the firebox is lower than that of the fully exposed surface of the straight tube. For this reason and because modern fabrication processes for forged return bends result in greater thickness at the inner radius, the higher stresses at the inner radius can be sustained without failure in most situations.

The hoop stress along the inner radius of the bend is given by

$$S_i = \frac{2R - r_m}{2(R - r_m)} S \tag{7}$$

Where:

S_i = stress, in pounds per square inch (megapascals).
R = center line radius of the bend, in inches (millimeters).
r_m = mean radius of the tube, in inches (millimeters).
S = stress given by Equation 1, in pounds per square inch (megapascals).

The hoop stress along the outer radius is given by

$$S_o = \frac{2R + r_m}{2(R + r_m)} S \tag{8}$$

Where:

S_o = stress, in pounds per square inch (megapascals).

Using the approximation that r_m is almost equal to $D_o/2$, Equation 7 can be solved for the stress thickness at the inner radius. For elastic design the stress thickness is as follows:

$$t_{si} = \frac{D_o P_e}{2N_i S_e + P_e} \tag{9}$$

Where:

t_{si} = stress thickness at inner radius, in inches (millimeters).

$$N_i = \frac{4\frac{R}{D_o} - 2}{4\frac{R}{D_o} - 1} \tag{10}$$

S_e = elastic allowable stress at the design metal temperature, in pounds per square inch (megapascals).

The design metal temperature shall be the estimated temperature at the inner radius plus an appropriate temperature allowance.

Using the approximation given above, Equation 8 can be solved for the stress thickness at the outer radius. For elastic design the stress thickness is as follows:

$$t_{so} = \frac{D_o P_e}{2N_o S_e + P_e} \tag{11}$$

Where:

t_{so} = stress thickness at outer radius, in inches (millimeters).

$$N_o = \frac{4\frac{R}{D_o} + 2}{4\frac{R}{D_o} + 1} \tag{12}$$

S_e = elastic allowable stress at the design metal temperature, in pounds per square inch (megapascals).

The design metal temperature shall be the estimated temperature at the outer radius plus an appropriate temperature allowance.

The minumum thickness at the inside radius, t_{si}, and outside radius, t_{so}, shall be calculated using Equations 9 and 11. The corrosion allowance, CA, shall be added to the minimum calculated thickness.

The minimum thickness along the neutral axis of the bend shall be the same as for a straight tube.

This design procedure is for return bends and elbows located in the firebox that may operate at temperatures close to that of the tubes. This procedure may not be applicable to these fittings if they are located in header boxes since they will operate at lower temperatures. Other considerations, such as hydrostatic test pressure, may govern the design of fittings located in header boxes.

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SECTION 3—ALLOWABLE STRESSES

3.1 General

The allowable stresses for various heater-tube alloys are plotted against design metal temperature in Figures 4A–4S. The values shown in the figures are recommended only for the design of heater tubes. These figures show two different allowable stresses, the elastic allowable stress and the rupture allowable stress. The bases for these allowable stresses are given in 3.2 and 3.3 (see also 1.3.3).

3.2 Elastic Allowable Stress

The elastic allowable stress (S_e) is two-thirds the yield strength at temperature for ferritic steels and 90 percent of the yield strength at temperature for austenitic steels. The data sources for the yield strength are given in Appendix A.

If a different design basis is desired for special circumstances, the user shall specify the basis, and the alternative elastic allowable stress shall be developed from the yield strength.

3.3 Rupture Allowable Stress

The rupture allowable stress (S_r) is 100 percent of the minimum rupture strength for a specified design life. Appendix A defines the minimum rupture strength and provides the data sources. The 20,000-, 40,000-, 60,000-, and 100,000-hour rupture allowable stresses were developed from the Larson-Miller parameter curves for the minimum rupture strength shown on the right-hand side of Figures 4A–4S. For a design life other than those shown, the corresponding rupture allowable stress shall be developed from the Larson-Miller parameter curves for the minimum rupture strength (see 3.6).

If a different design basis is desired, the user shall specify the basis, and the alternative rupture allowable stress shall be developed from the Larson-Miller parameter curves for the minimum or average rupture strength. If the resulting rupture allowable stress is greater than the minimum rupture strength for the design life, the effects of creep on the tube design equation shall be considered.

3.4 Rupture Exponent

Figures 4A–4S show the rupture exponent (n) as a function of the design metal temperature. The rupture exponent is used for design in the creep-rupture range (see 2.4). The meaning of the rupture exponent is discussed in A.4.

3.5 Yield and Tensile Strengths

Figures 4A–4S also show the yield and tensile strengths. These curves are included only for reference. Their sources are given in Appendix A.

Table 4—Limiting Design Metal Temperature for Heater-Tube Alloys

Materials	Type or Grade	Limiting Design Metal Temperature		Lower Critical Temperature	
		Degrees Fahrenheit	Degrees Celsius	Degrees Fahrenheit	Degrees Celsius
Carbon steel	B	1000	540	1325	720
C-½Mo steel	T1 or P1	1100	595	1325	720
1 ¼Cr-½Mo steel	T11 or P11	1100	595	1430	775
2 ¼Cr-1Mo steel	T22 or P22	1200	650	1480	805
3Cr-1Mo steel	T21 or P21	1200	650	1500	815
5Cr-½Mo steel	T5 or P5	1200	650	1510	820
5Cr-½Mo-Si steel	T5b or P5b	1300	705	1550	845
7Cr-½Mo steel	T7 or P7	1300	705	1515	825
9Cr-1Mo steel	T9 or P9	1300	705	1515	825
9Cr-1Mo-Va steel	T91 or P91	1200[a]	650	1525	830
18Cr-8Ni steel	304 or 304H	1500	815	—	—
16Cr-12Ni-2Mo steel	316 or 316H	1500	815	—	—
16Cr-12Ni-2Mo steel	316L	1500	815	—	—
18Cr-10Ni-Ti steel	321 or 321H	1500	815	—	—
18Cr-10NiCb steel	347 or 347H	1500	815	—	—
Ni-Fe-Cr	Alloy 800H	1800[a]	985	—	—
25CR-20Ni	HK-40	1850[a]	1010	—	—

[a]This is the upper limit on the reliability of the rupture strength data (see Appendix A); however, these materials are commonly used for heater tubes at higher temperatures in applications where the internal pressure is so low that rupture strength does not govern the design.

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3.6 Larson-Miller Parameter Curves

On the right-hand side of Figures 4A–4S are plots of the minimum and average 100,000-hour rupture strengths against the Larson-Miller parameter. The Larson-Miller parameter is calculated from the design metal temperature (T_d) and the design life (L_d) as follows: When T_d is expressed in degrees Fahrenheit,

$$(T_d + 460)\,(C + \log_{10} L_d) \times 10^{-3}$$

When T_d is expressed in degrees Celsius,

$$(T_d + 273)\,(C + \log_{10} L_d) \times 10^{-3}$$

The Larson-Miller constant C is stated on the curves. (See A.3 for a detailed description of these curves).

The plot of the minimum rupture strength against the Larson-Miller parameter is included so that the rupture allowable stress can be determined for any design life. The curves shall not be used to determine rupture allowable stresses for temperatures higher than the limiting design metal temperatures shown in Table 4 and Figures 4A–4S. In addition, the curves may give inaccurate rupture allowable stresses for times less than 20,000 hours or greater than 200,000 hours (see A.3).

The curves for minimum and average rupture strength can be used to calculate remaining tube life, as shown in Appendix E.

3.7 Limiting Design Metal Temperature

The limiting design metal temperature for each heater-tube alloy is given in Table 4. The limiting design metal temperature is the upper limit of the reliability of the rupture strength data. Higher temperatures—up to 50°F (30°C) below the lower critical temperature—may be permitted for short-term operating conditions, such as those that exist during steam-air decoking or regeneration. Operation at higher temperatures may result in changes in the alloy's microstructure. Lower critical temperatures for ferritic steels are shown in Table 4. Austenitic steels do not have lower critical temperatures. Other considerations that may require lower operating-temperature limits such as oxidation, graphitization, carburization, and hydrogen attack. These factors shall be considered when furnace tubes are designed.

3.8 Allowable Stress Curves

Figures 4A–4S provide the elastic allowable stress and the rupture allowable stress for most common heater-tube alloys. The sources for these curves are provided in Appendix A. The figure number for each alloy is shown in Table 5.

Table 5—Index to Allowable Stress Curves

Figure Number	Alloy
	Ferritic Steels
4A	Low-carbon steel (A 161, A 192)
4B	Medium-carbon steel (A 53B, A 106B)
4C	C-$^1/_2$ Mo
4D[a]	1 $^1/_4$Cr-$^1/_2$Mo
4E[a]	2 $^1/_4$Cr-1Mo
4F[a]	3Cr-1Mo
4G[a]	5Cr-$^1/_2$Mo
4H	5Cr-$^1/_2$Mo-Si
4I[a]	7Cr-$^1/_2$Mo
4J[a]	9Cr-1Mo
4K	9Cr-1Mo-Va
	Austenitic Steels
4L	18Cr-8Ni (304 and 304H)
4M	16Cr-12Ni-2Mo (316 and 316H)
4N	16Cr-12Ni-2Mo (316L)
4O	18Cr-10Ni-Ti (321)
4P	18Cr-10Ni-Ti (321H)
4Q	18Cr-10Ni-Cb (347 and 347H)
4R	Ni-Fe-Cr (Alloy 800H)
4S	25Cr-20Ni (HK-40)

[a]Broken lines on these figures indicate the elastic allowable stresses for the A 200 grades. These figures do not show the yield strengths of the A 200 grades. The yield strengths of the A 200 grades are 83 percent of the yield strengths shown. The tensile strengths, rupture allowable stresses, rupture strengths, and rupture exponents for the A 200 grades are the same as for the A 213 and A 335 grades.

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Future Stress Curves (Customary Units)

New stress curves may be added to this standard sometime in the future. As new stress curves are added, revision packages will be prepared and distributed.

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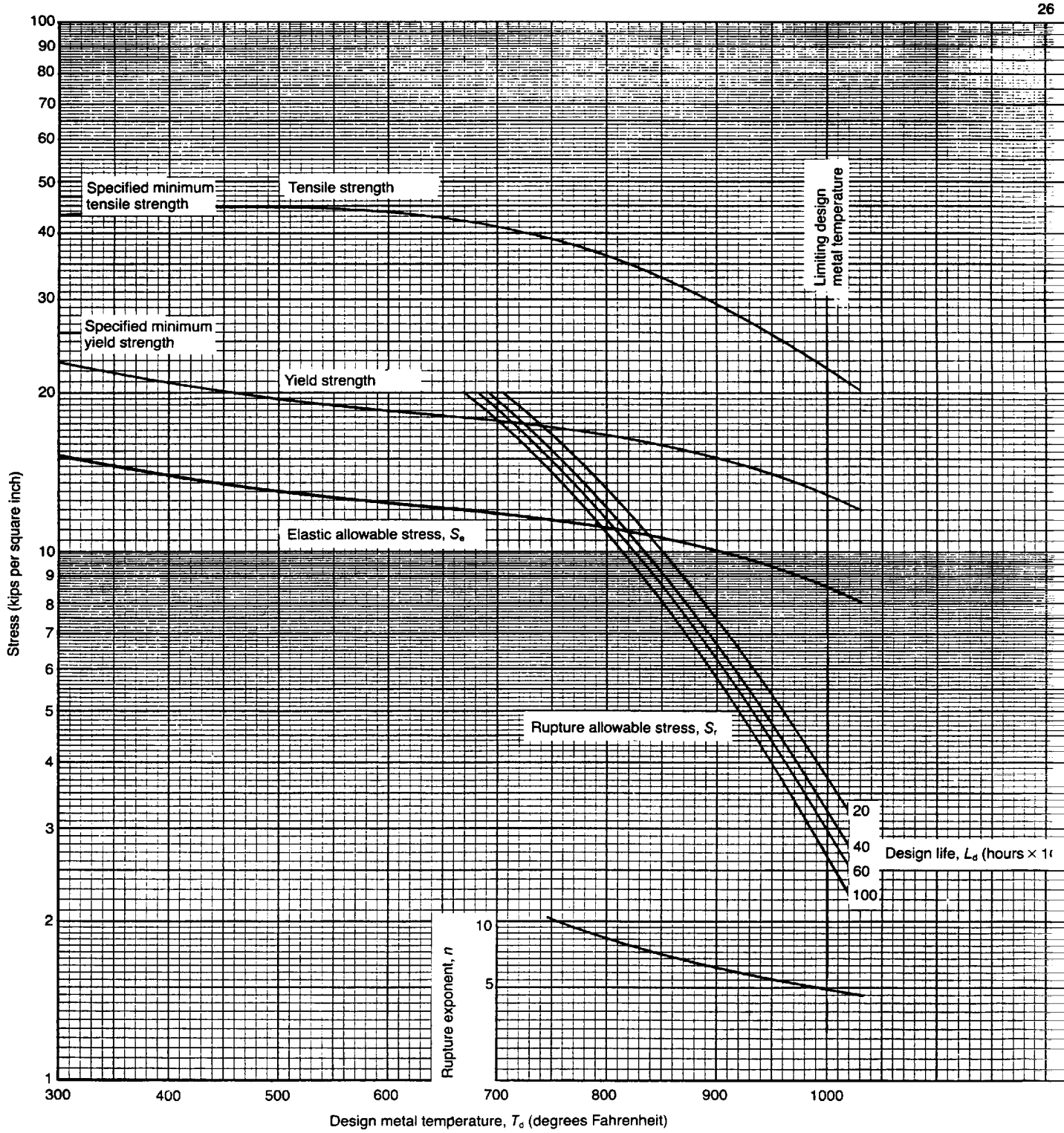


Figure 4A—Low-Carbon Steel, ASTM A 161,

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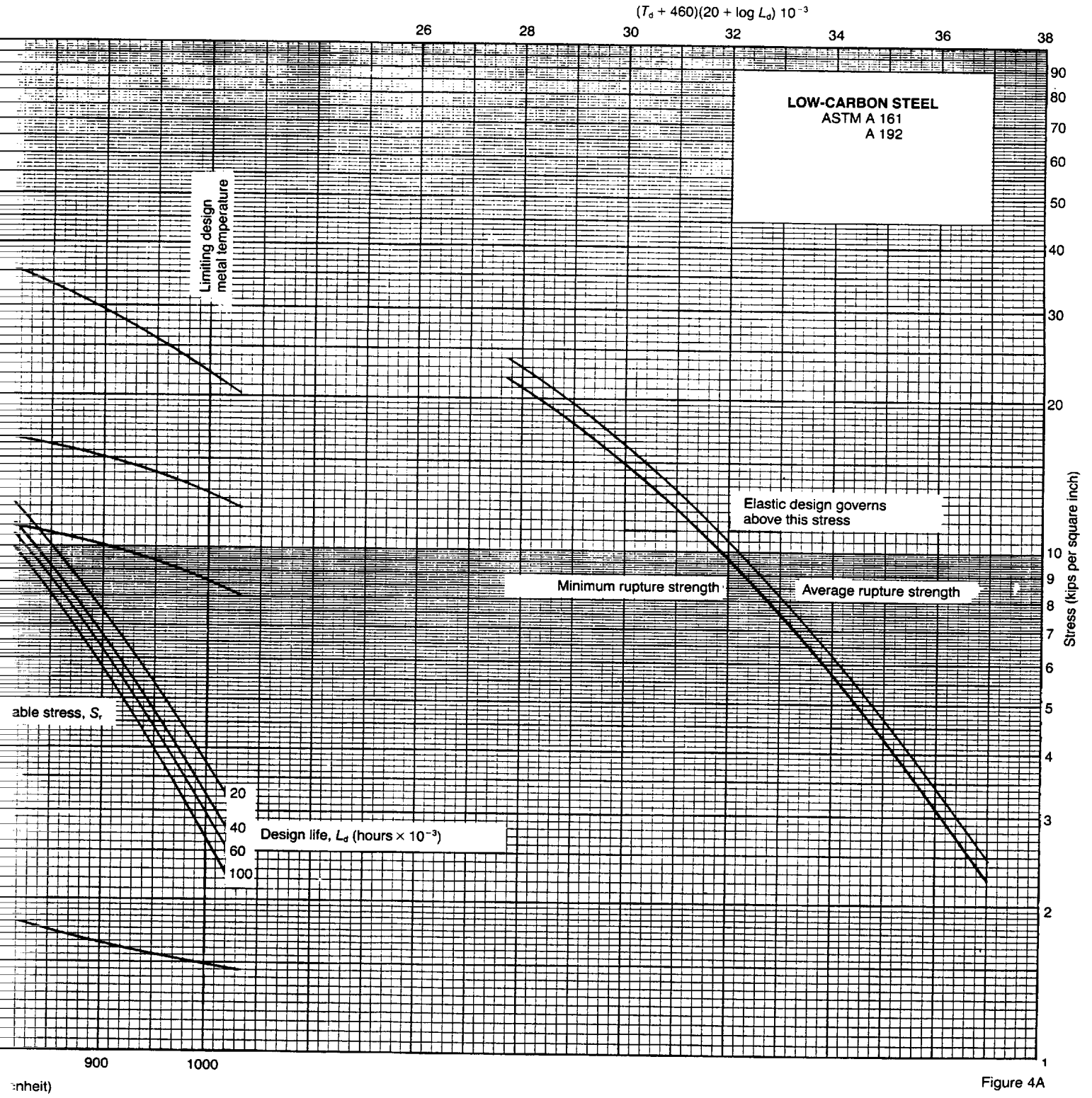


Figure 4A—Low-Carbon Steel, ASTM A 161, A 192

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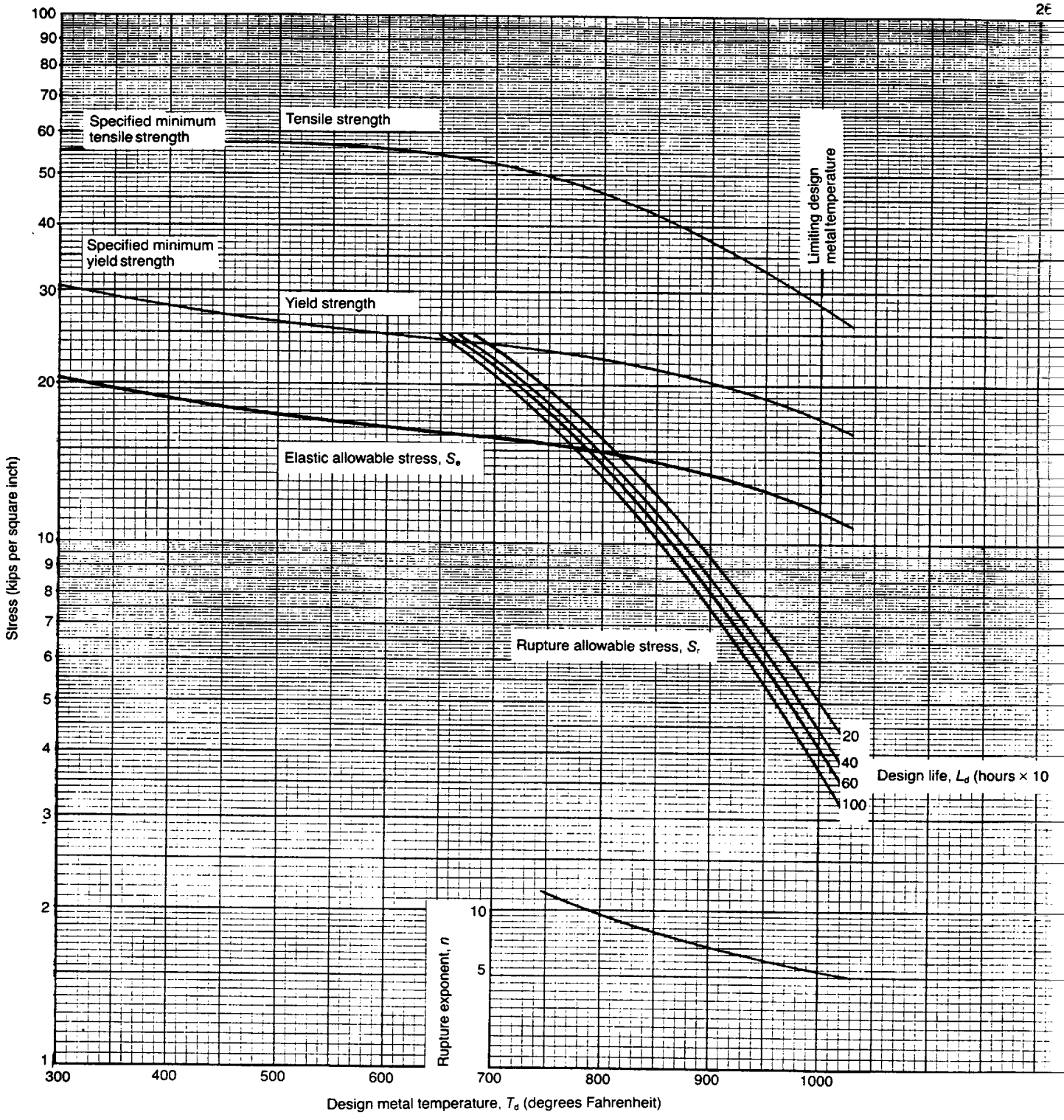


Figure 4B—Medium-Carbon Steel, ASTM A 53 Grade B (Seamless), A 1(

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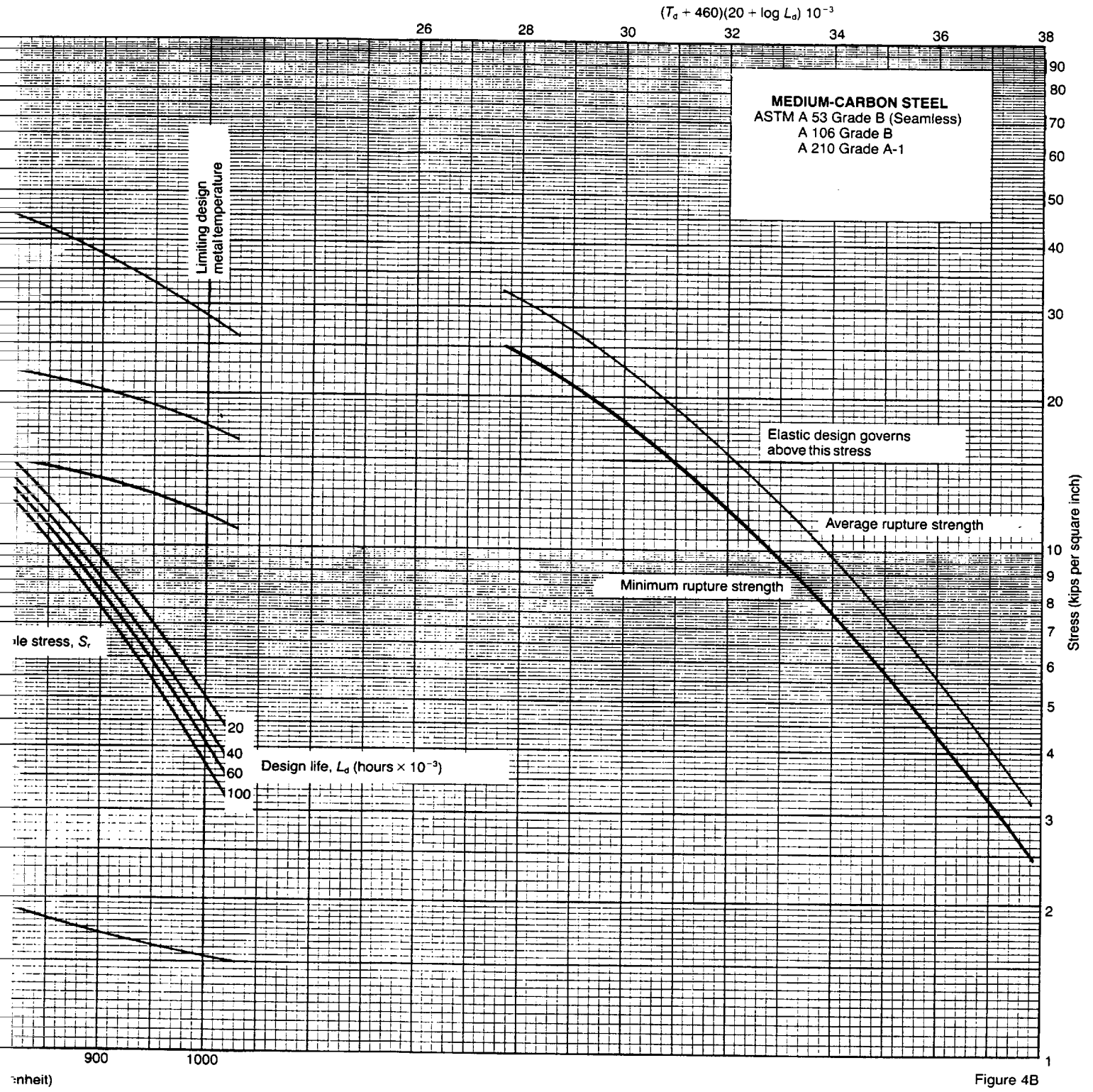


Figure 4B

rbon Steel, ASTM A 53 Grade B (Seamless), A 106 Grade B, A 210 Grade A-1

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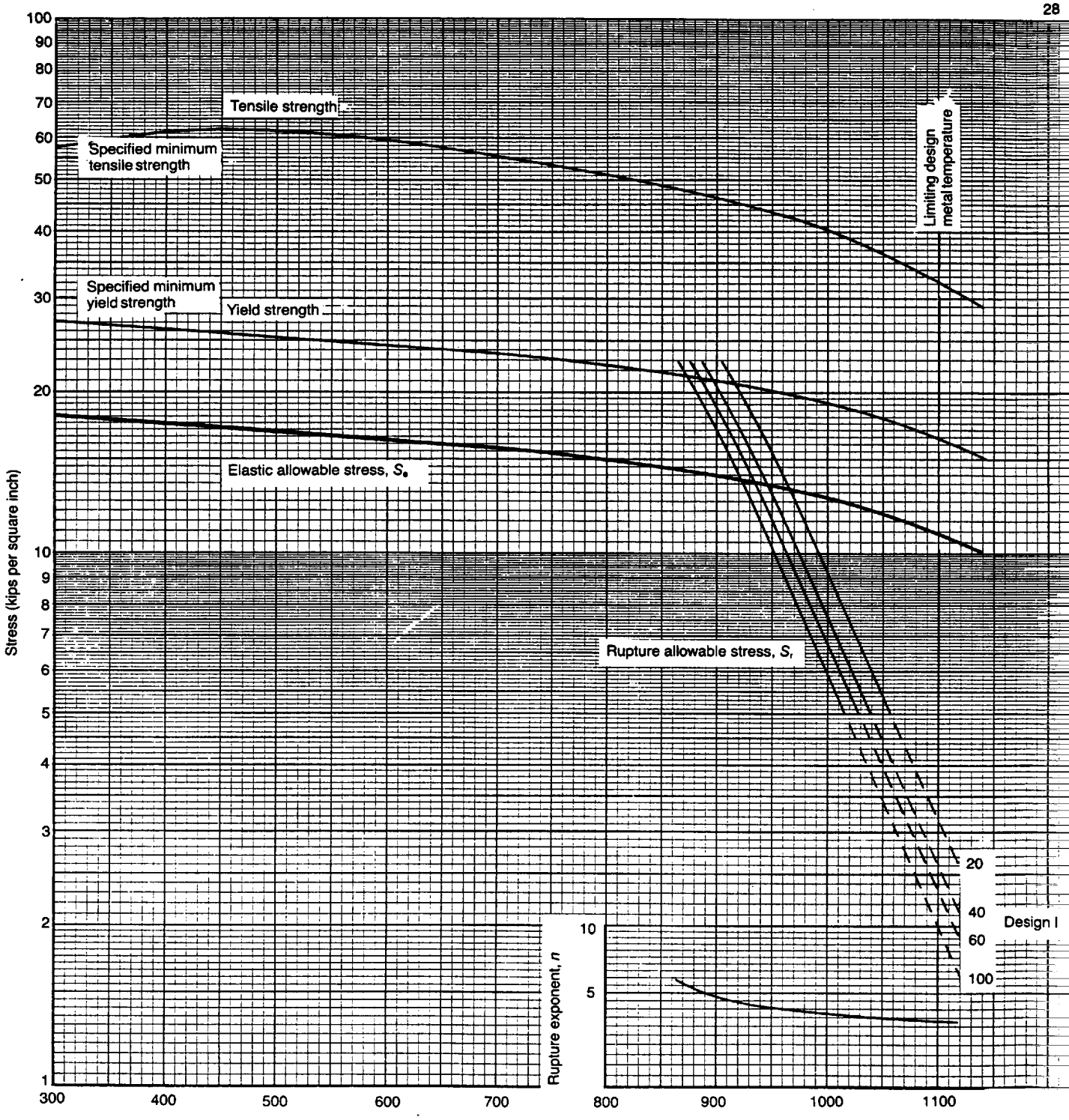


Figure 4C—C-1/2-Mo Steel, ASTM A 161 T1, A209 T

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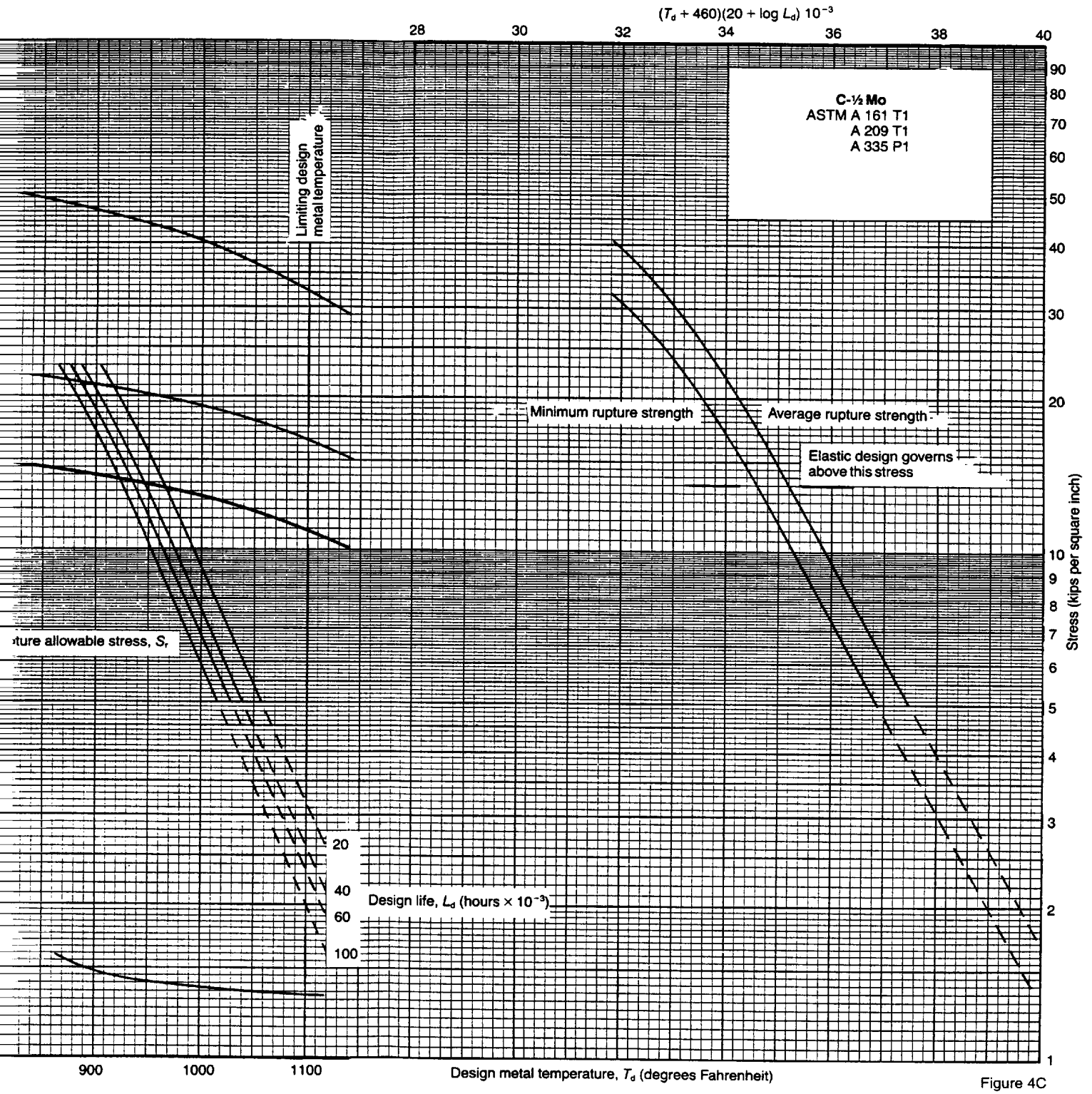


Figure 4C

ıre 4C—C-1/2-Mo Steel, ASTM A 161 T1, A209 T1, A335 P1

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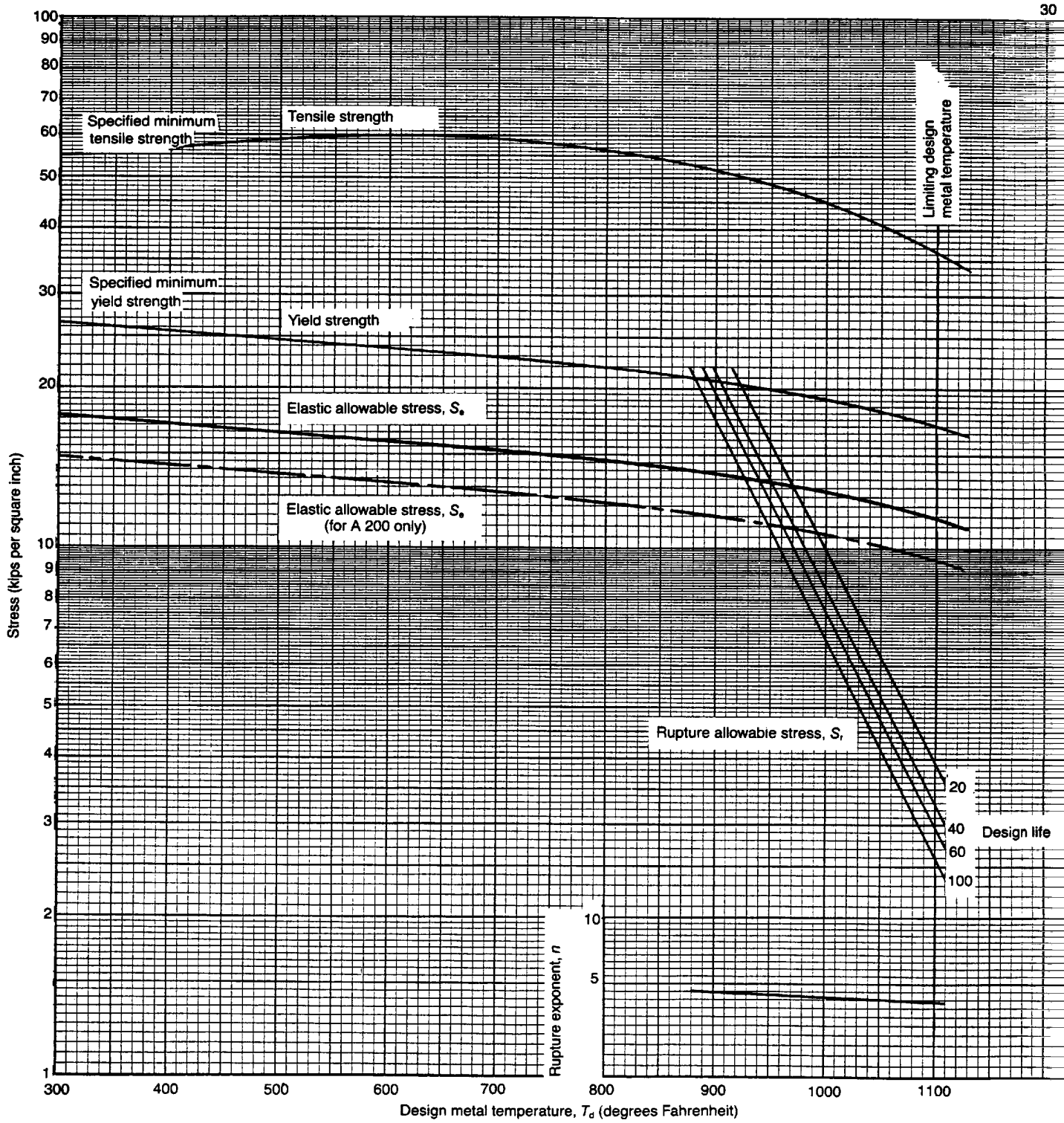


Figure 4D—1 1/4Cr–1/2Mo Steel, ASTM A 213 T11, A 335

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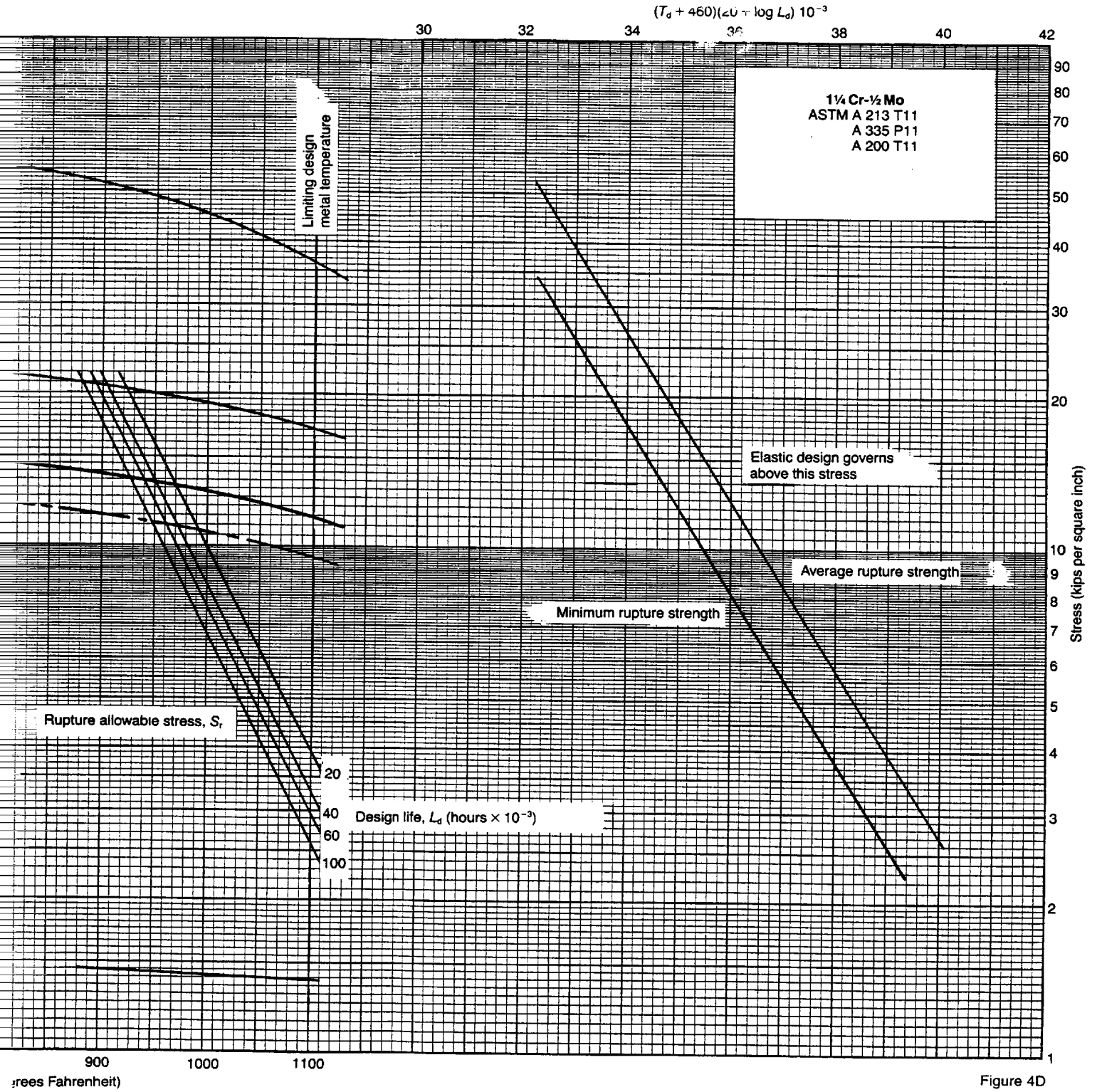


Figure 4D

4D—1¼Cr–½Mo Steel, ASTM A 213 T11, A 335 P11, A 200 T11

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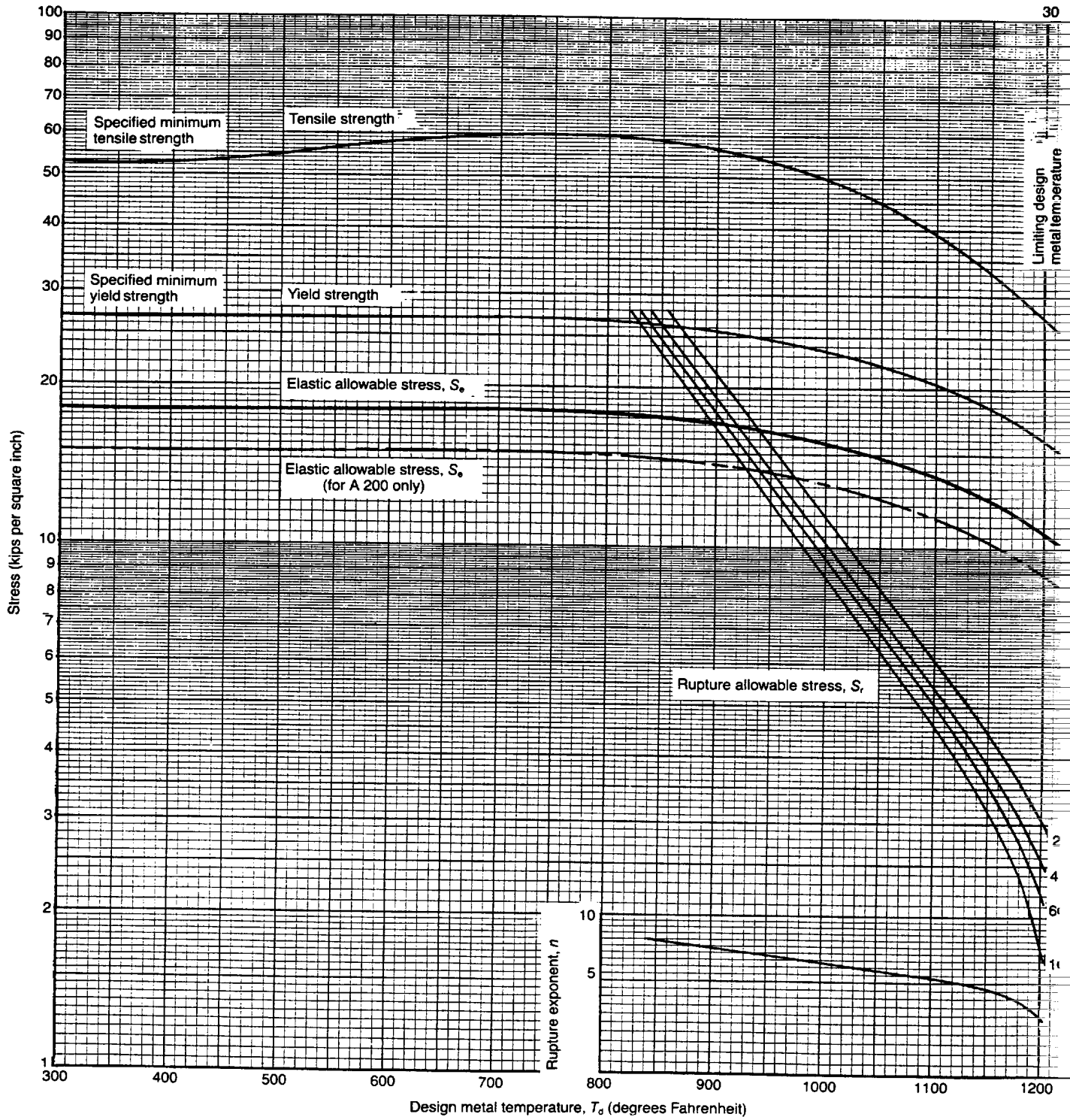


Figure 4E—2¼Cr–1Mo Steel, ASTM A 213 T22, A 335 P

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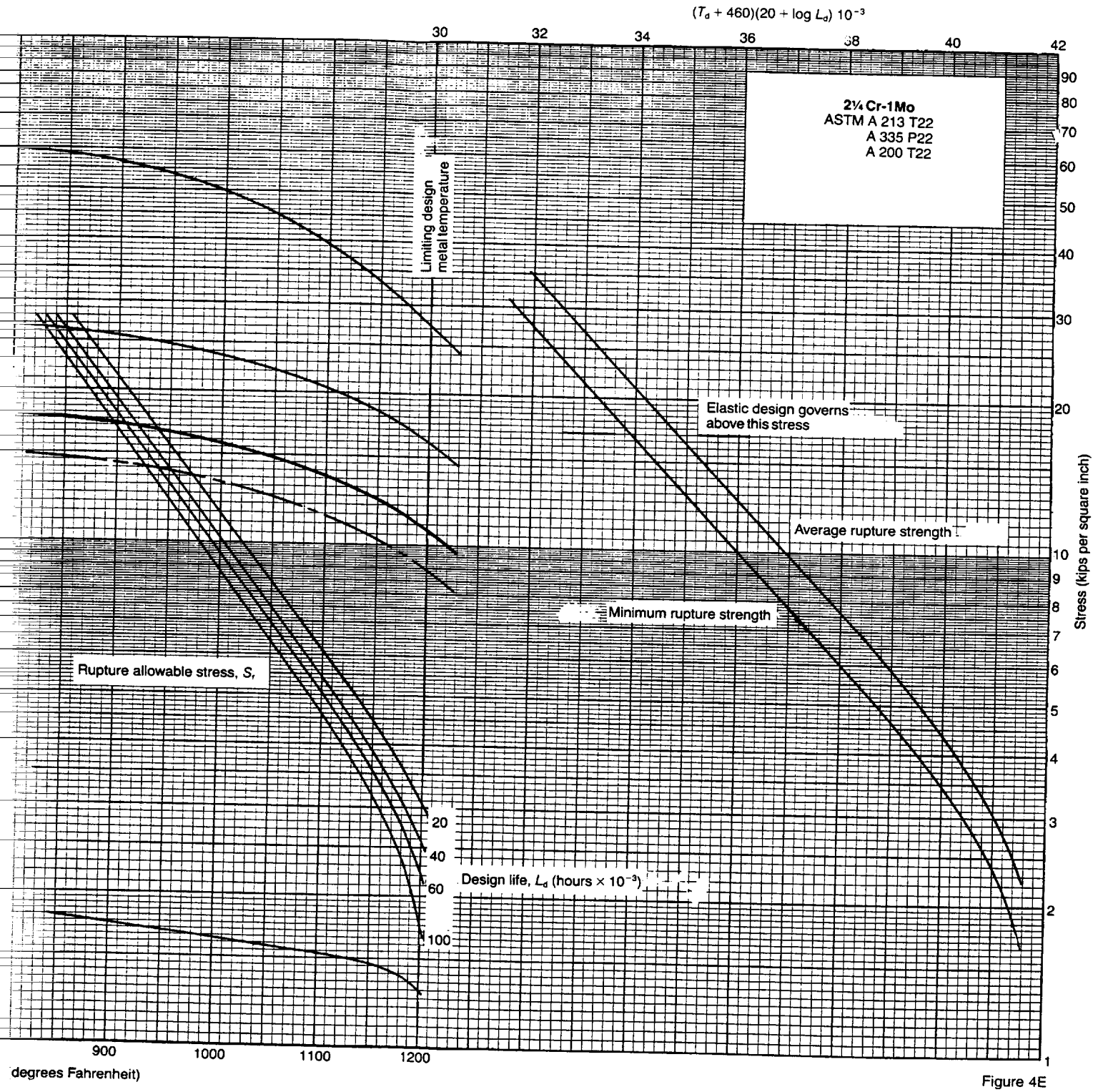


Figure 4E

4E—2¼Cr–1Mo Steel, ASTM A 213 T22, A 335 P22, A 200 T22

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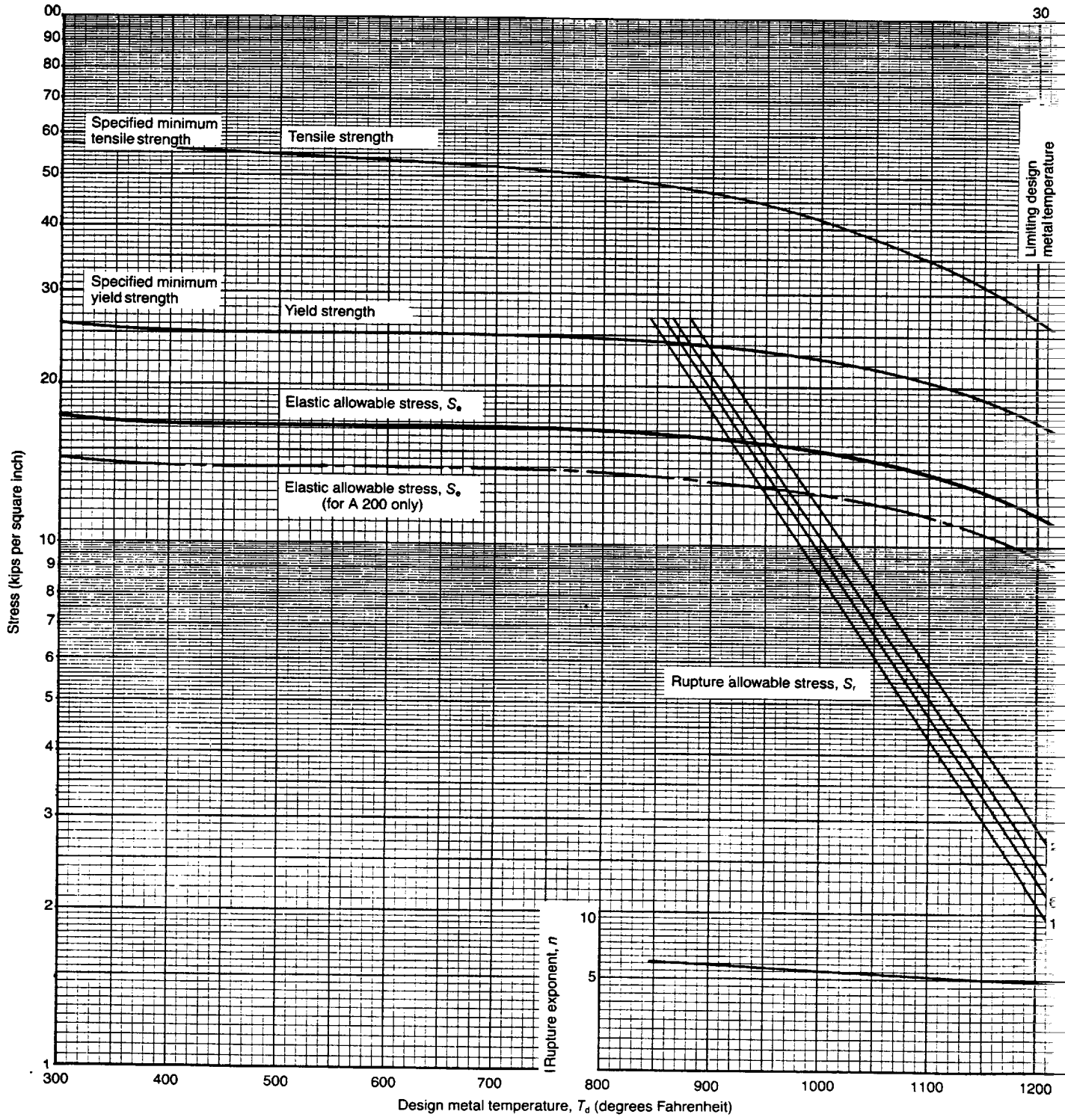


Figure 4F—3Cr–1Mo Steel, ASTM A 213 T21, A 335 P

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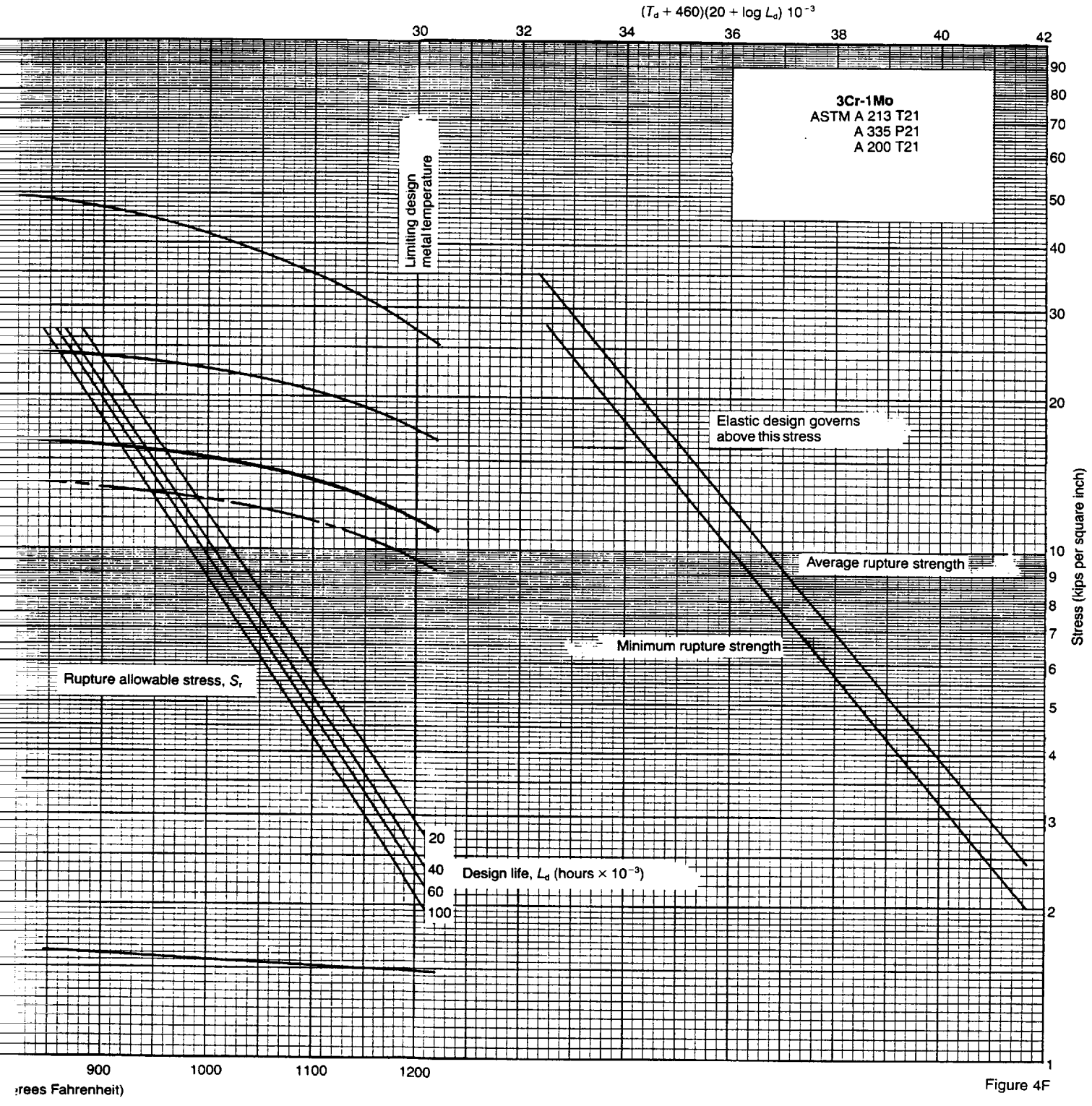


Figure 4F

e 4F—3Cr–1Mo Steel, ASTM A 213 T21, A 335 P21, A 200 T21

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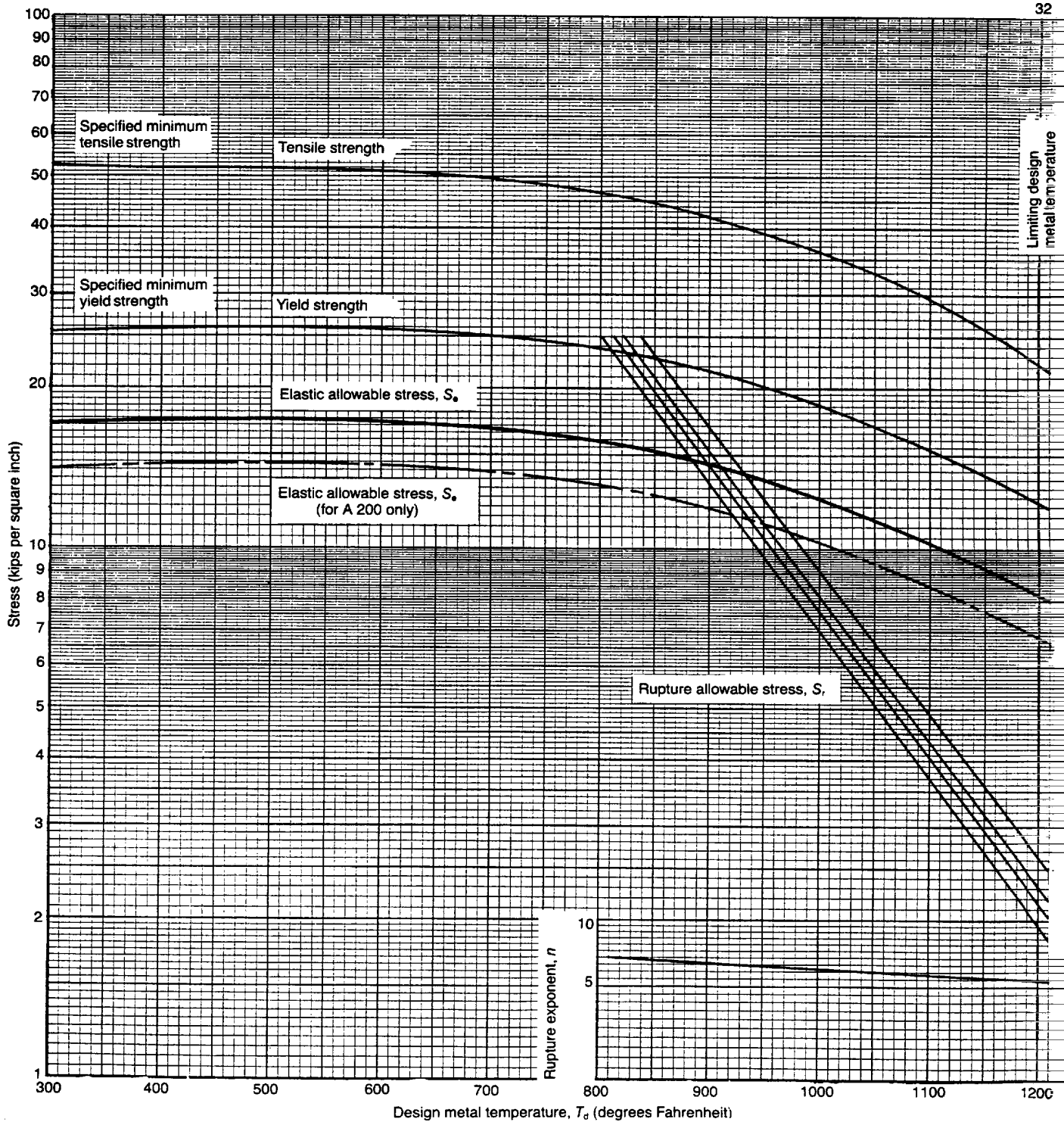


Figure 4G—5Cr–½Mo Steel, ASTM A 213 T5, A 335

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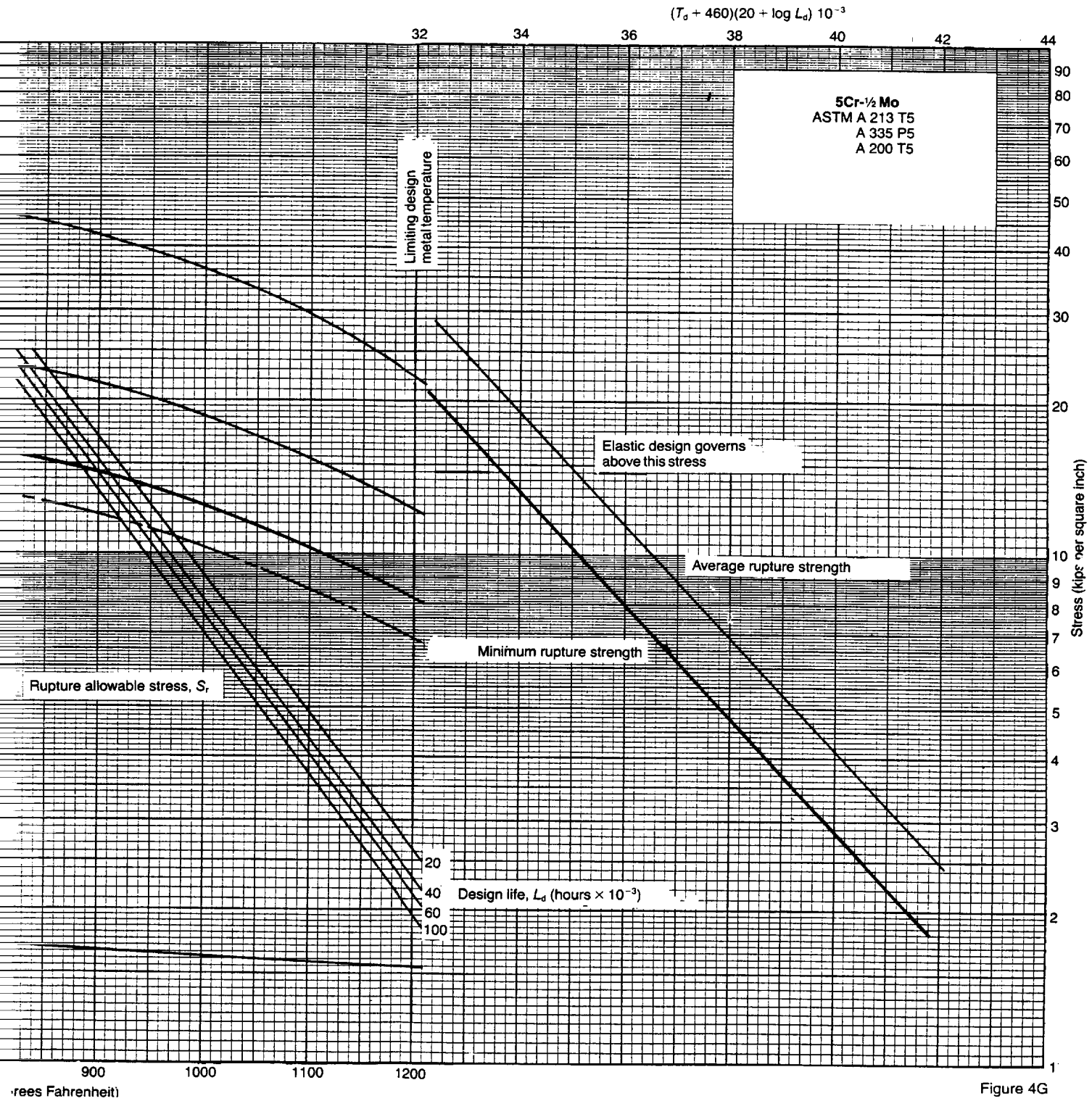


Figure 4G

e 4G—5Cr–½Mo Steel, ASTM A 213 T5, A 335 P5, A 200 T5

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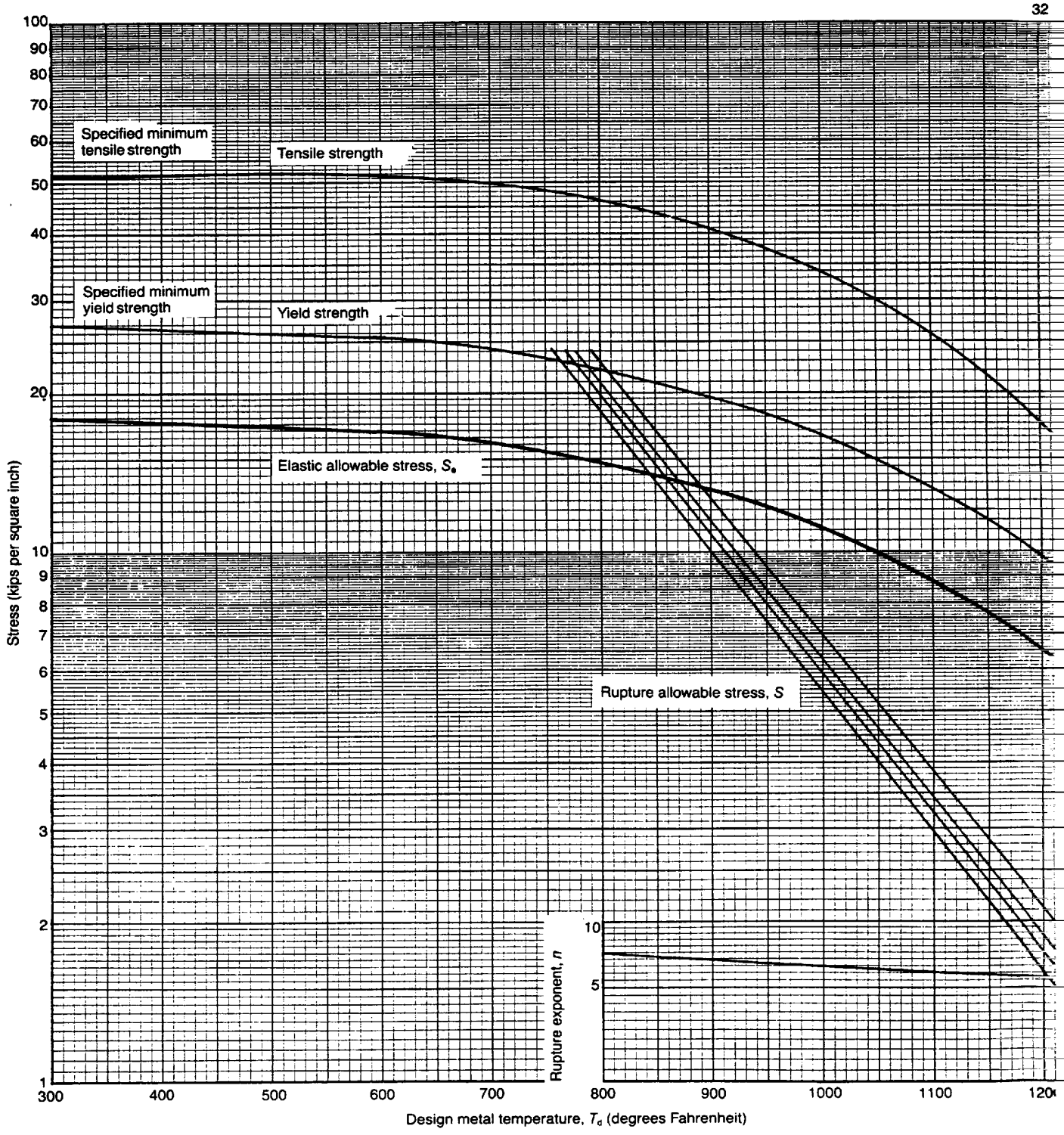
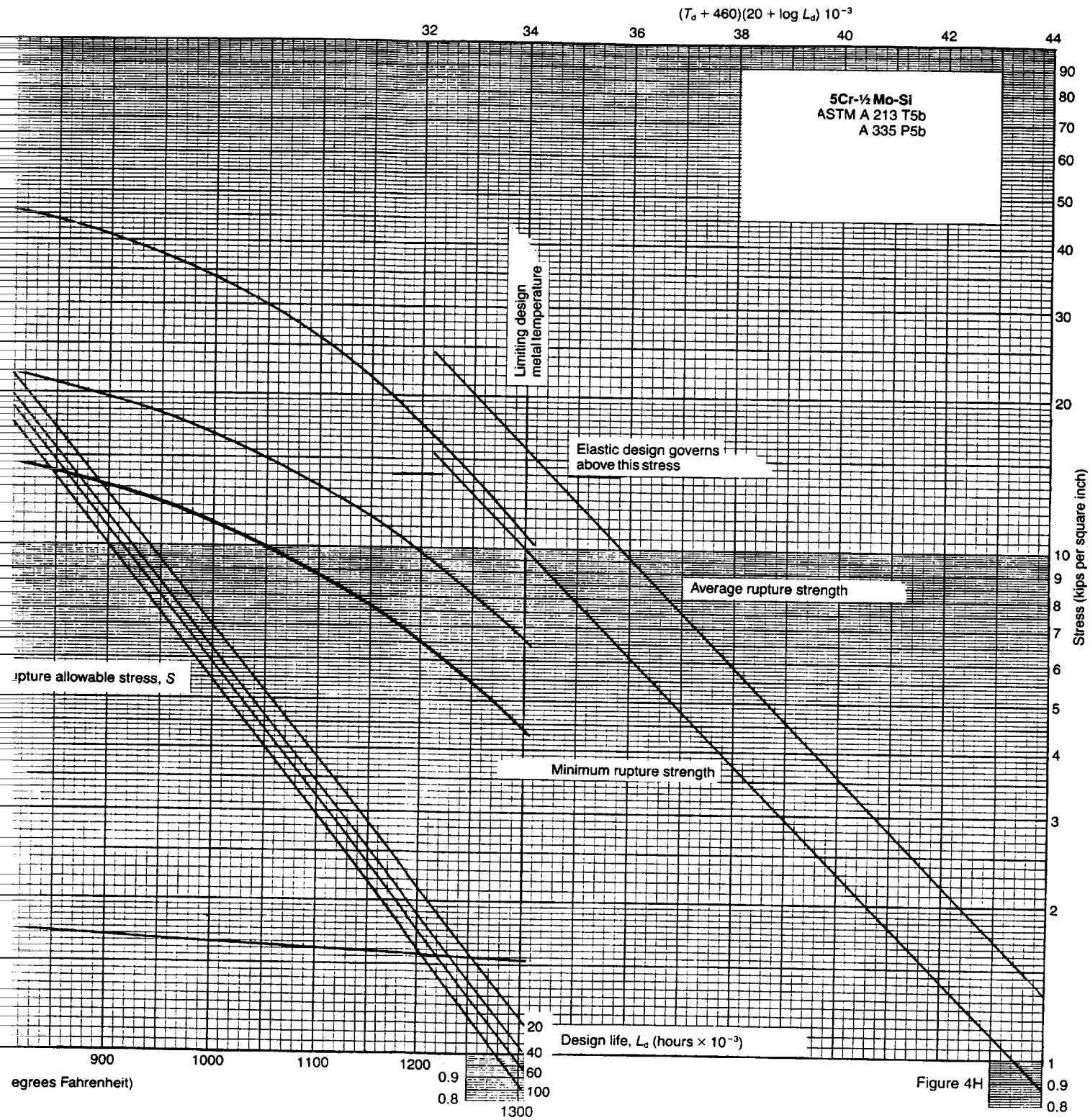


Figure 4H—5Cr–½Mo Si Steel, ASTM A 213 T5b,

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upture allowable stress, *S*

egrees Fahrenheit)

Figure 4H

igure 4H—5Cr–½Mo Si Steel, ASTM A 213 T5b, A 335 P5b

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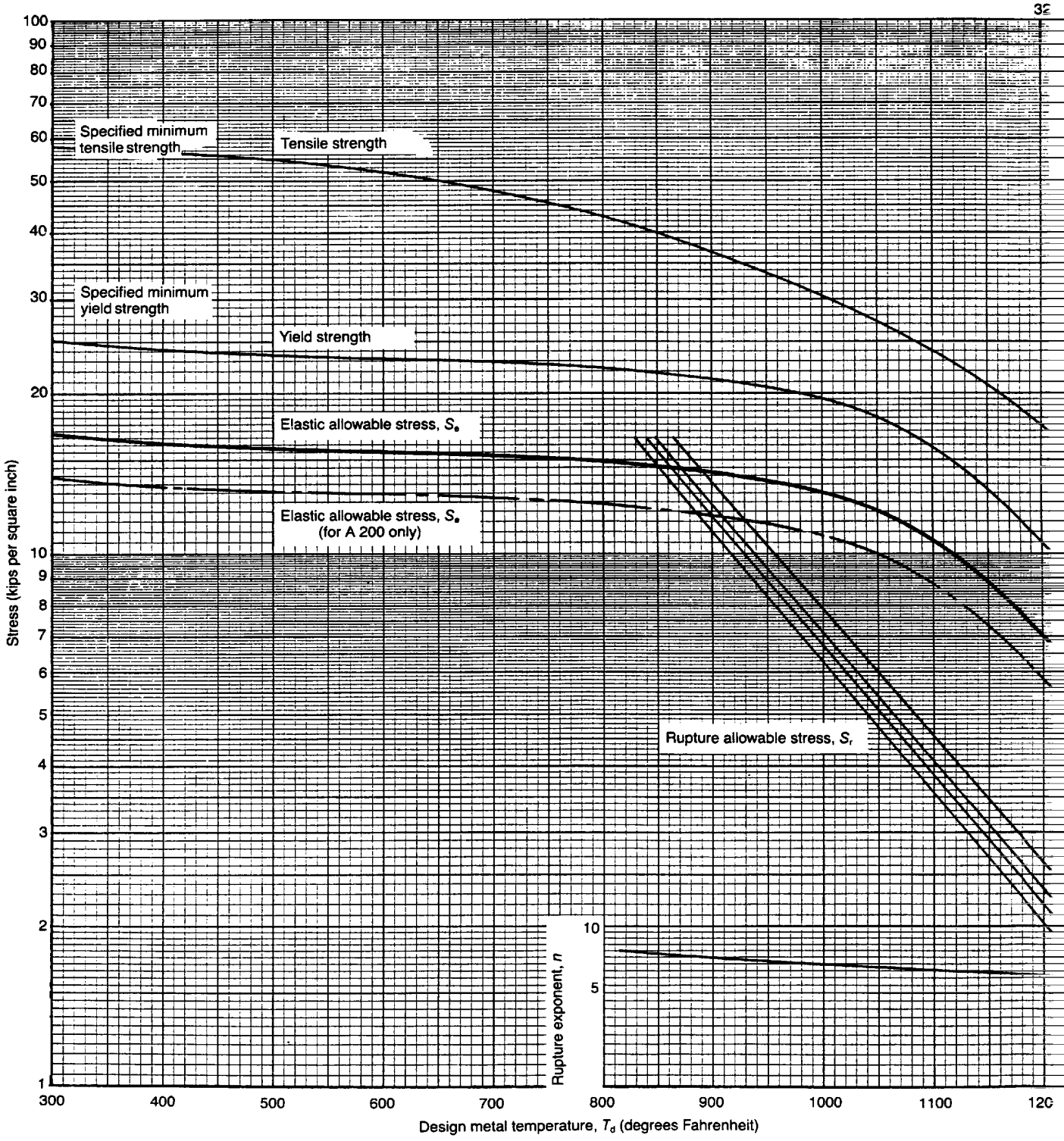


Figure 4l—7Cr–½Mo Steel, ASTM A 213 T7, A 335

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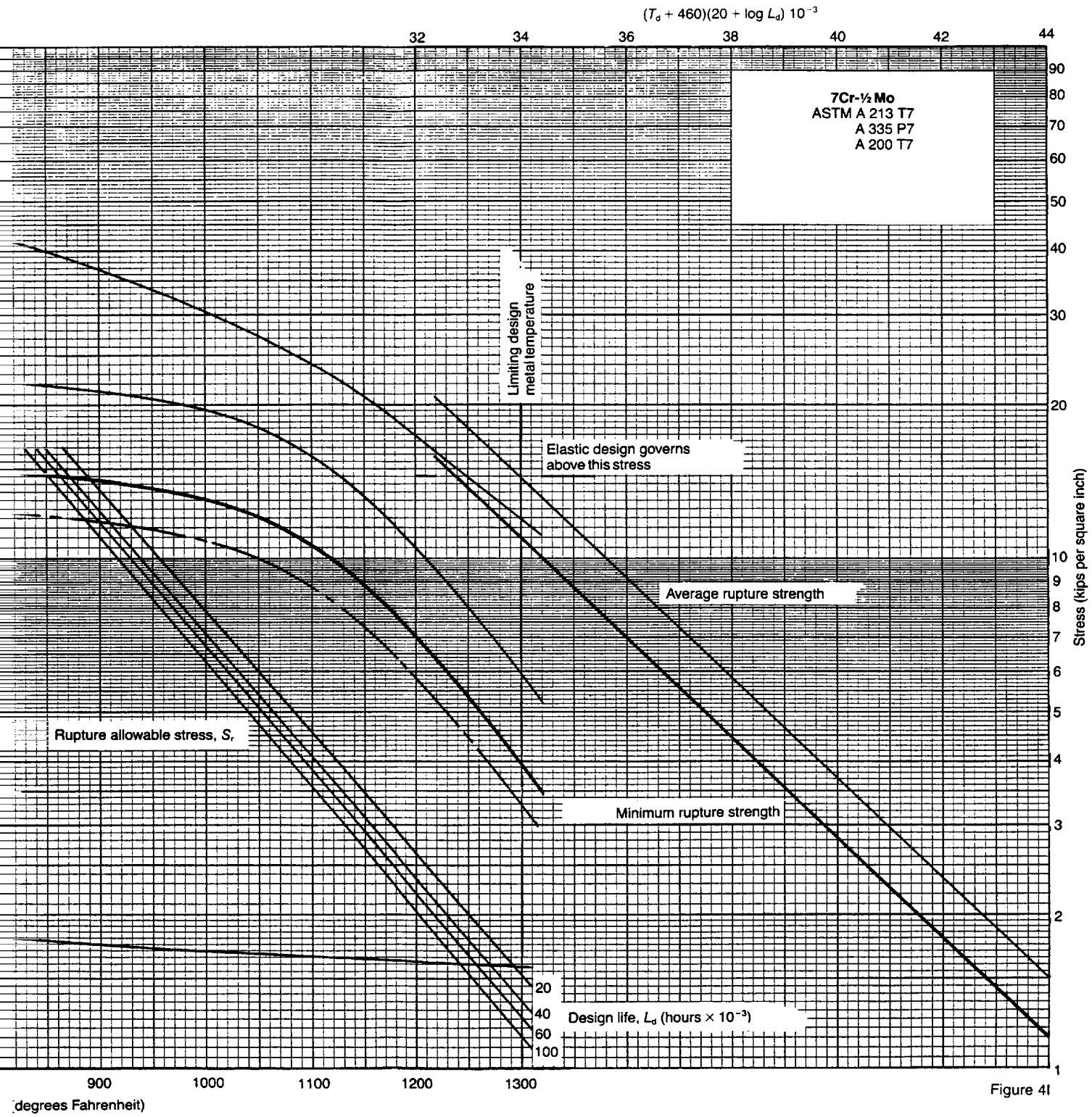


Figure 4I

ure 4I—7Cr–½Mo Steel, ASTM A 213 T7, A 335 P7, A 200 T7

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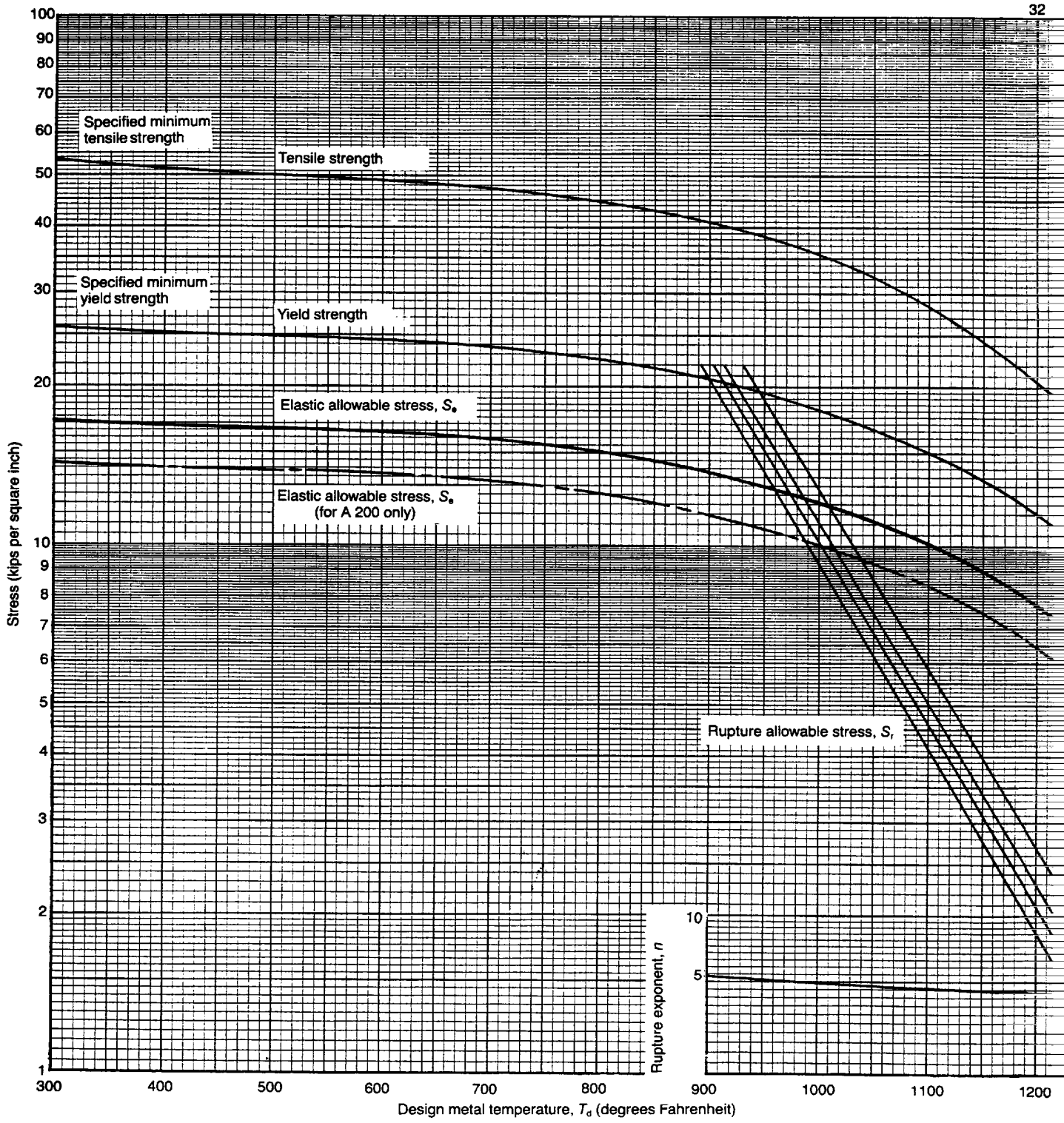


Figure 4J—9Cr–1Mo Steel, ASTM A 213 T9, A 335 F

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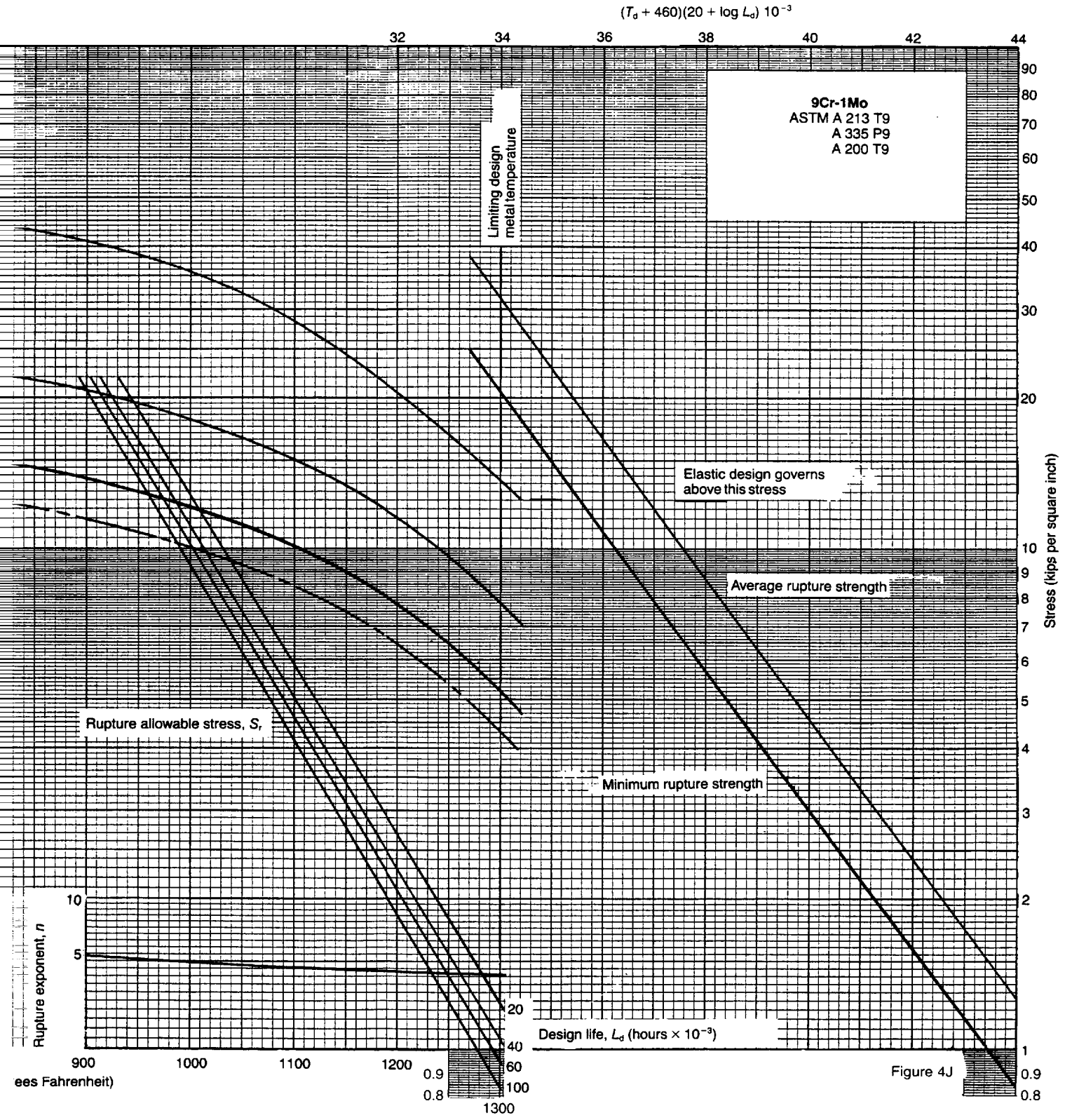


Figure 4J

re 4J—9Cr–1Mo Steel, ASTM A 213 T9, A 335 P9, A 200 T9

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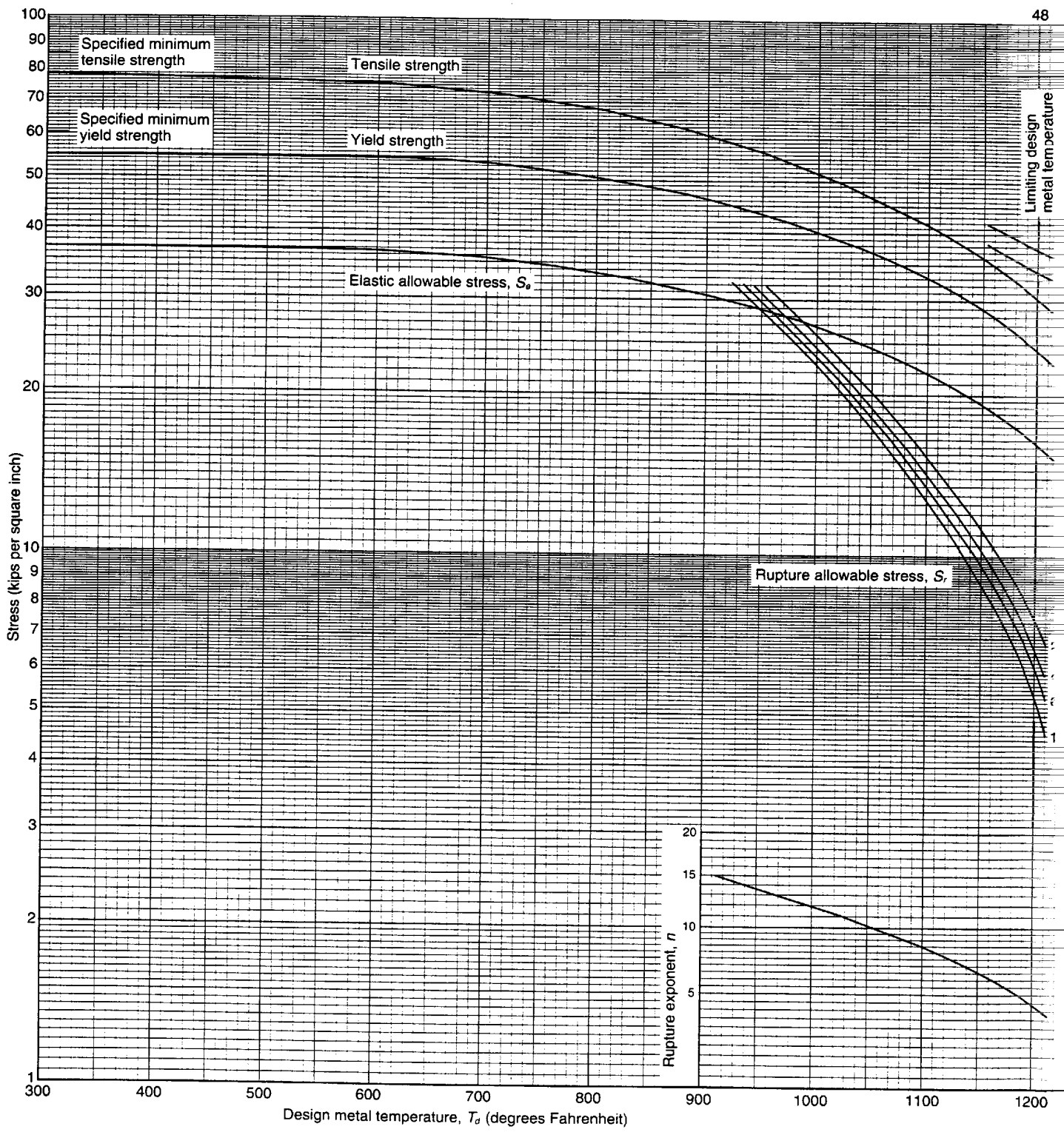


Figure 4K—9Cr–1Mo–Va Steel, ASTM A 213 T91, A 335 P

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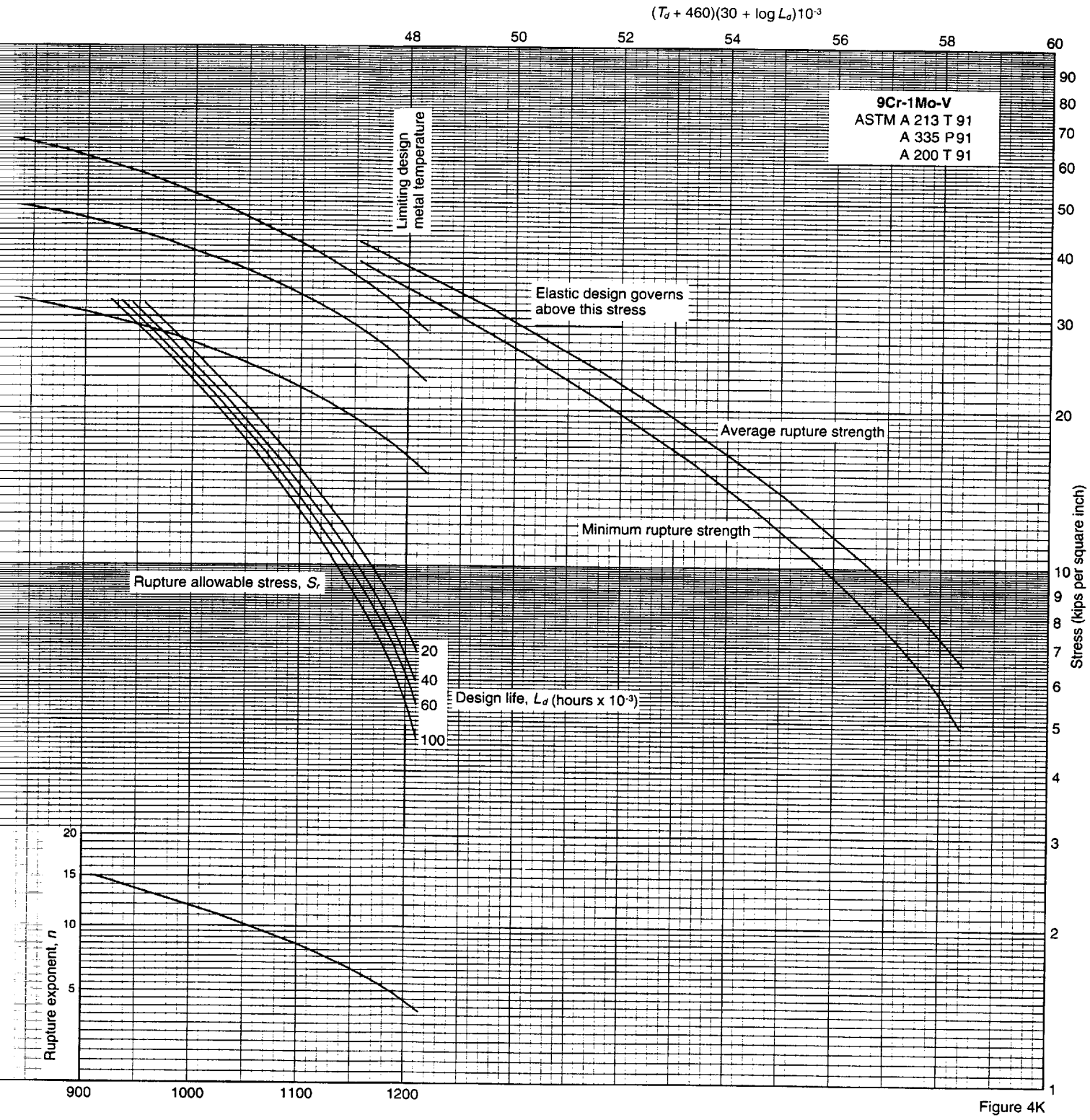


Figure 4K

—9Cr–1Mo–Va Steel, ASTM A 213 T91, A 335 P91, A 200 T91

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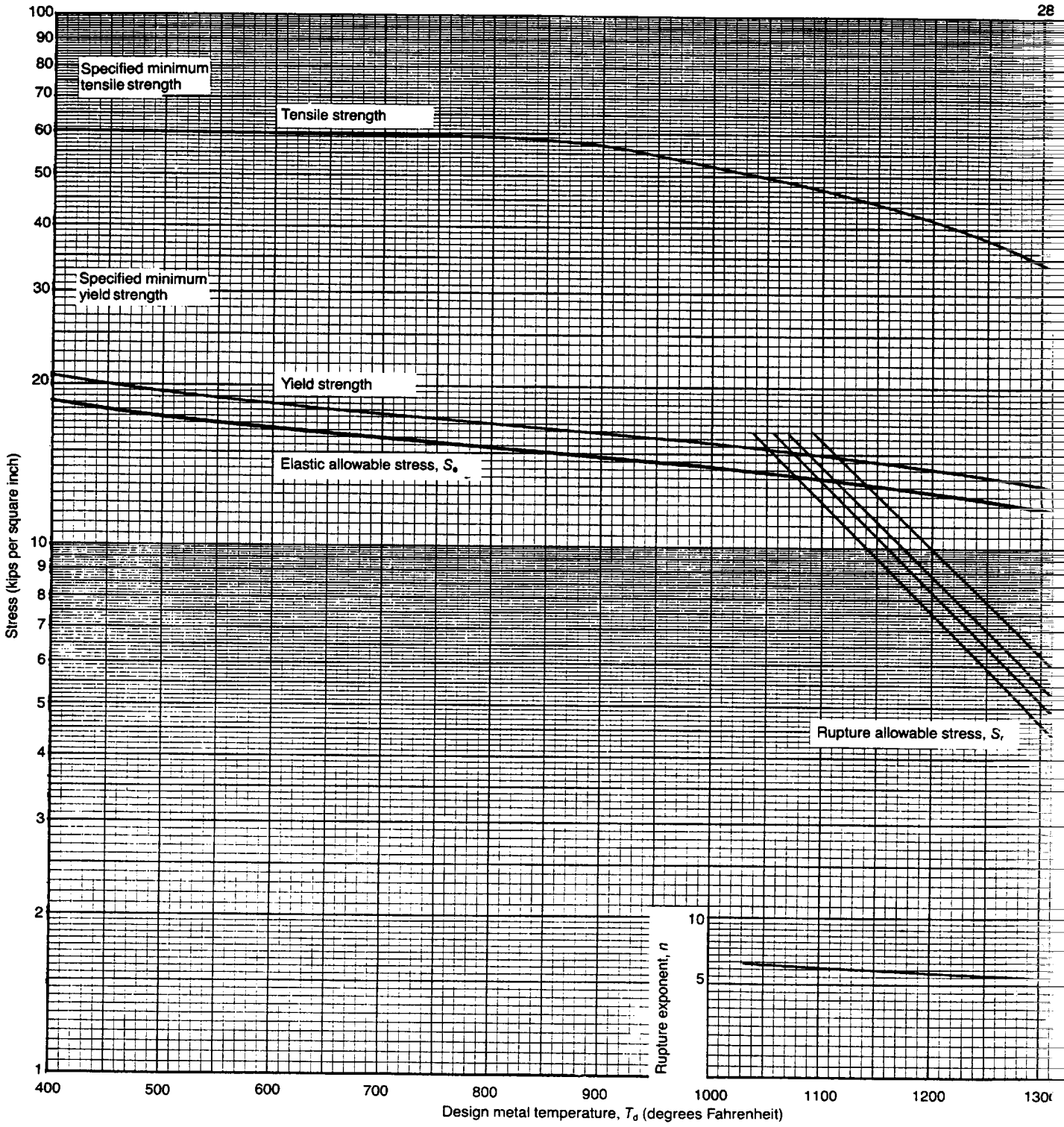


Figure 4L—Types 304 and 304H Stainless Steel, ASTM A 213,

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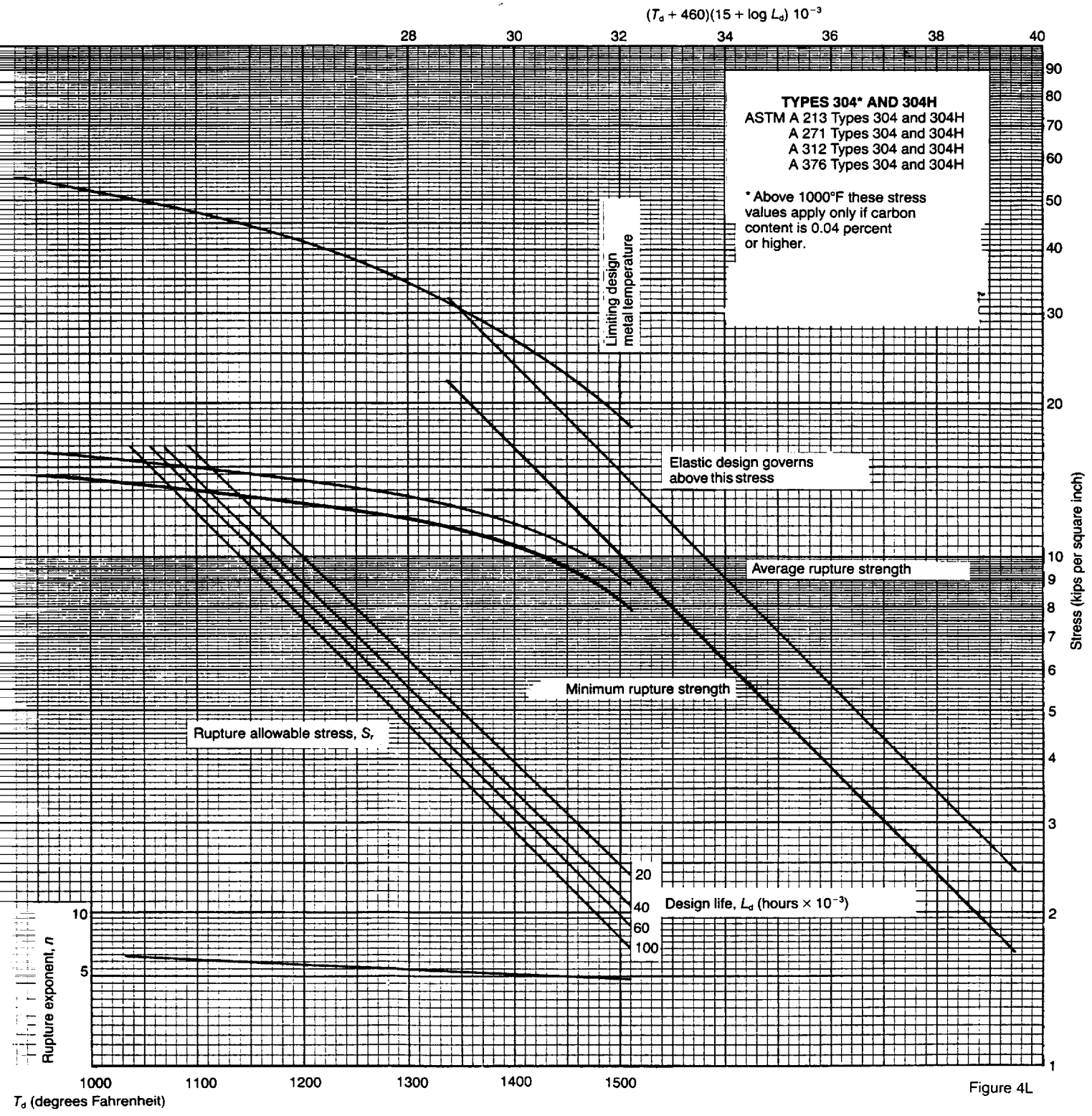


Figure 4L

ypes 304 and 304H Stainless Steel, ASTM A 213, A 271, A 312, A 376

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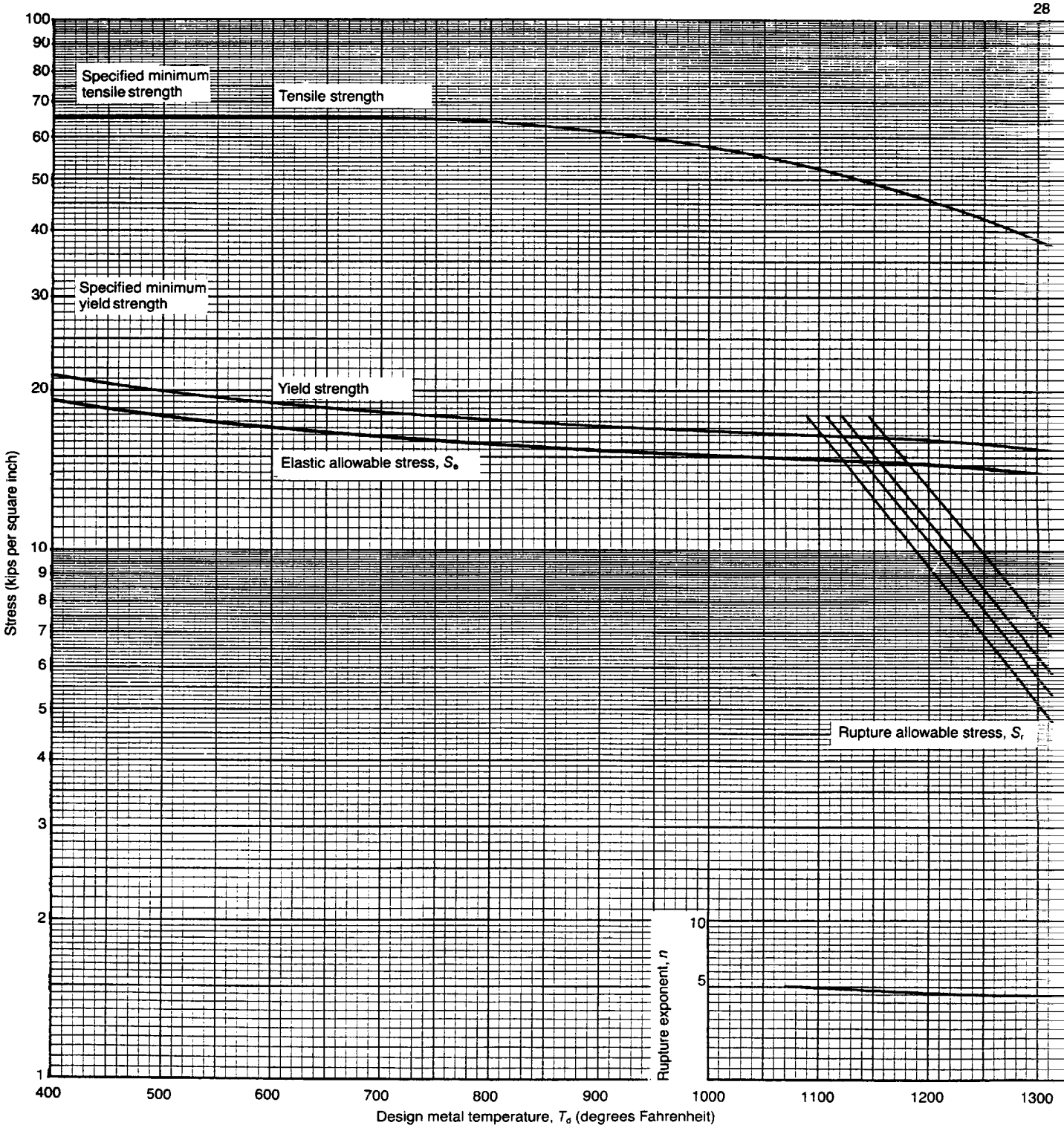


Figure 4M—Types 316 and 316H Stainless Steel, ASTM A 213,

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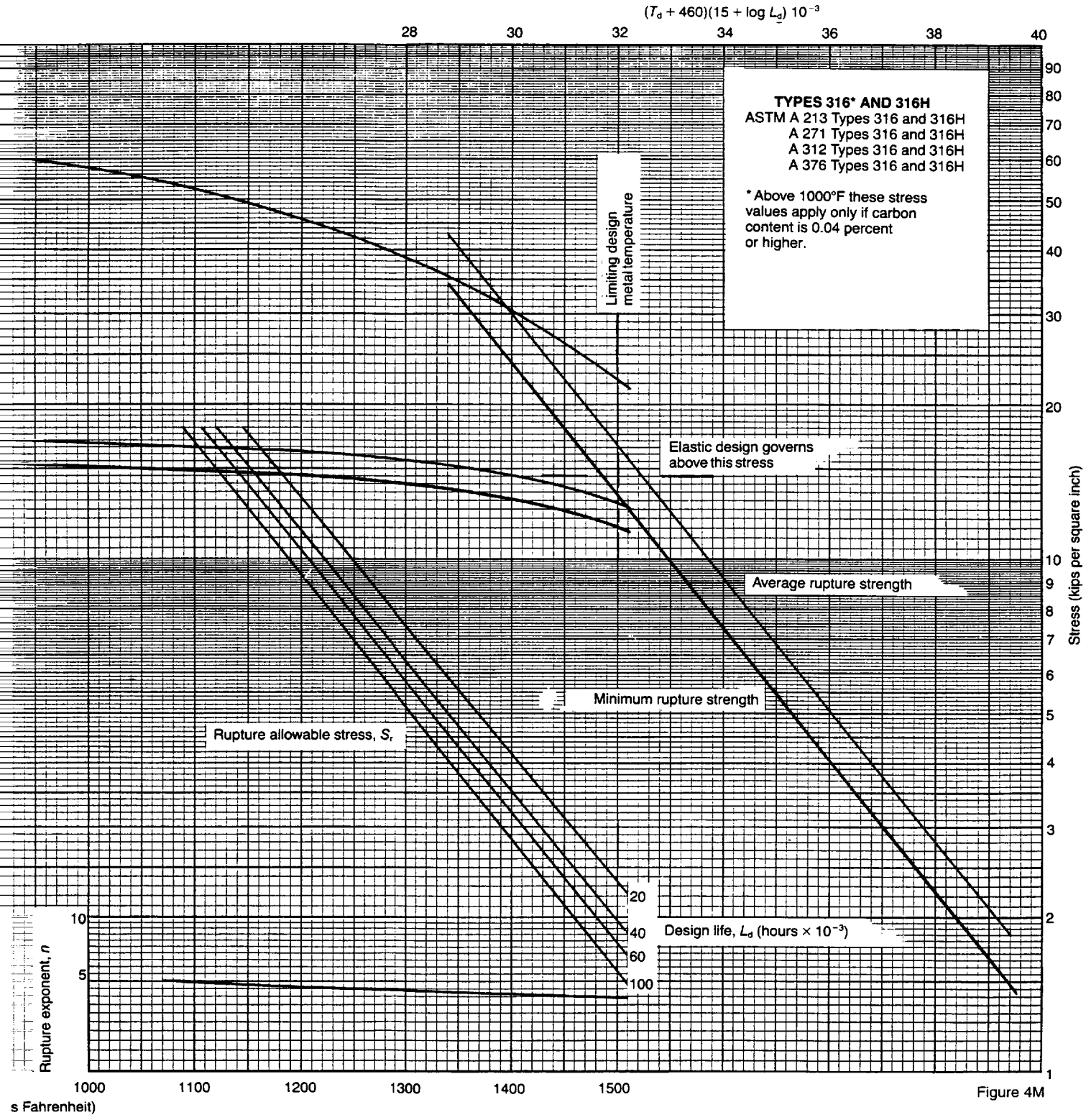


Figure 4M

pes 316 and 316H Stainless Steel, ASTM A 213, A 271, A 312, A 376

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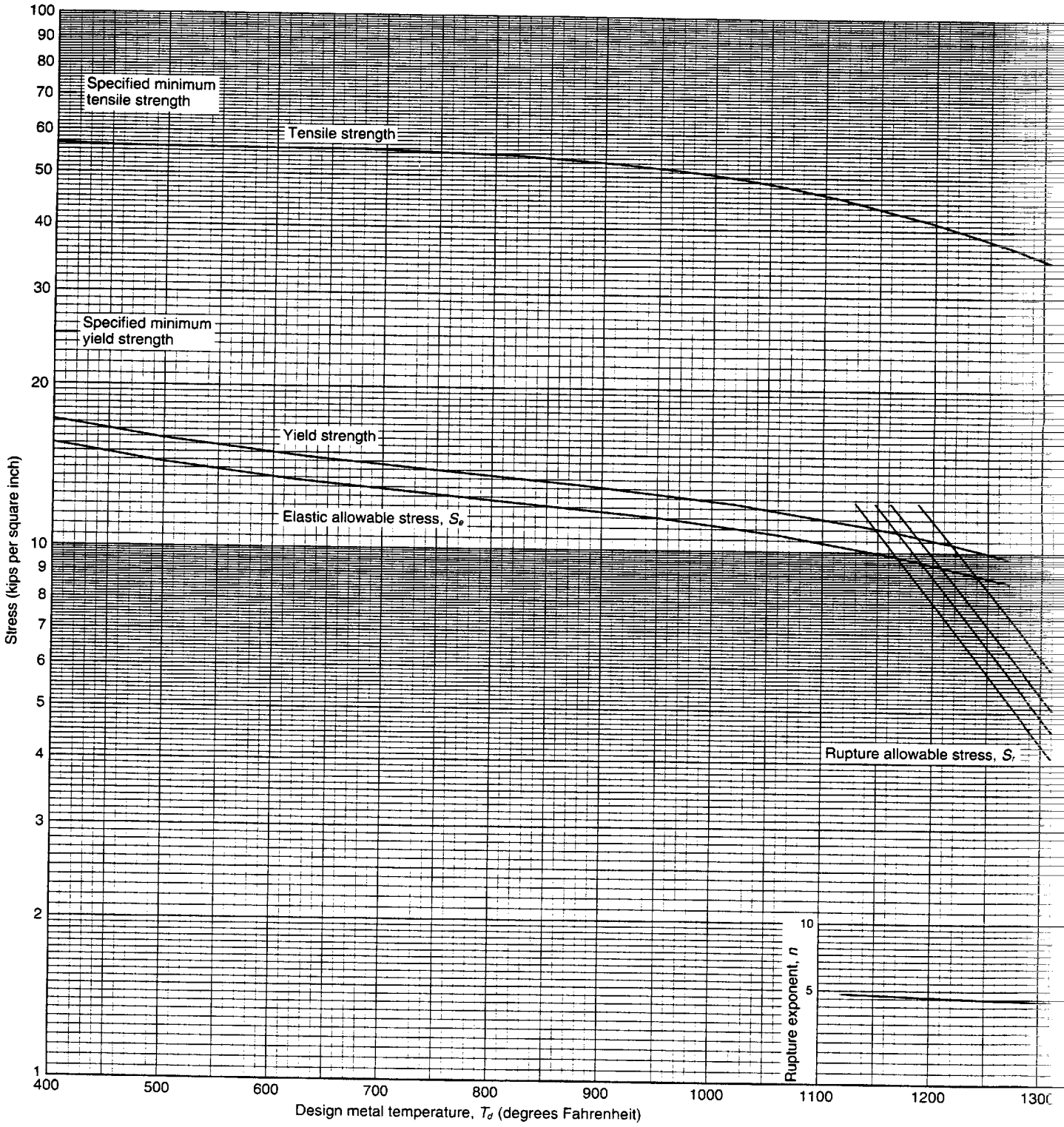


Figure 4N—Type 316L Stainless Steel

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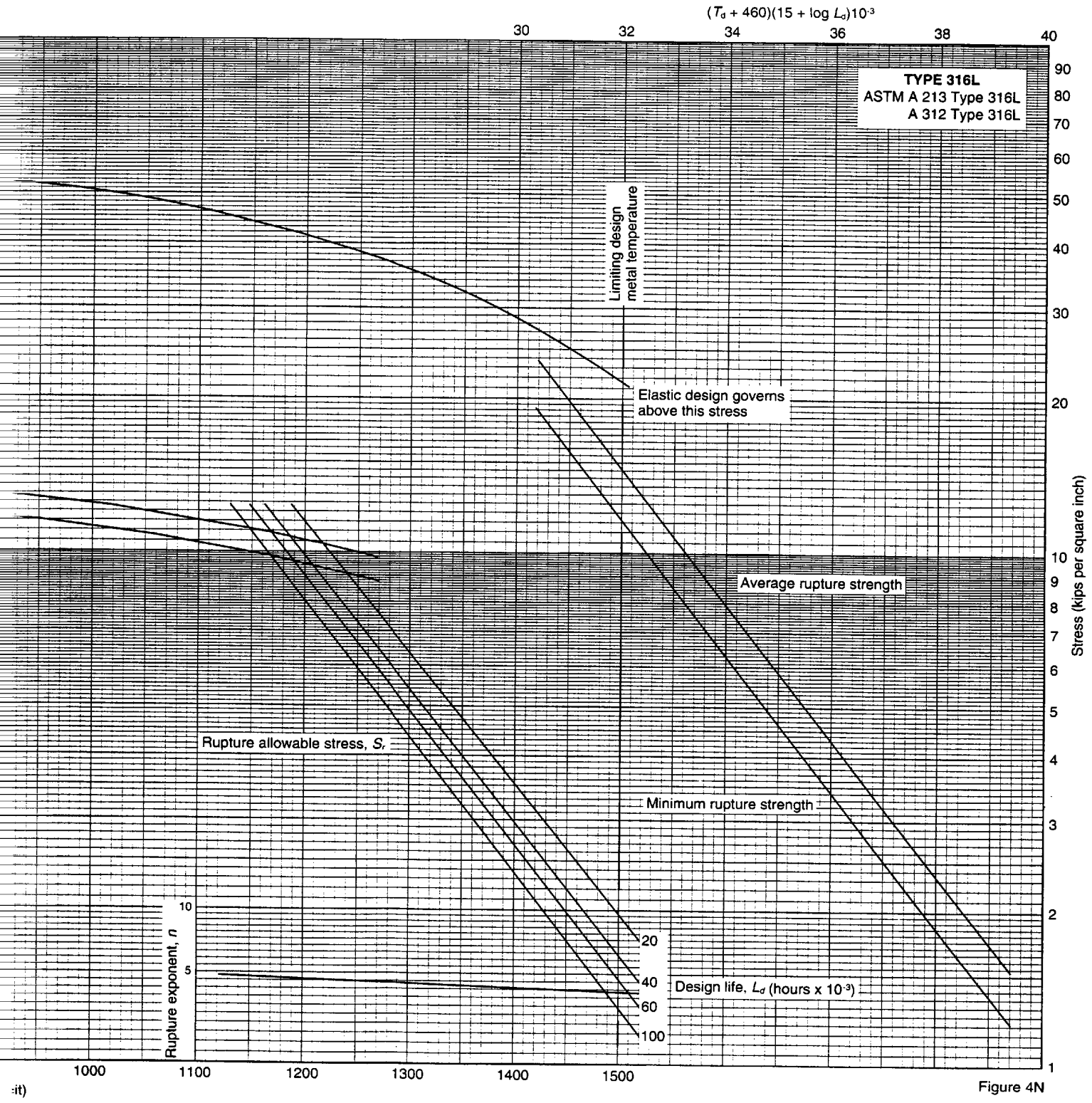


Figure 4N—Type 316L Stainless Steel

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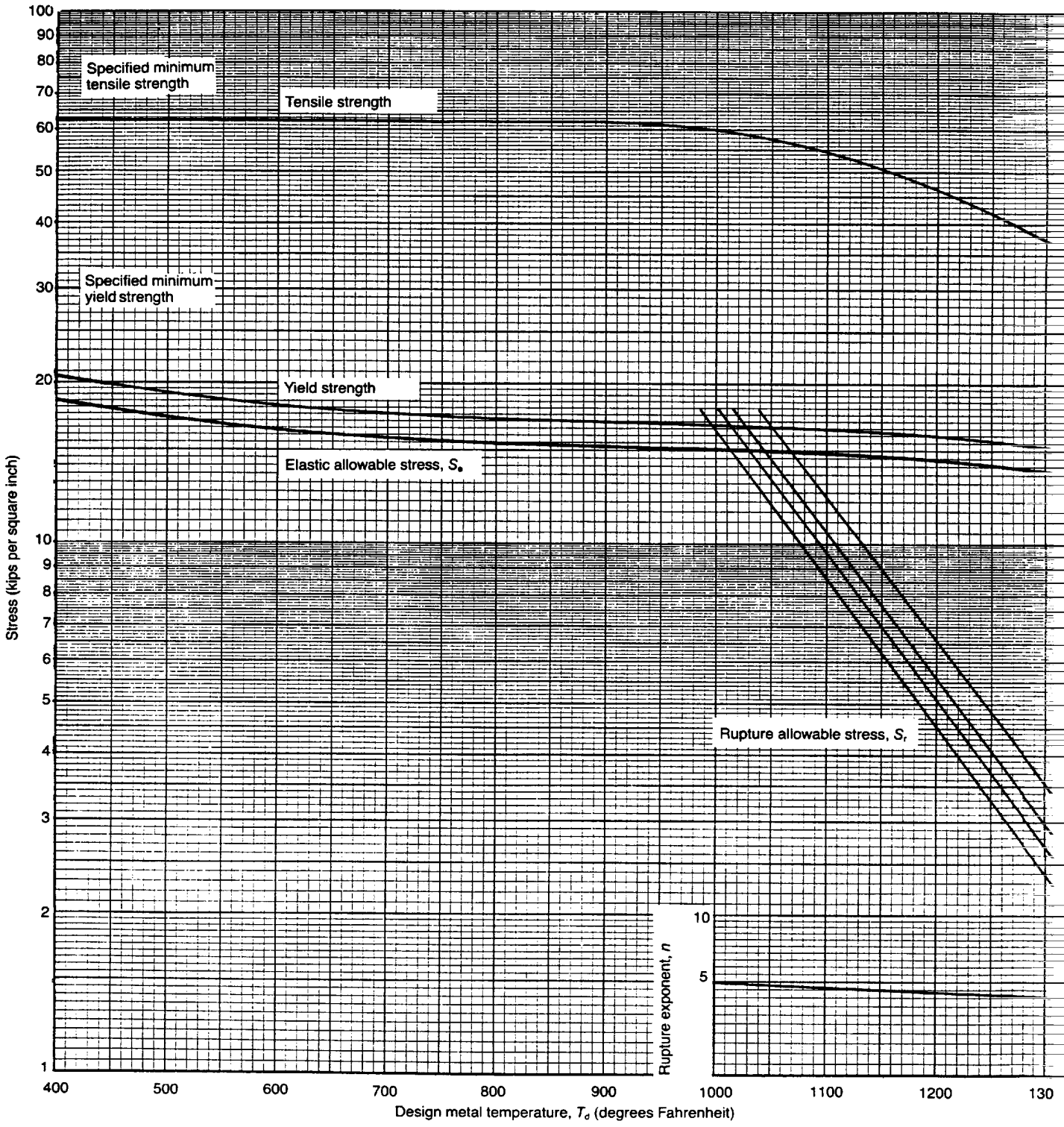


Figure 4O—Type 321 Stainless Steel, ASTM A 213, A 271,

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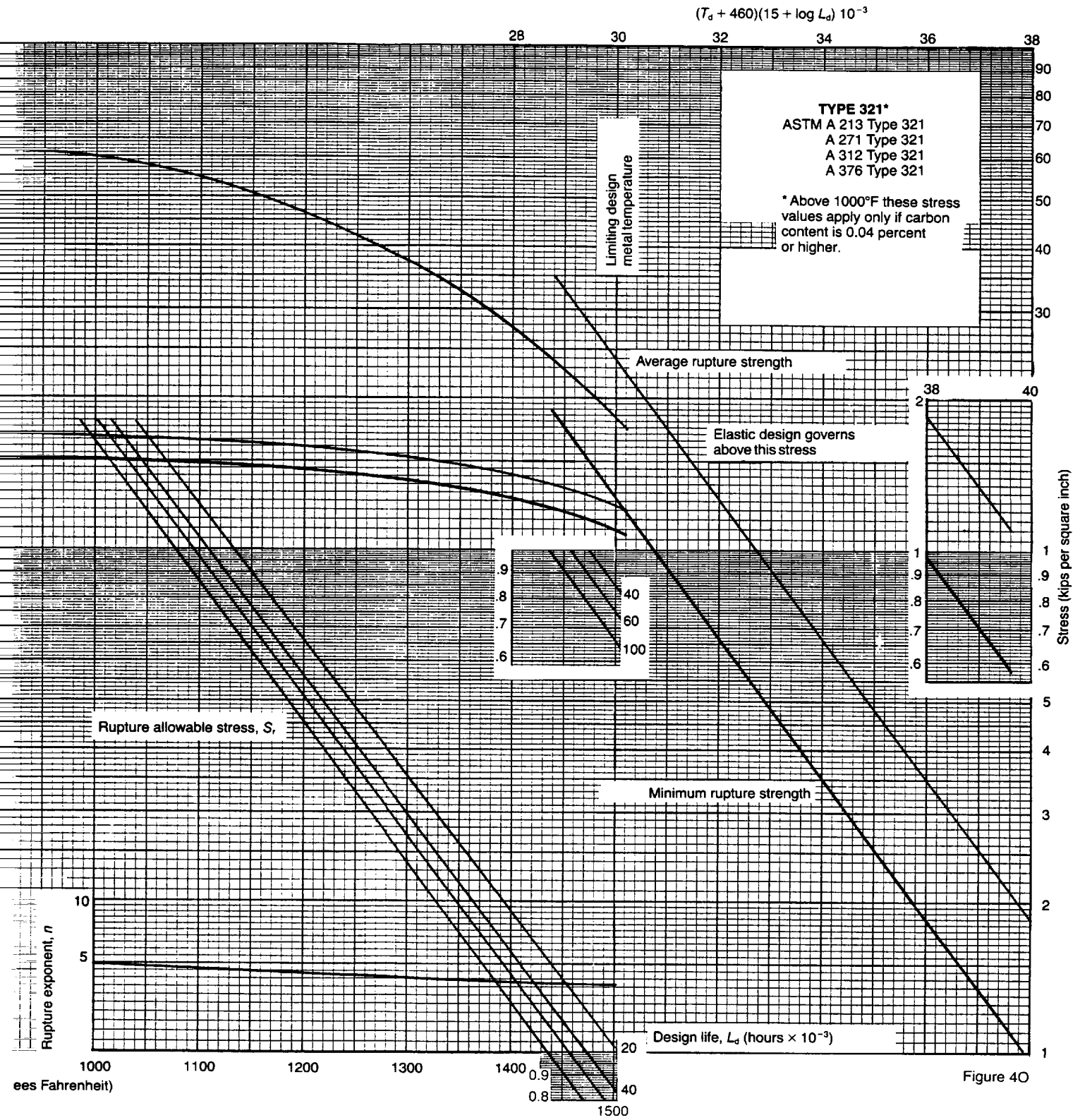


Figure 4O

—Type 321 Stainless Steel, ASTM A 213, A 271, A 312, A 376

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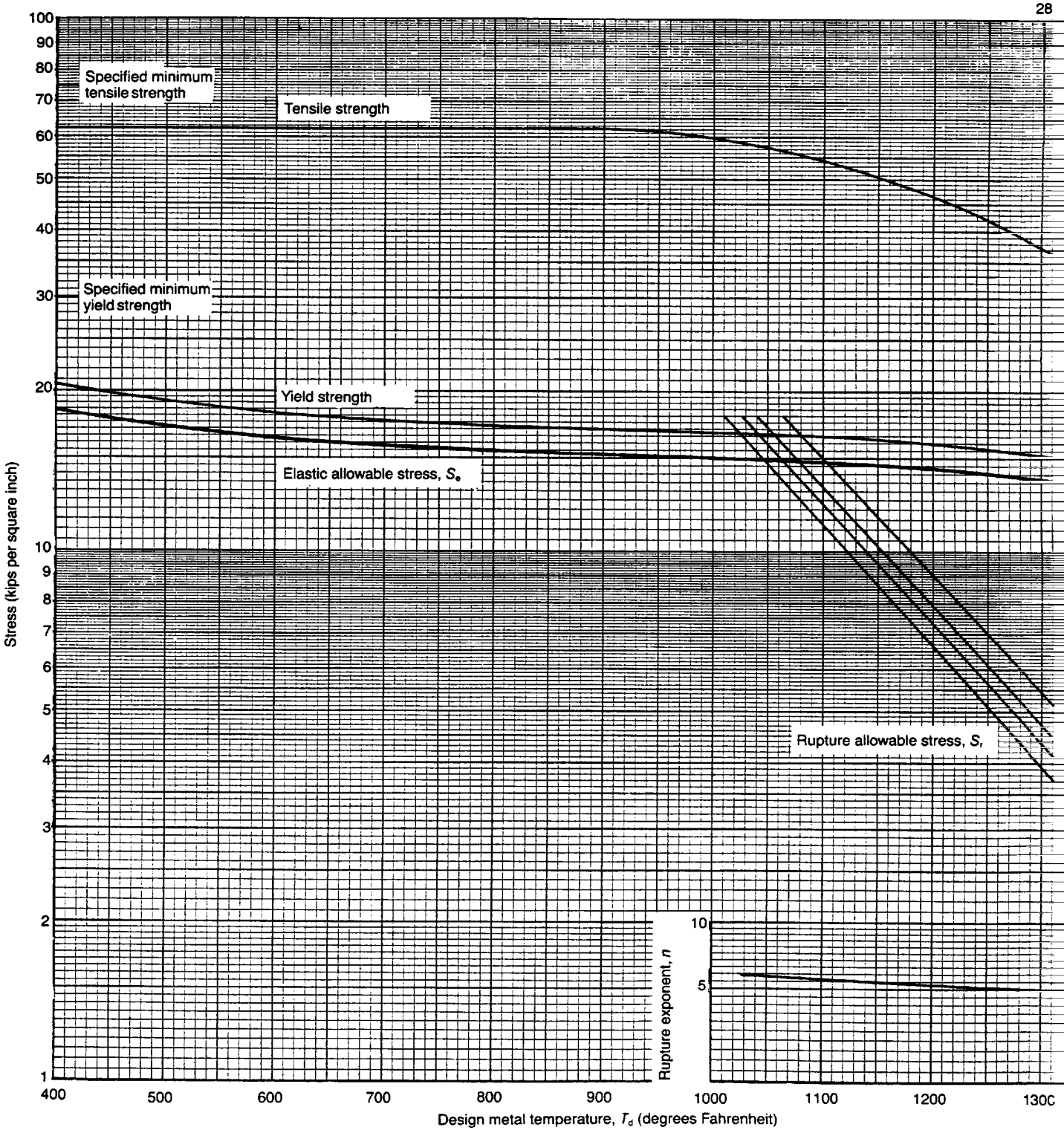


Figure 4P—Type 321H Stainless Steel, ASTM A 213, A 2

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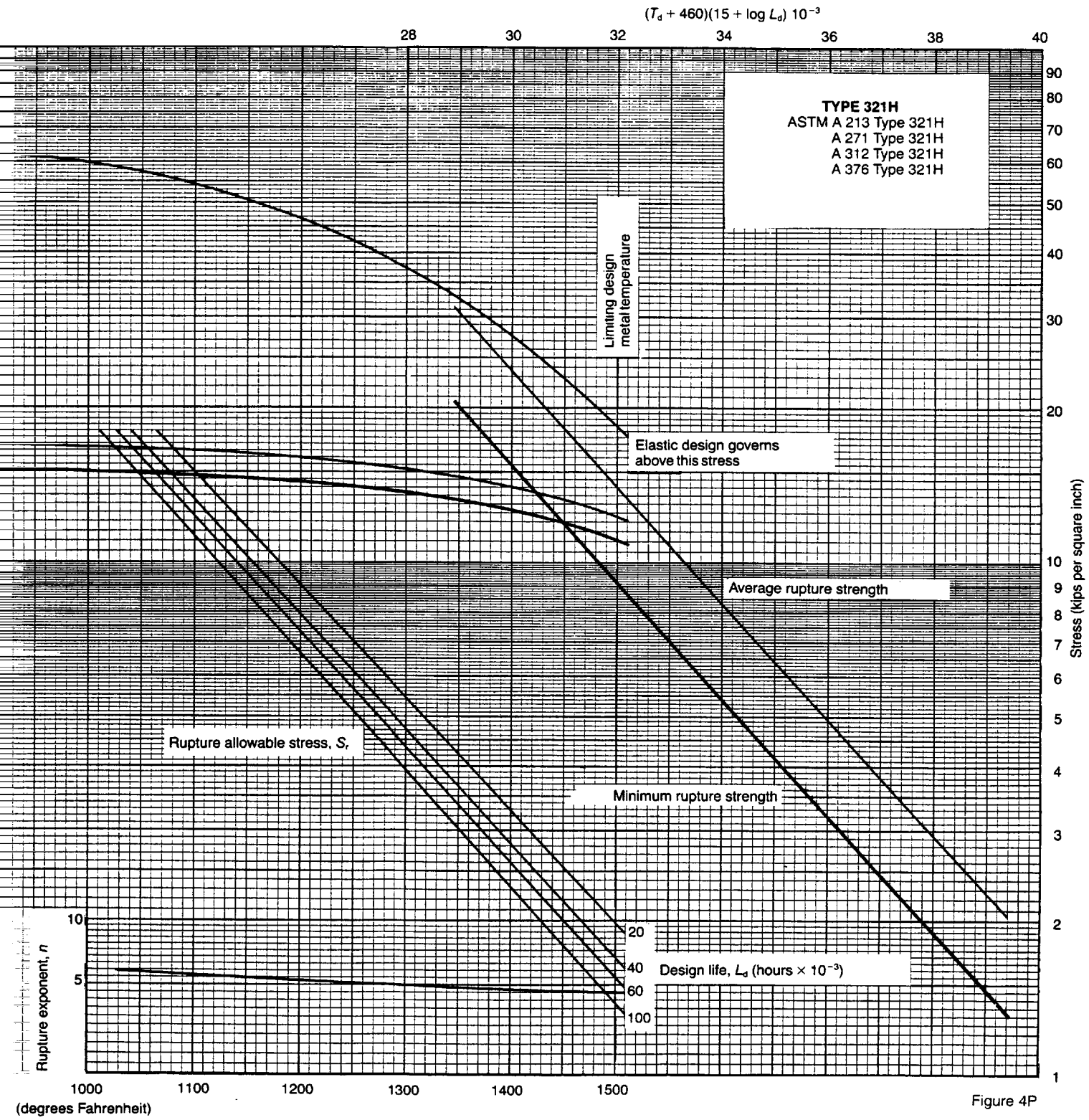


Figure 4P

'—Type 321H Stainless Steel, ASTM A 213, A 271, A 312, A 376

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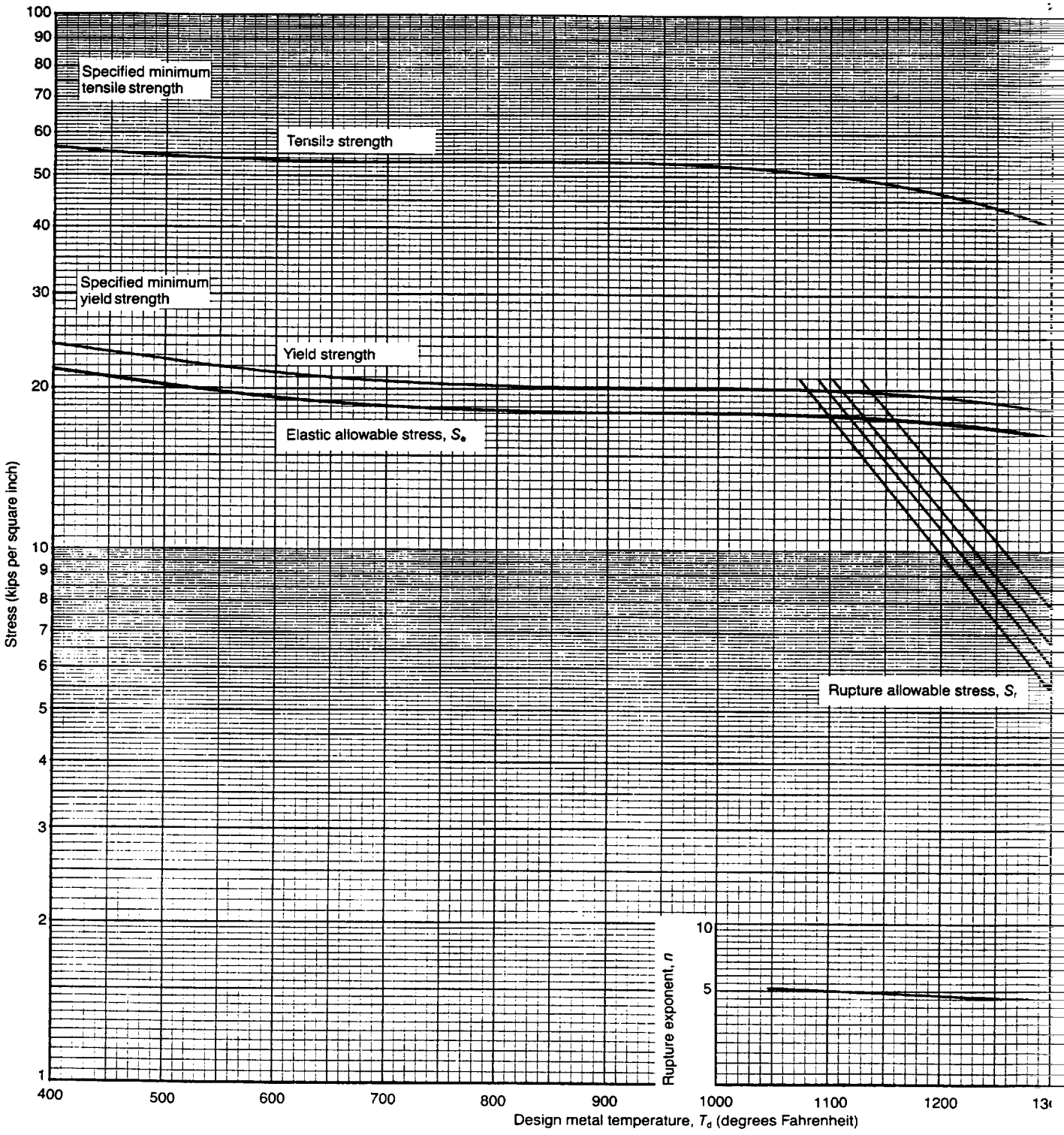


Figure 4Q—Type 347 and 347H Stainless Steel, ASTM A 213

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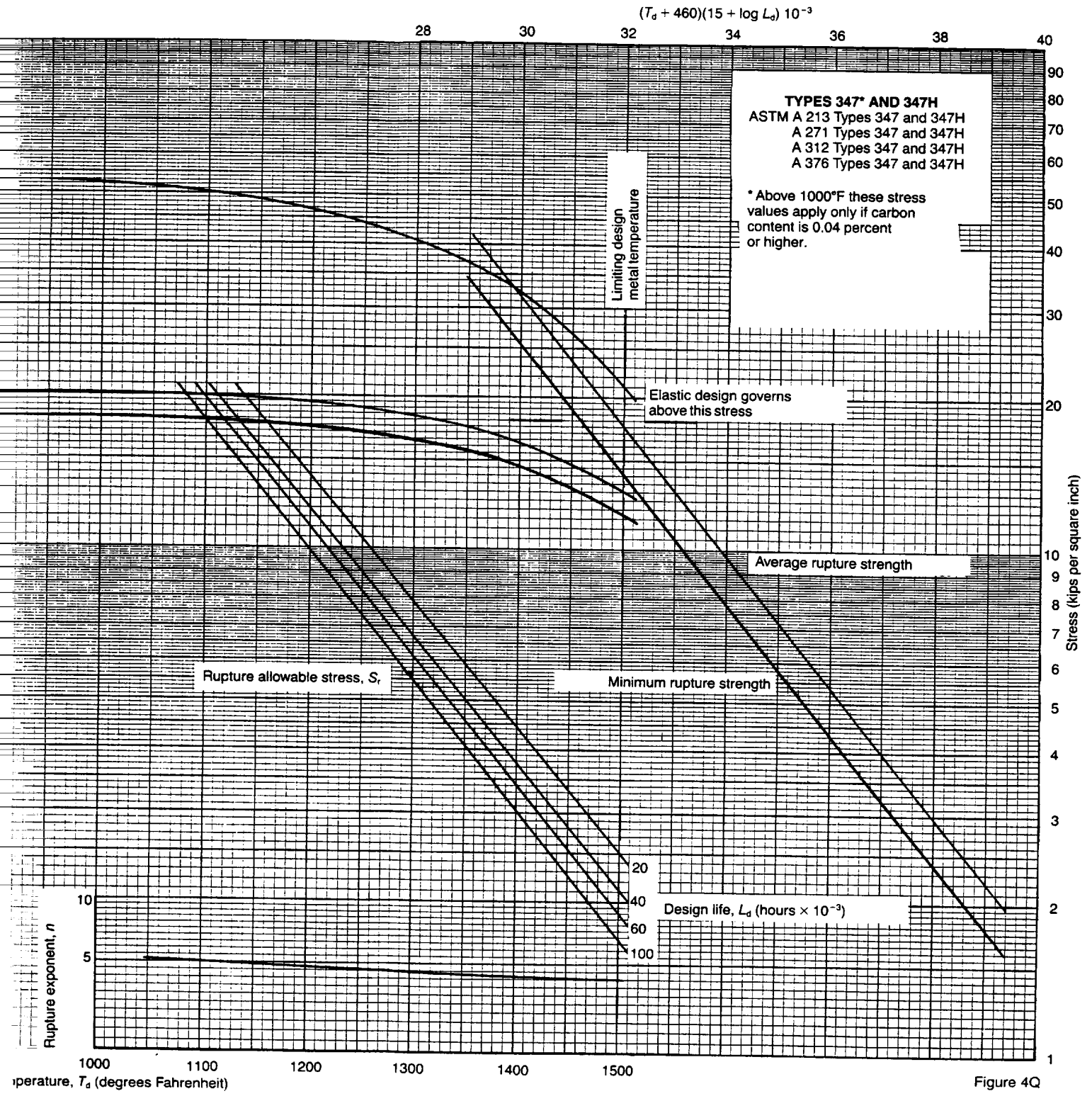


Figure 4Q

ype 347 and 347H Stainless Steel, ASTM A 213, A 271, A 312, A 376

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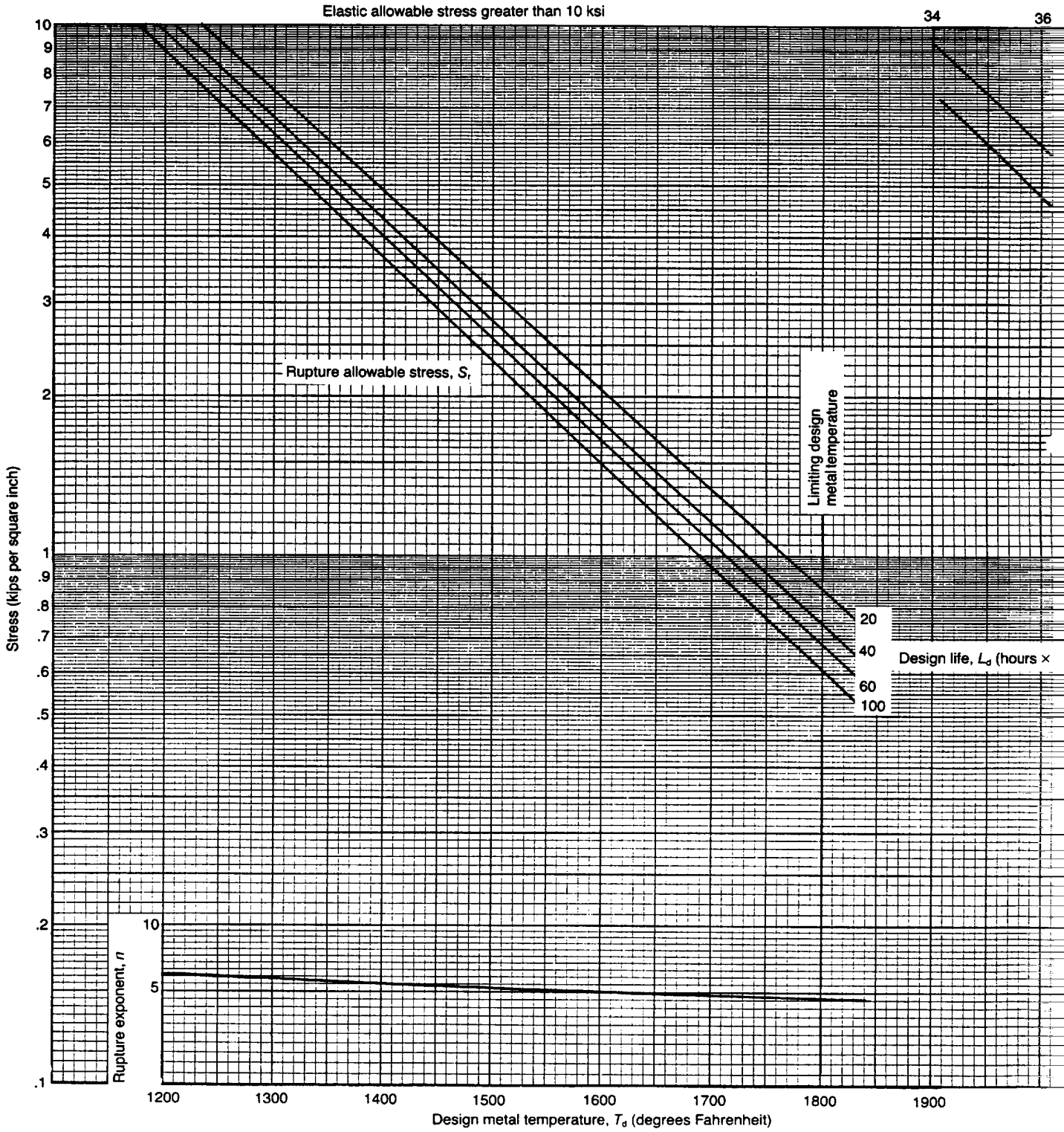


Figure 4R—Alloy 800H, ASTM B 407 UNS NC

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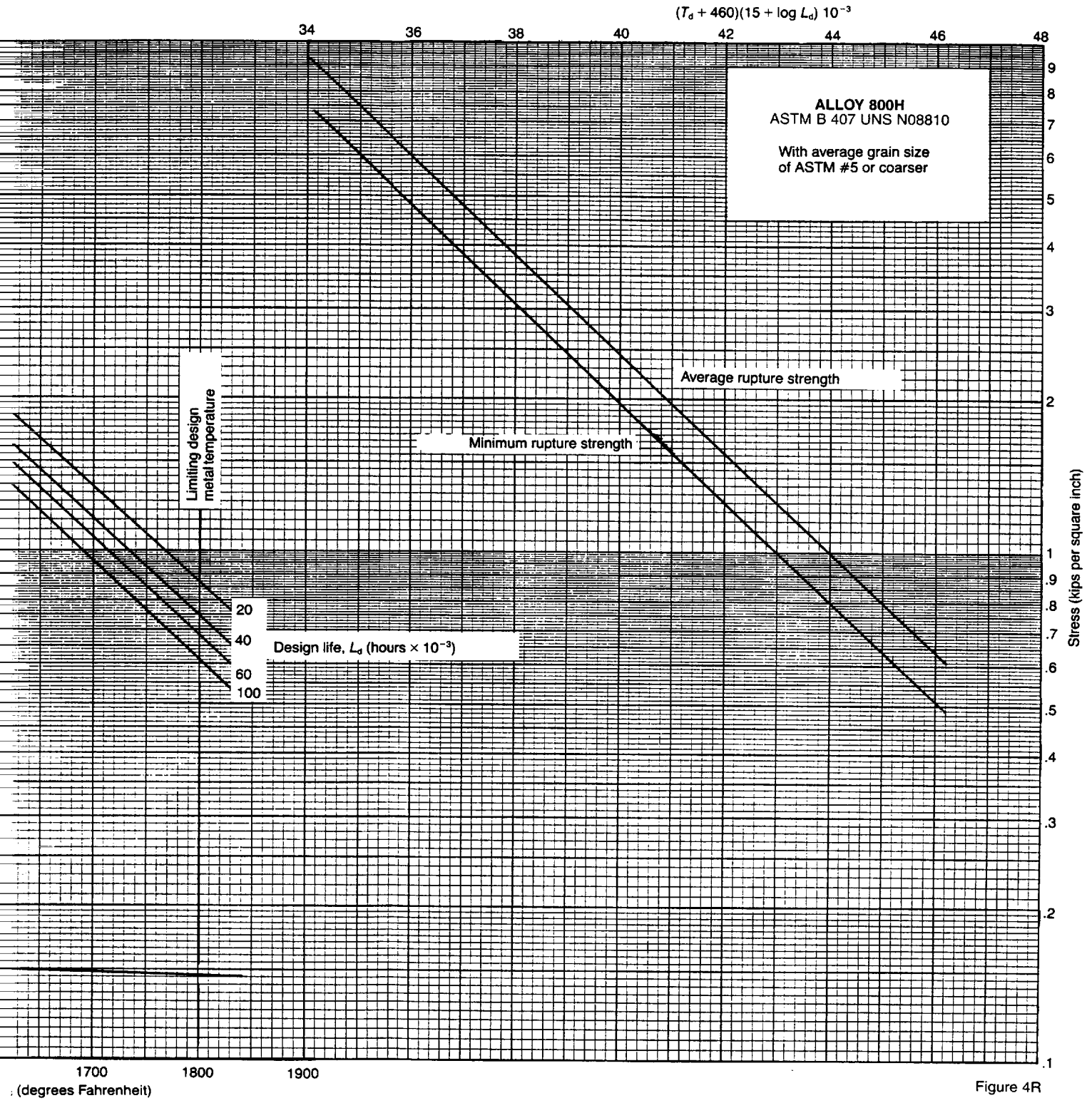


Figure 4R

Figure 4R—Alloy 800H, ASTM B 407 UNS N08810

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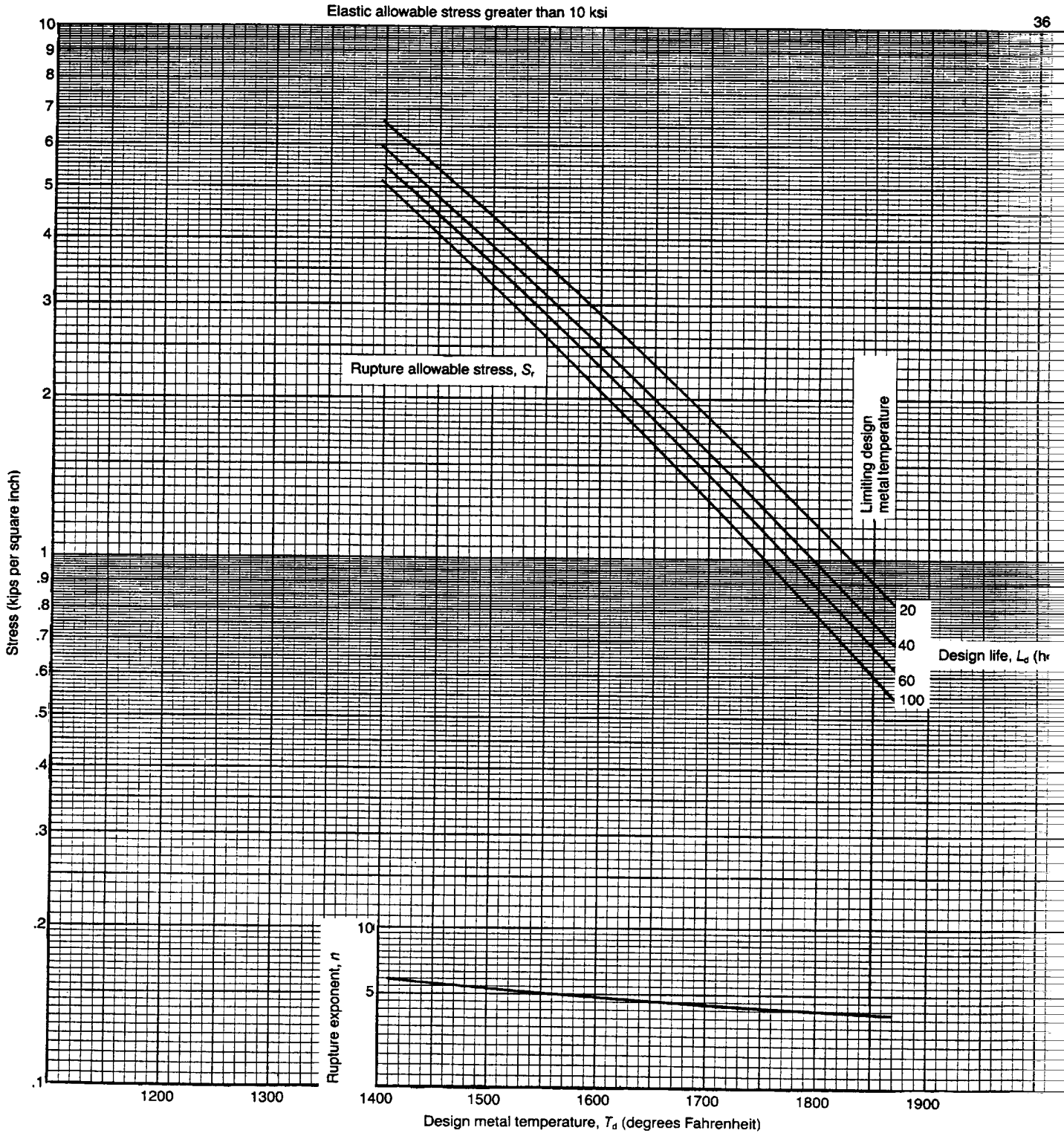


Figure 4S —HK-40, ASTM A 608 Grade HK

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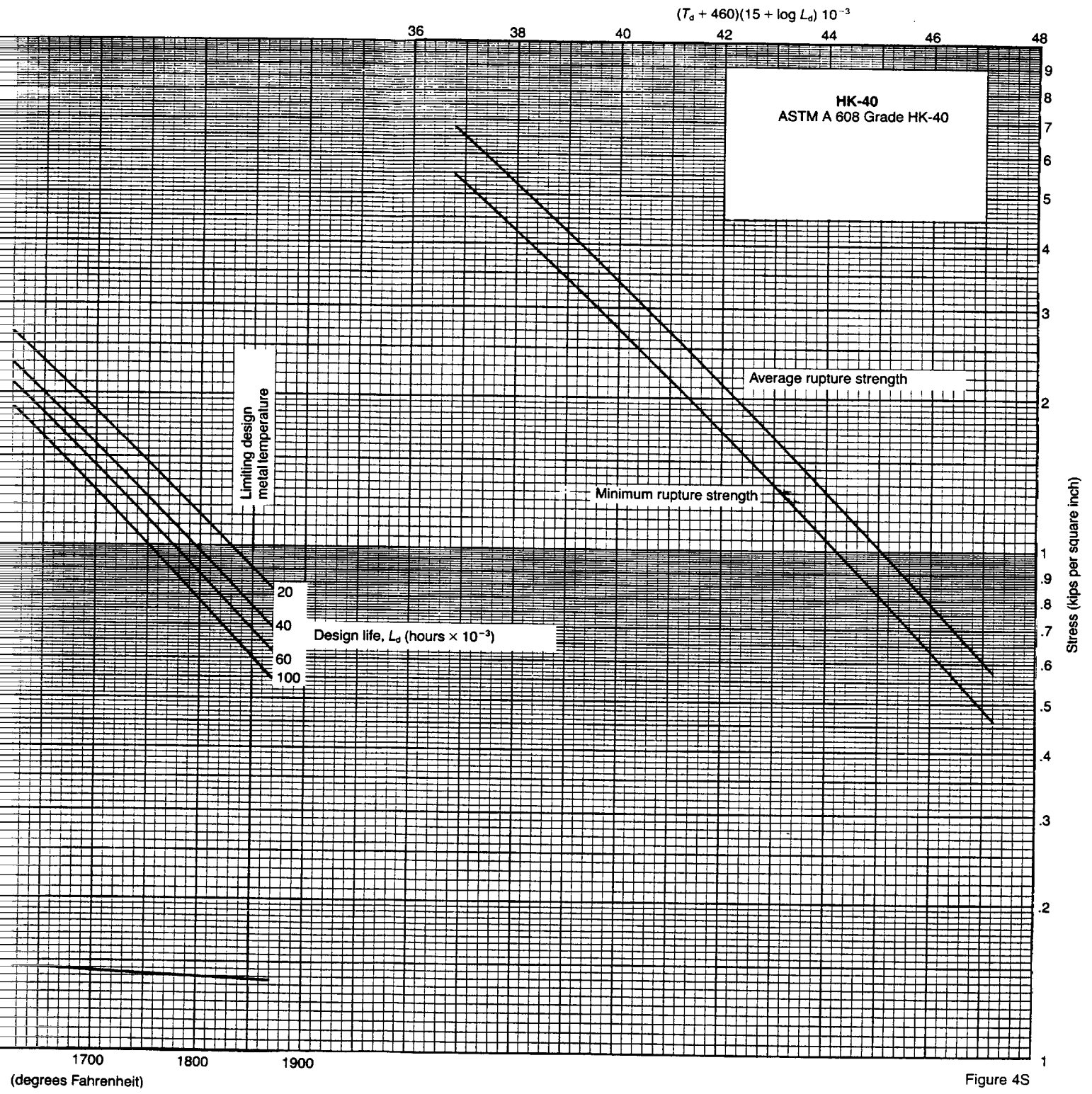


Figure 4S —HK-40, ASTM A 608 Grade HK-40

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Future Stress Curves (SI Units)

New stress curves may be added to this standard sometime in the future. As new stress curves are added, revision packages will be prepared and distributed.

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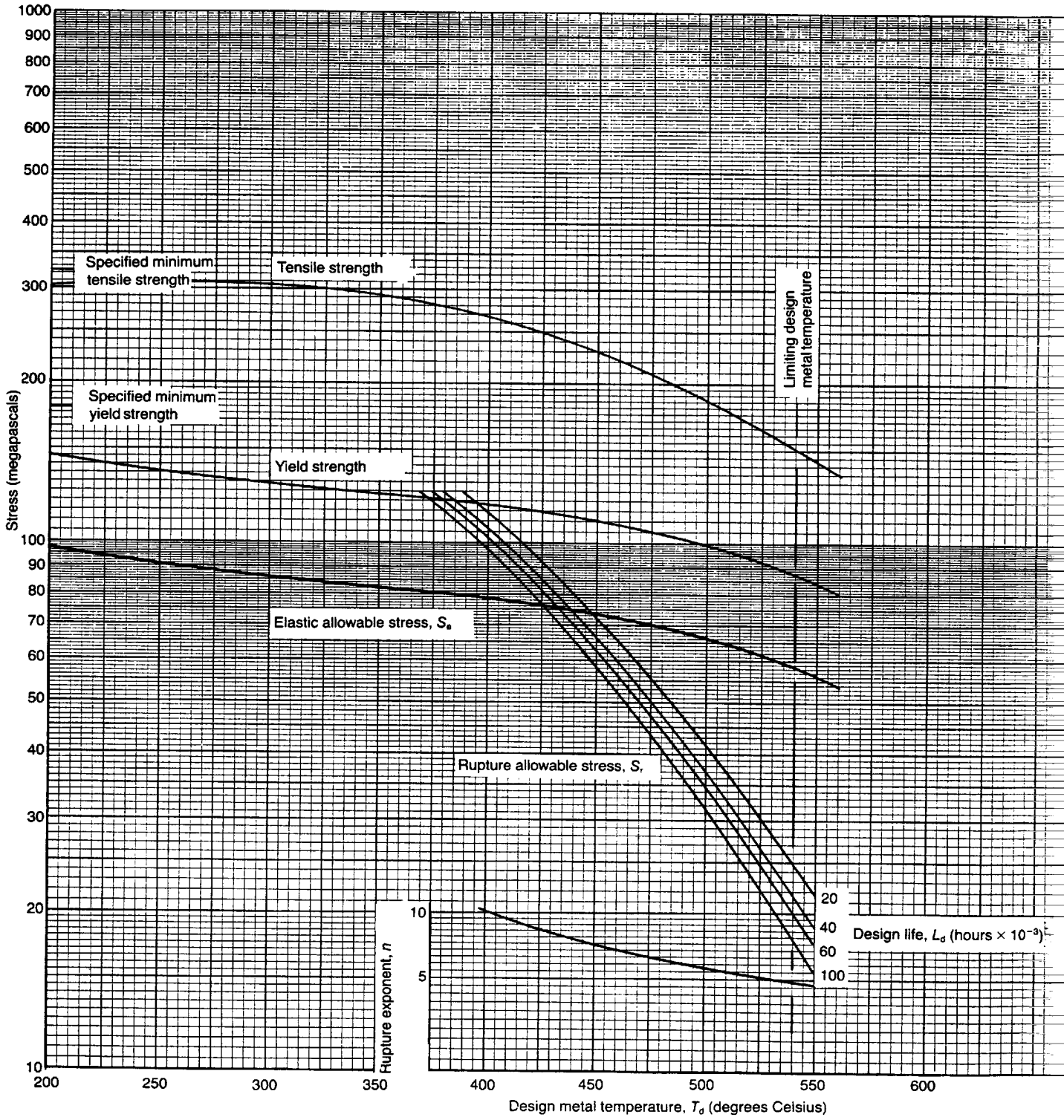


Figure 4A (SI)—Low-Carbon Steel, ASTM A 161,

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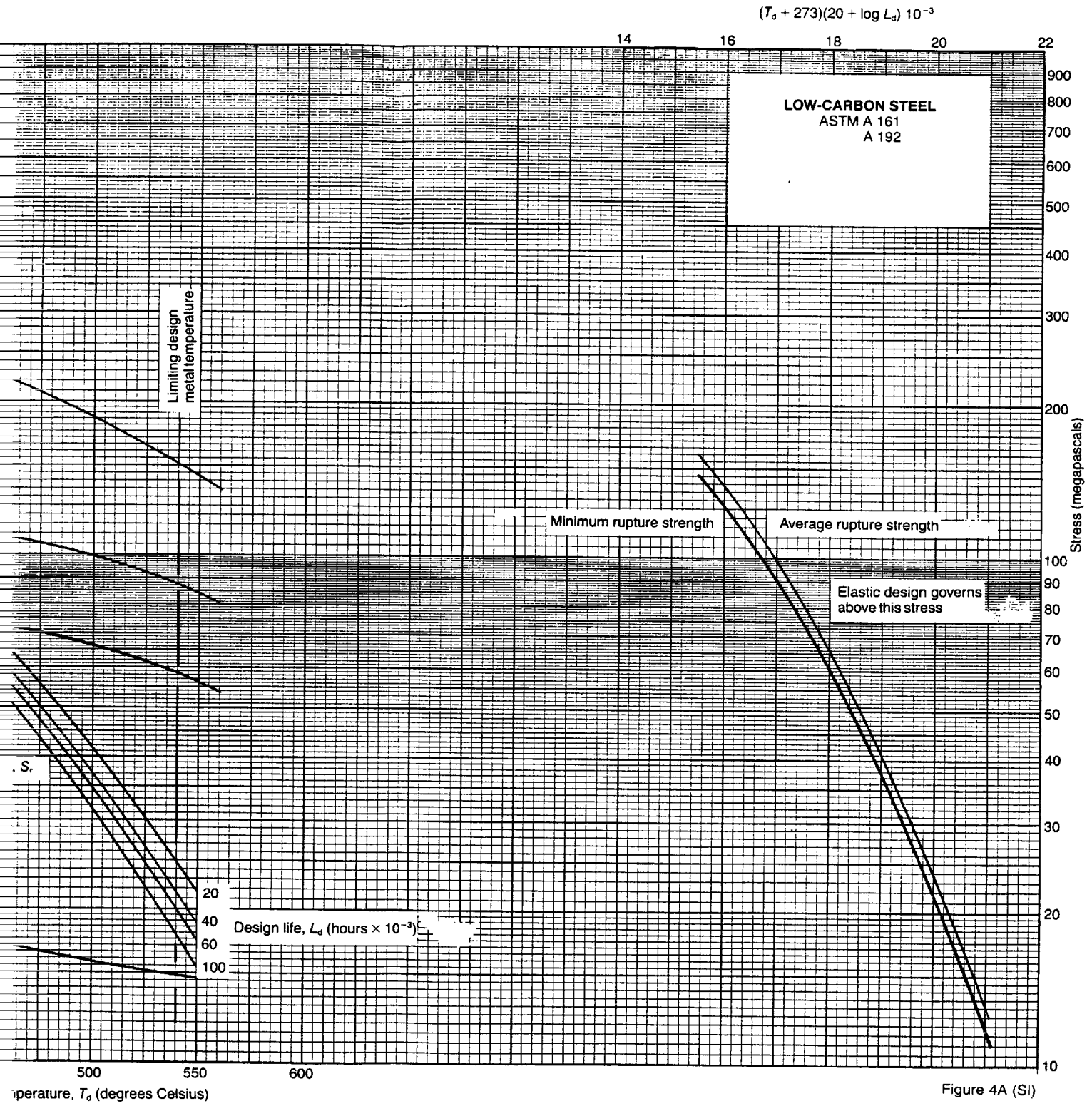


Figure 4A (SI)

Figure 4A (SI)—Low-Carbon Steel, ASTM A 161, A 192

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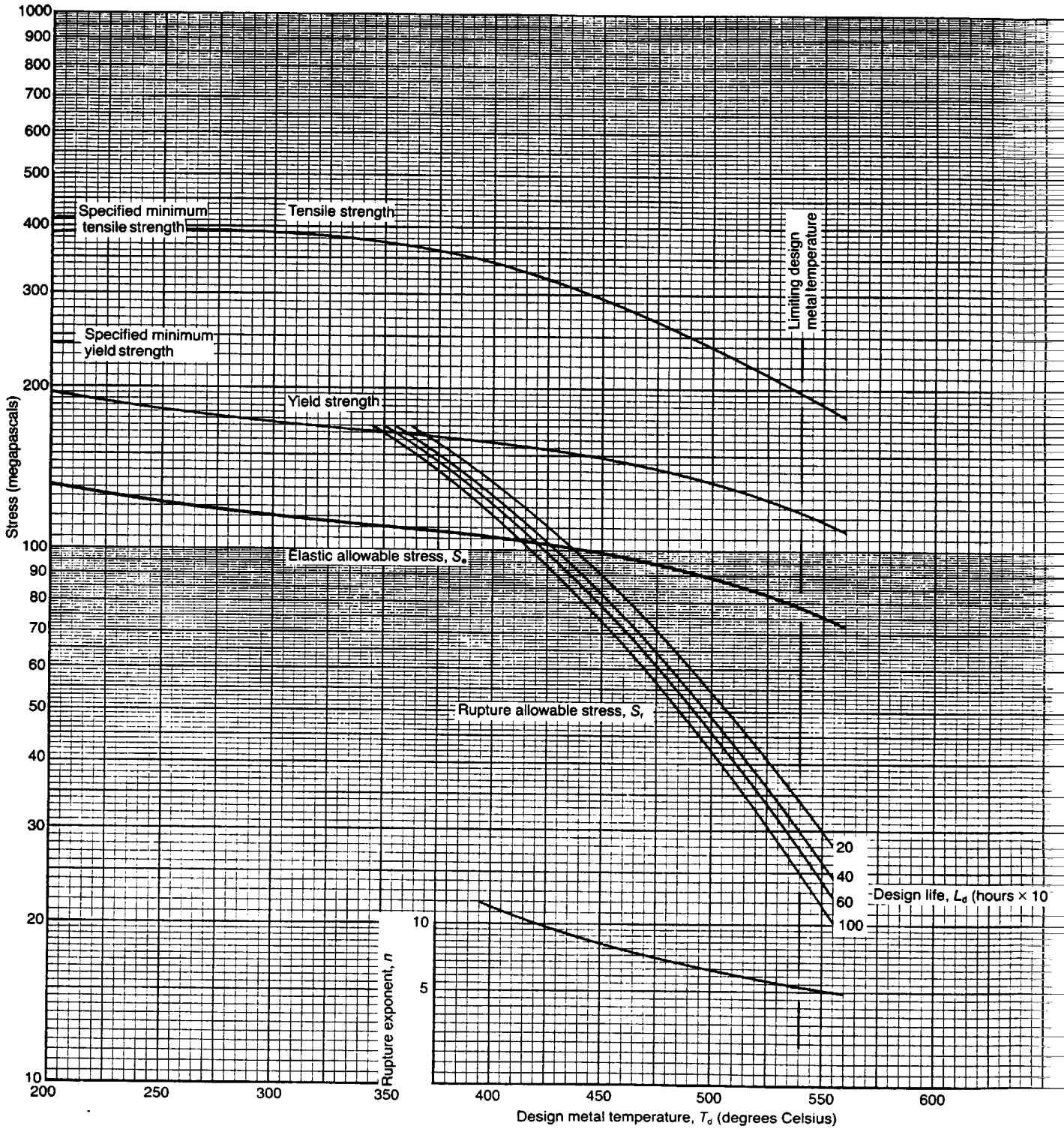


Figure 4B (SI)—Medium-Carbon Steel, ASTM A 53 Grade B (Seamless),

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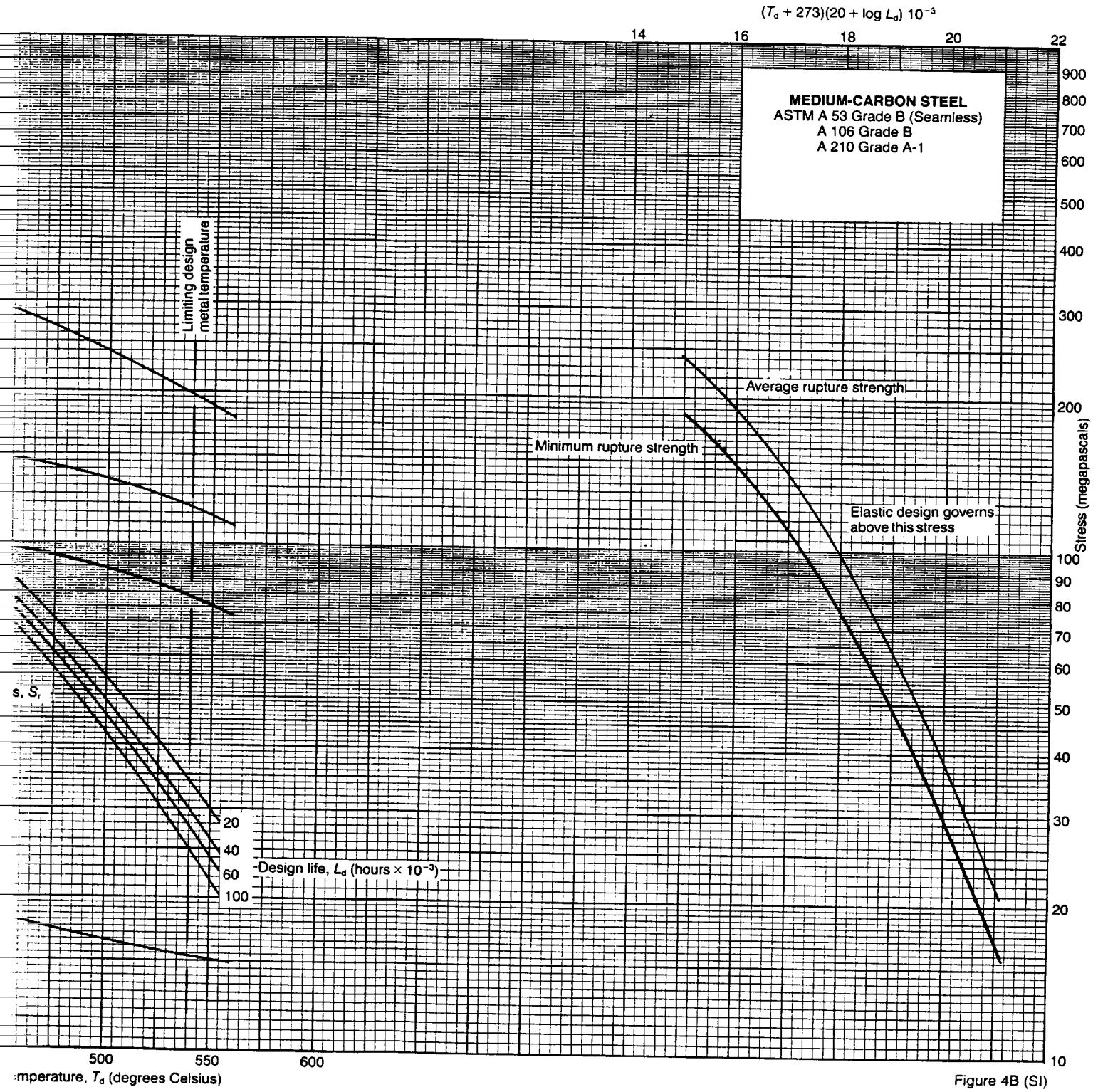


Figure 4B (SI)

-Carbon Steel, ASTM A 53 Grade B (Seamless), A 106 Grade B, A 210 Grade A-1

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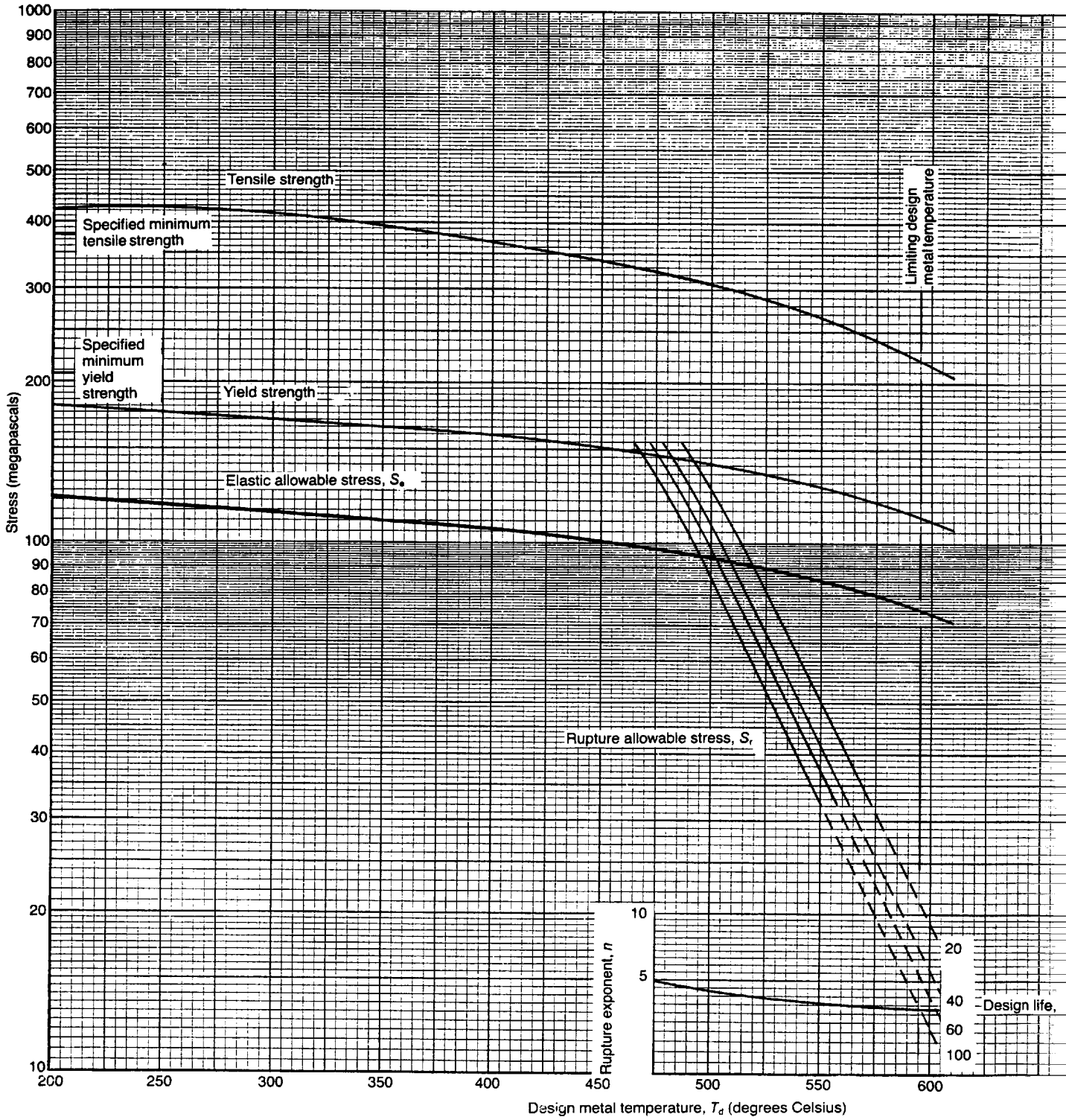


Figure 4C (SI)—C-½-Mo Steel, ASTM A 161 T1, A209 T

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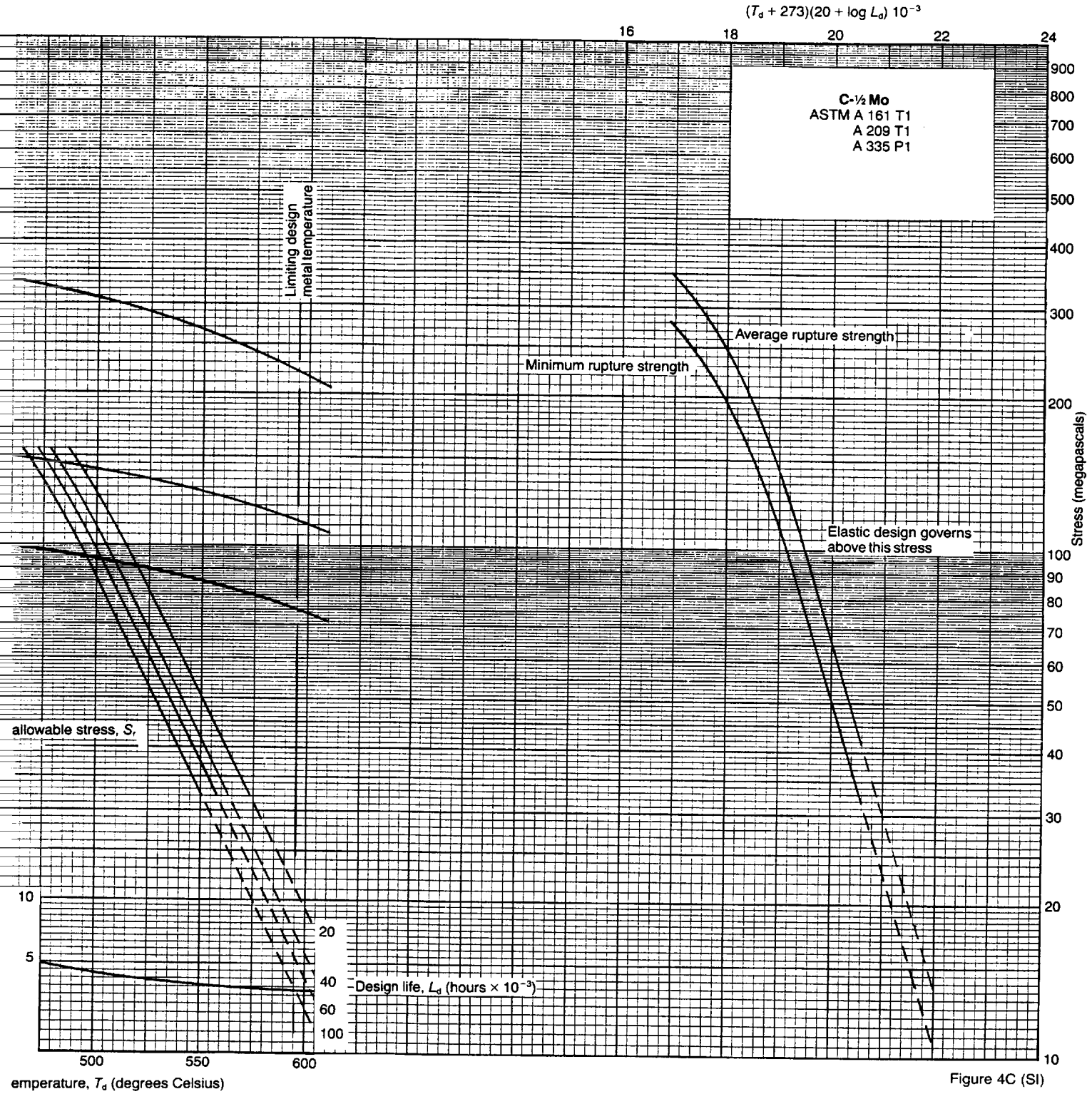


Figure 4C (SI)

:C (SI)—C-½-Mo Steel, ASTM A 161 T1, A209 T1, A335 P1

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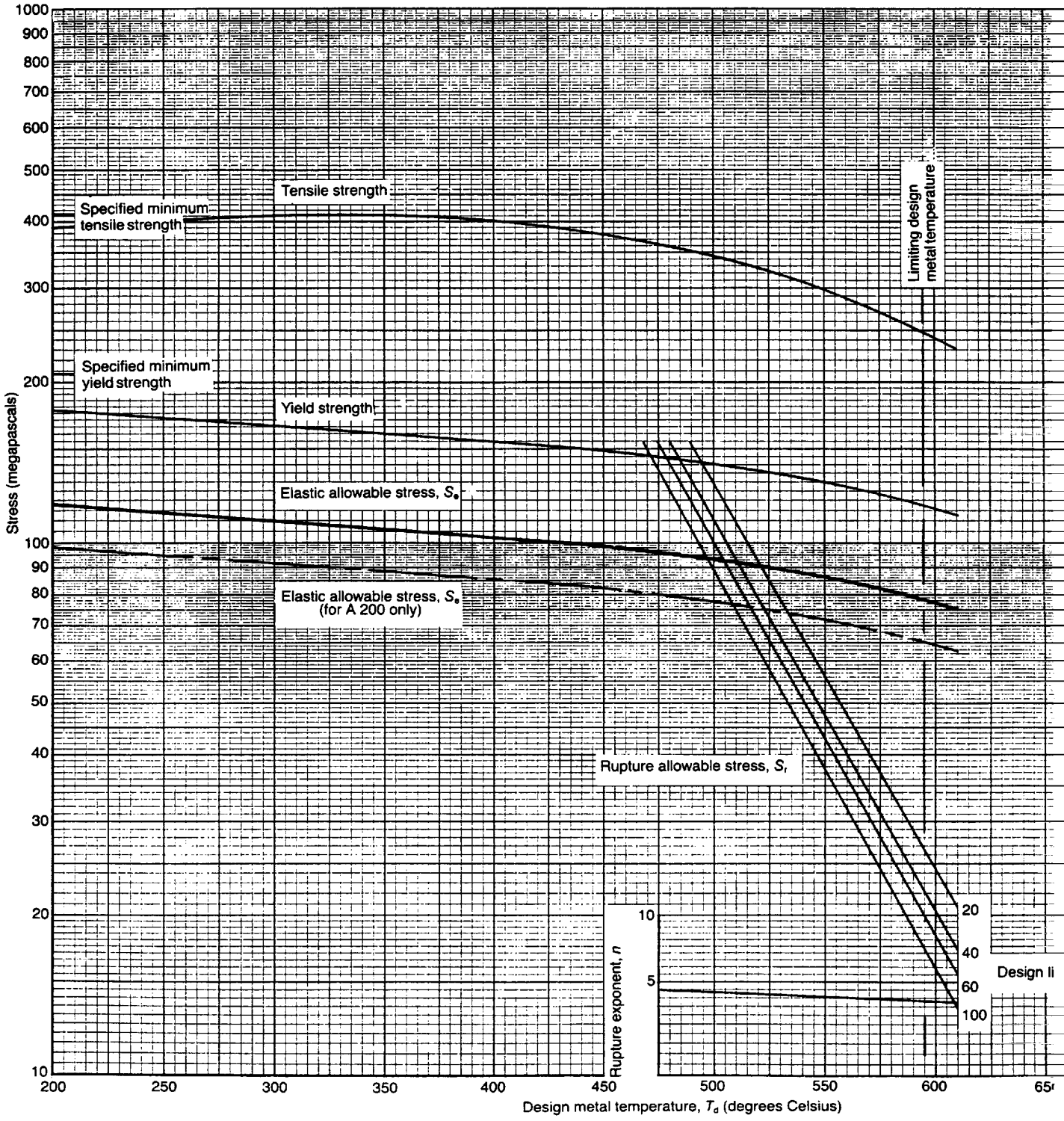


Figure 4D (SI)—1 1/4Cr–1/2Mo Steel, ASTM A 213 T11, A 3:

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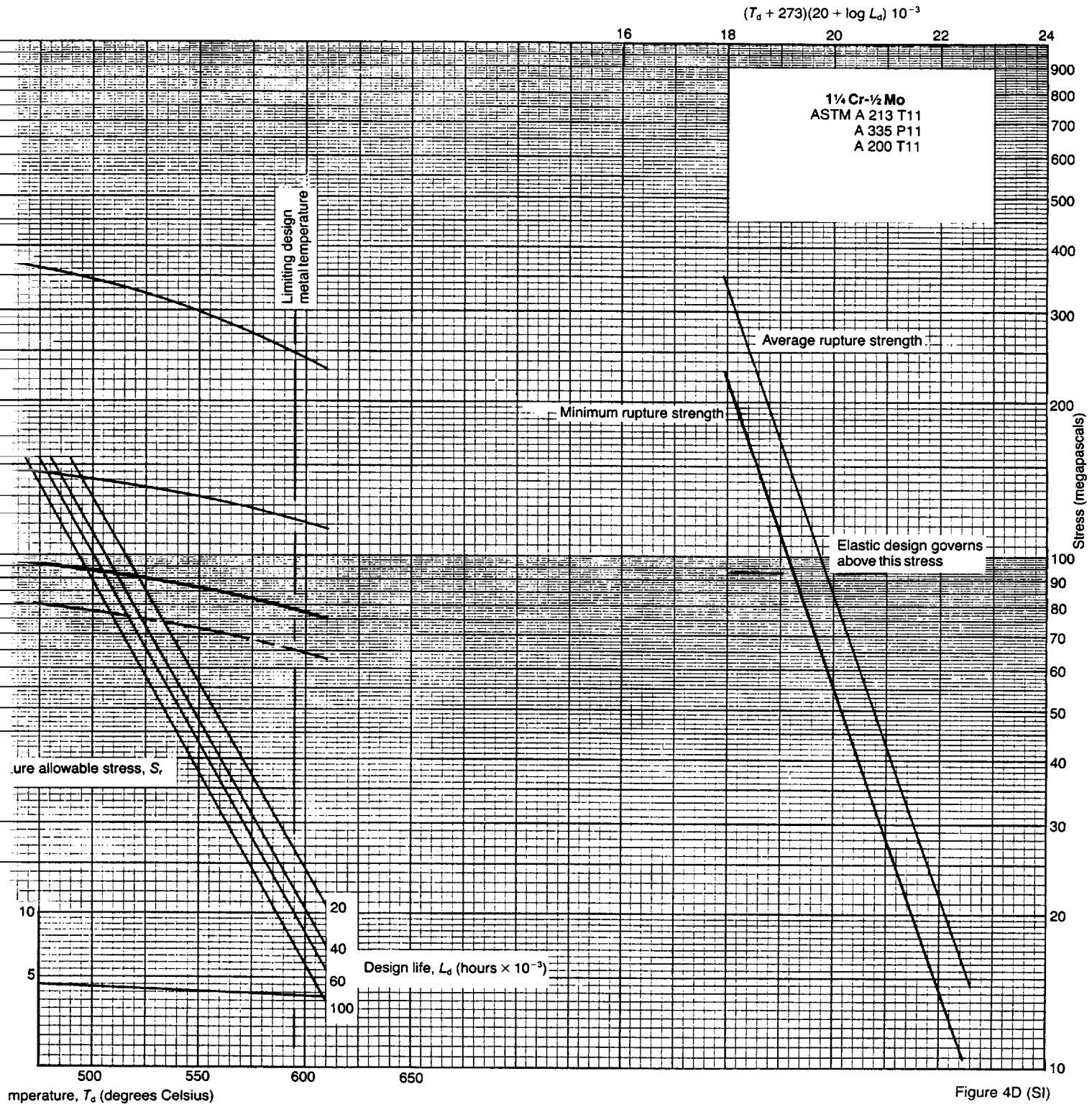


Figure 4D (SI)

(SI)—1 1/4Cr–1/2Mo Steel, ASTM A 213 T11, A 335 P11, A 200 T11

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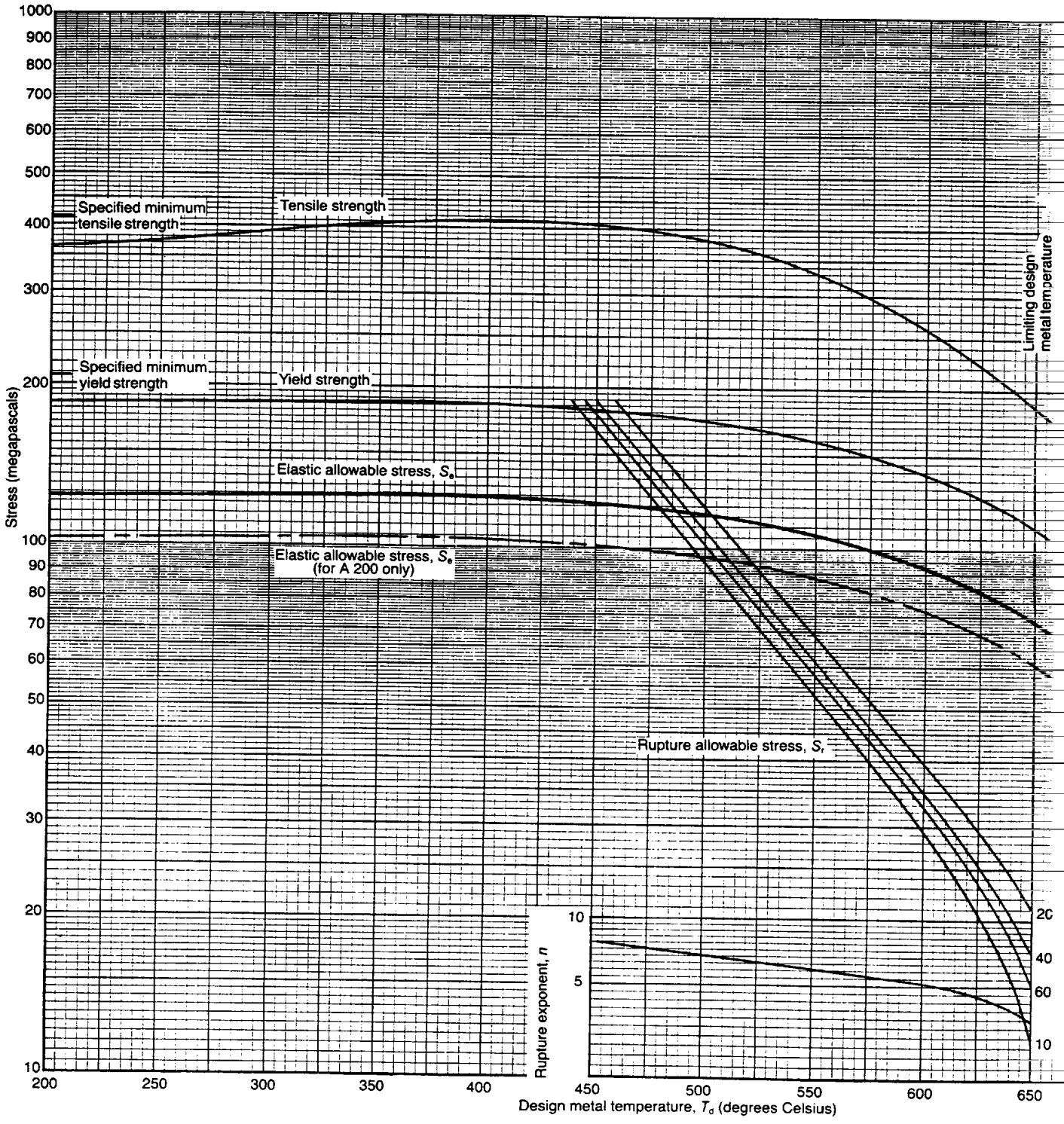


Figure 4E (SI)—2¼Cr–1Mo Steel, ASTM A 213 T22, A 33

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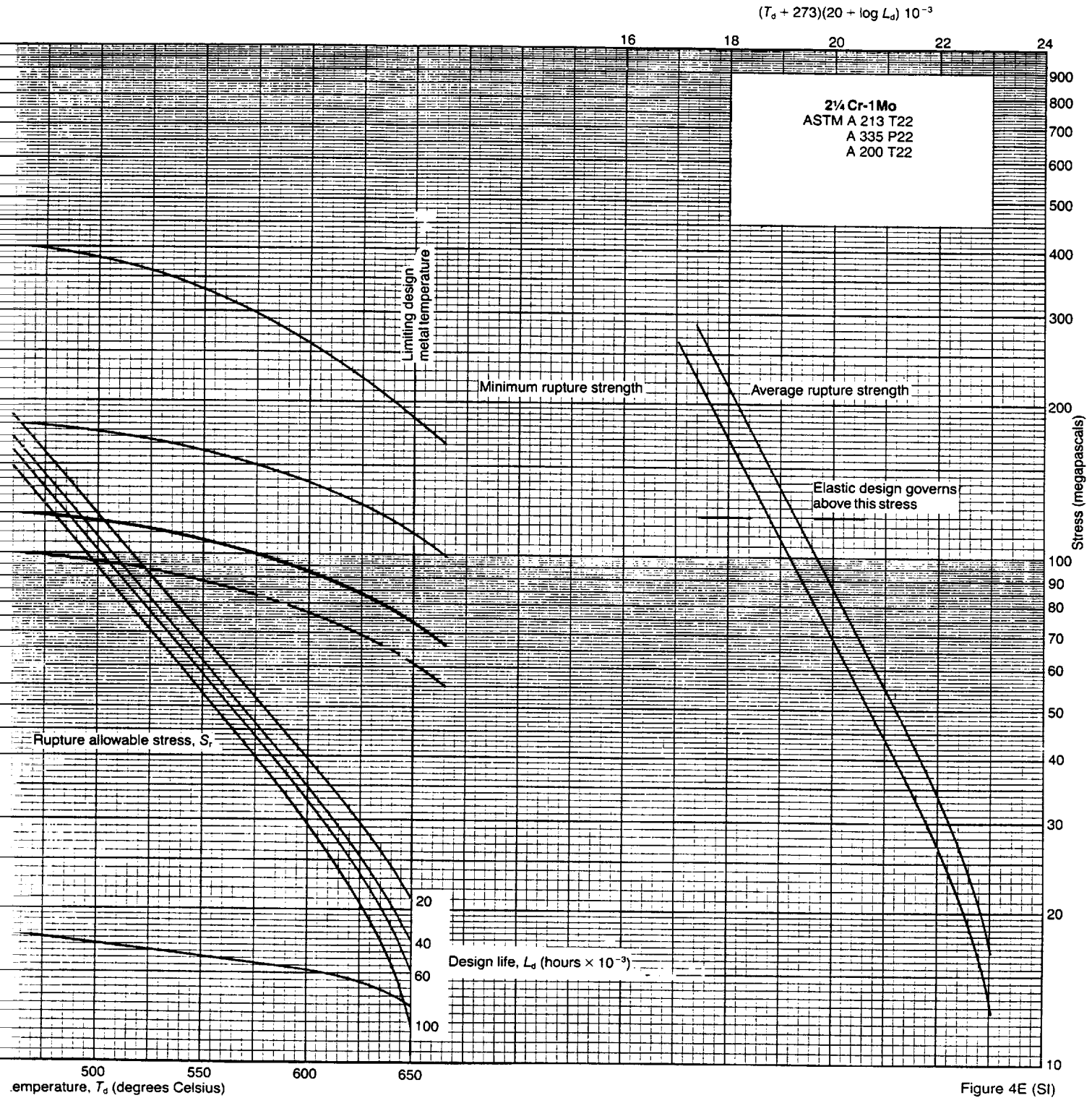


Figure 4E (SI)

(SI)—2¼Cr–1Mo Steel, ASTM A 213 T22, A 335 P22, A 200 T22

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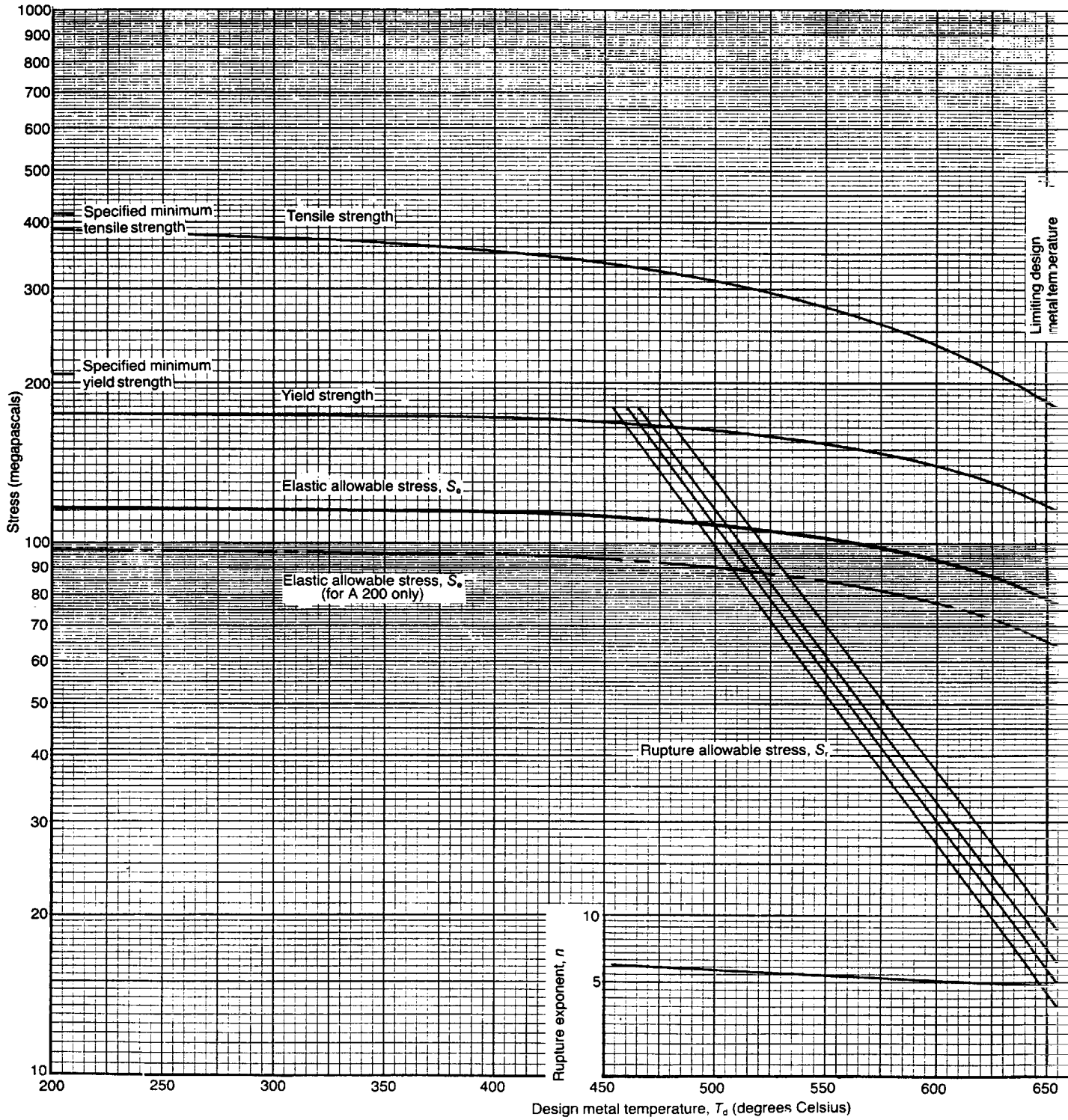


Figure 4F (SI)—3Cr–1Mo Steel, ASTM A 213 T21, A 335

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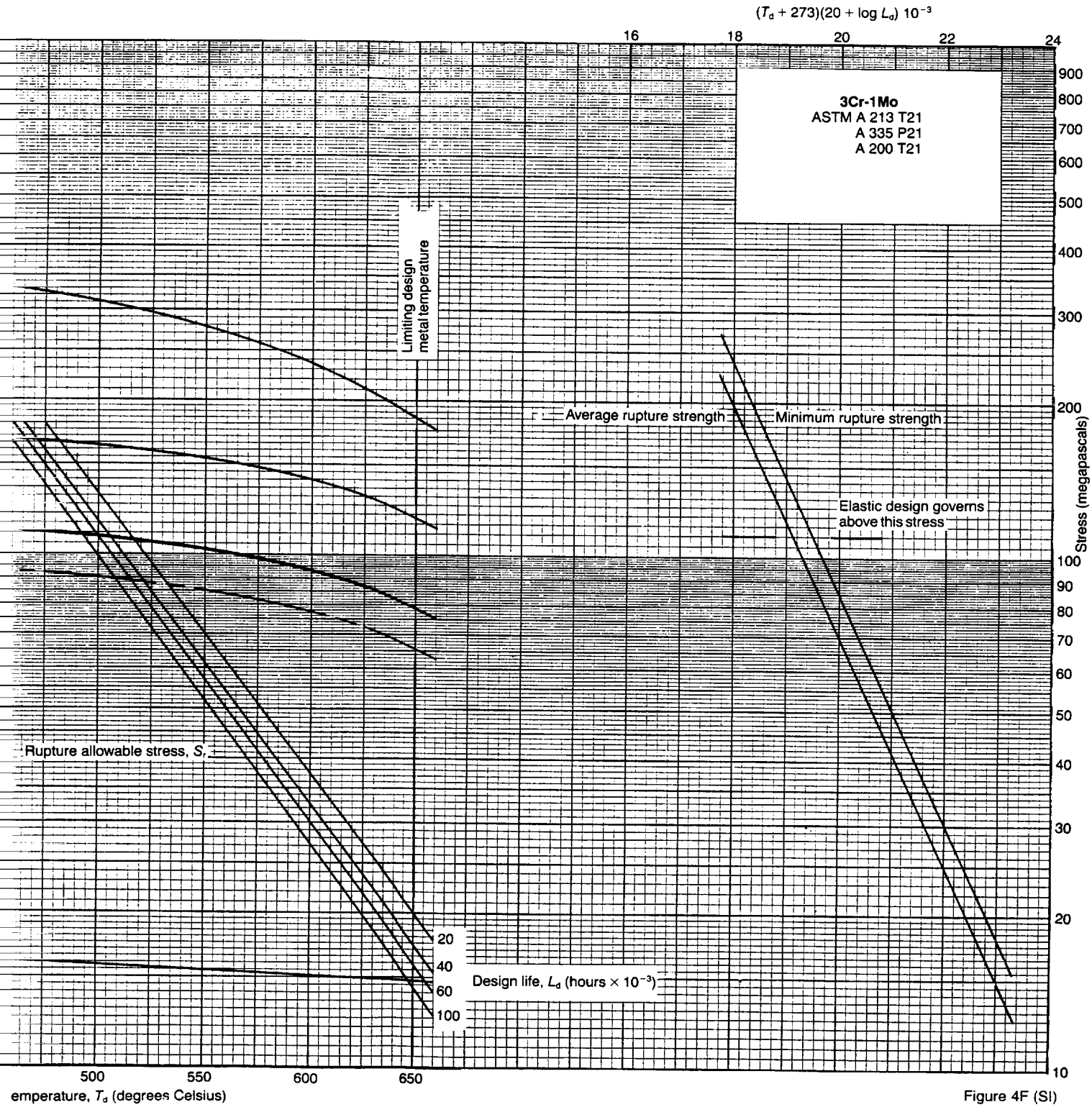


Figure 4F (SI)

F (SI)—3Cr–1Mo Steel, ASTM A 213 T21, A 335 P21, A 200 T21

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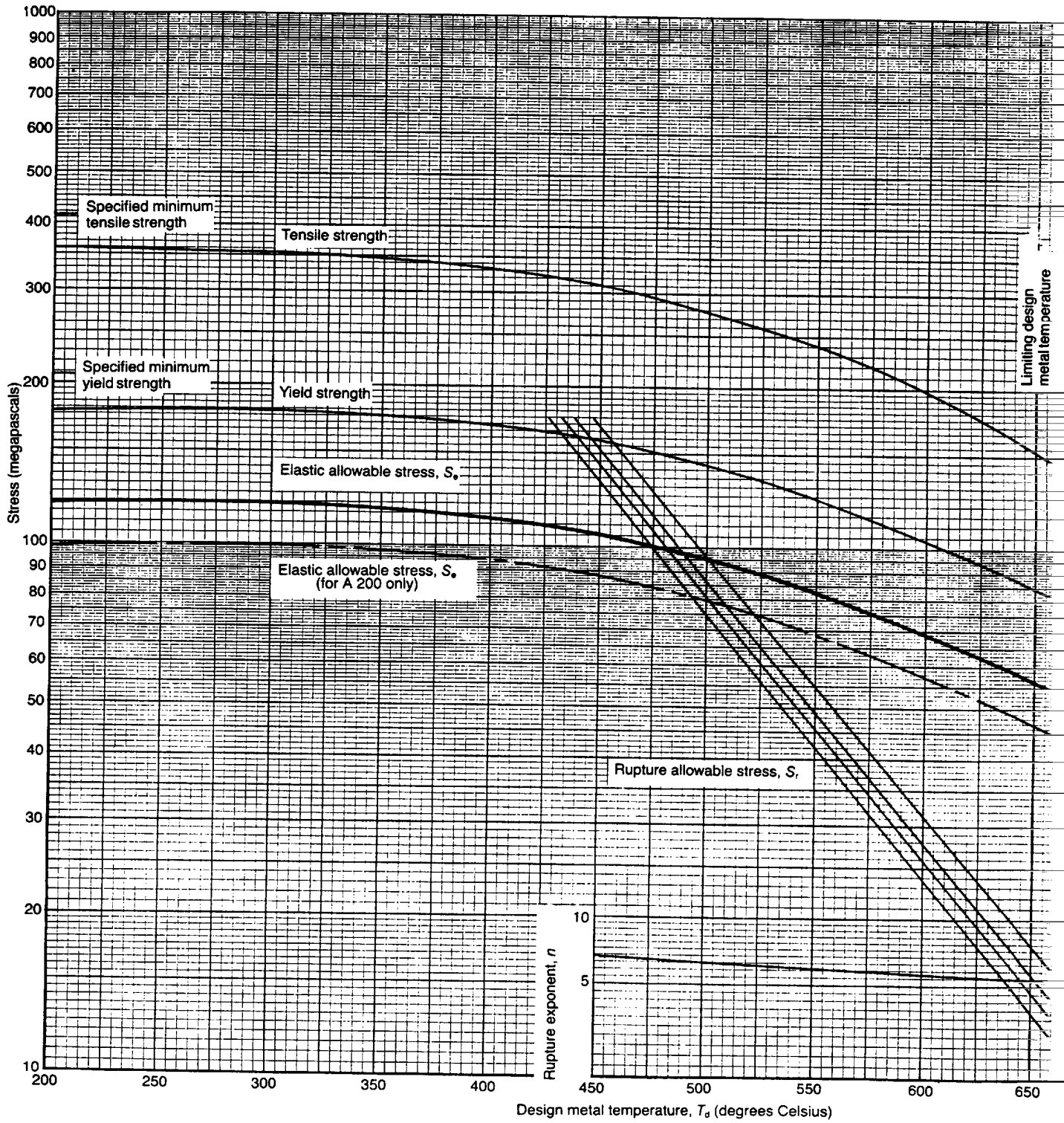


Figure 4G (SI)—5Cr–½Mo Steel, ASTM A 213 T5, A 335

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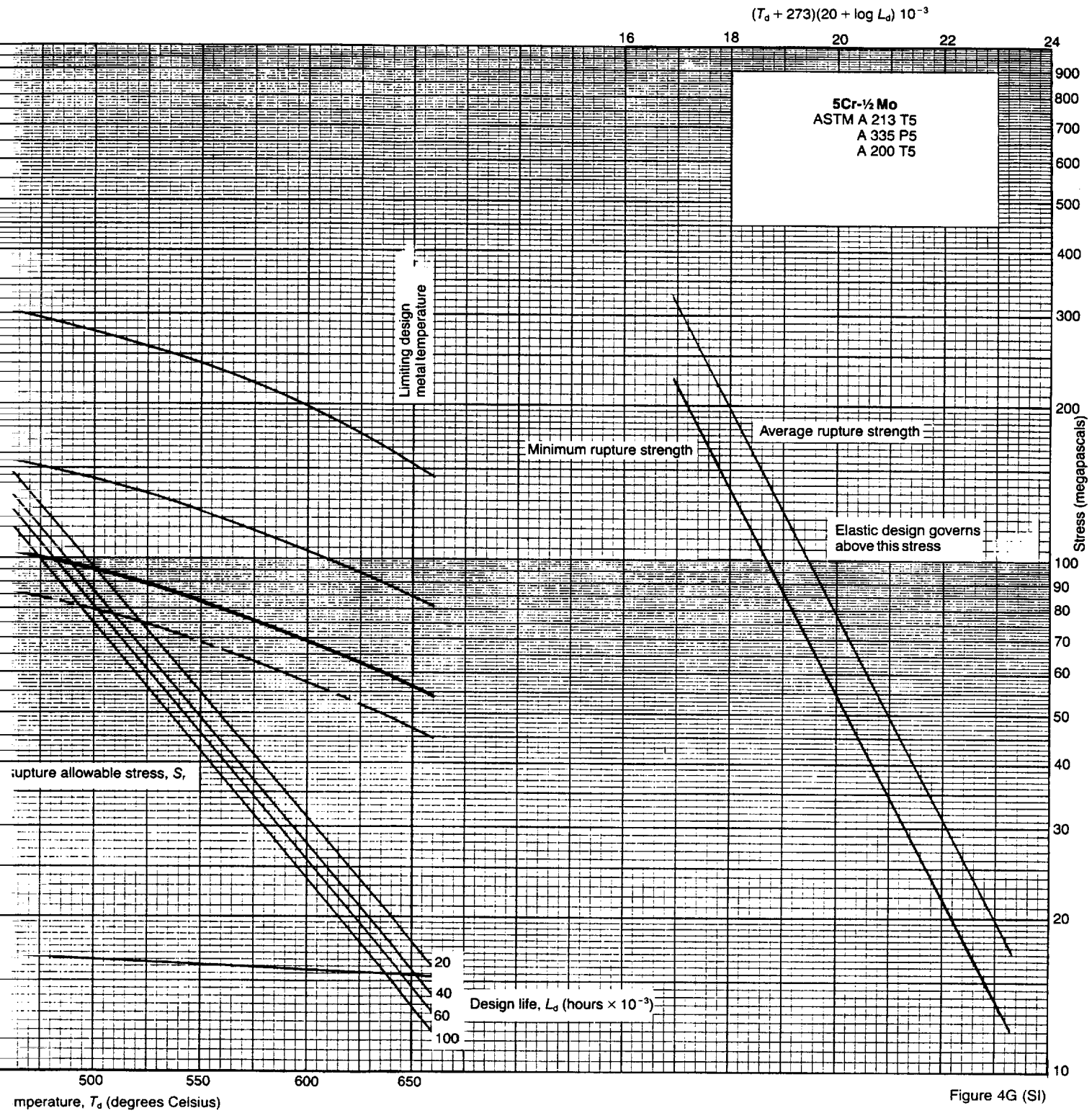


Figure 4G (SI)

4G (SI)—5Cr–½Mo Steel, ASTM A 213 T5, A 335 P5, A 200 T5

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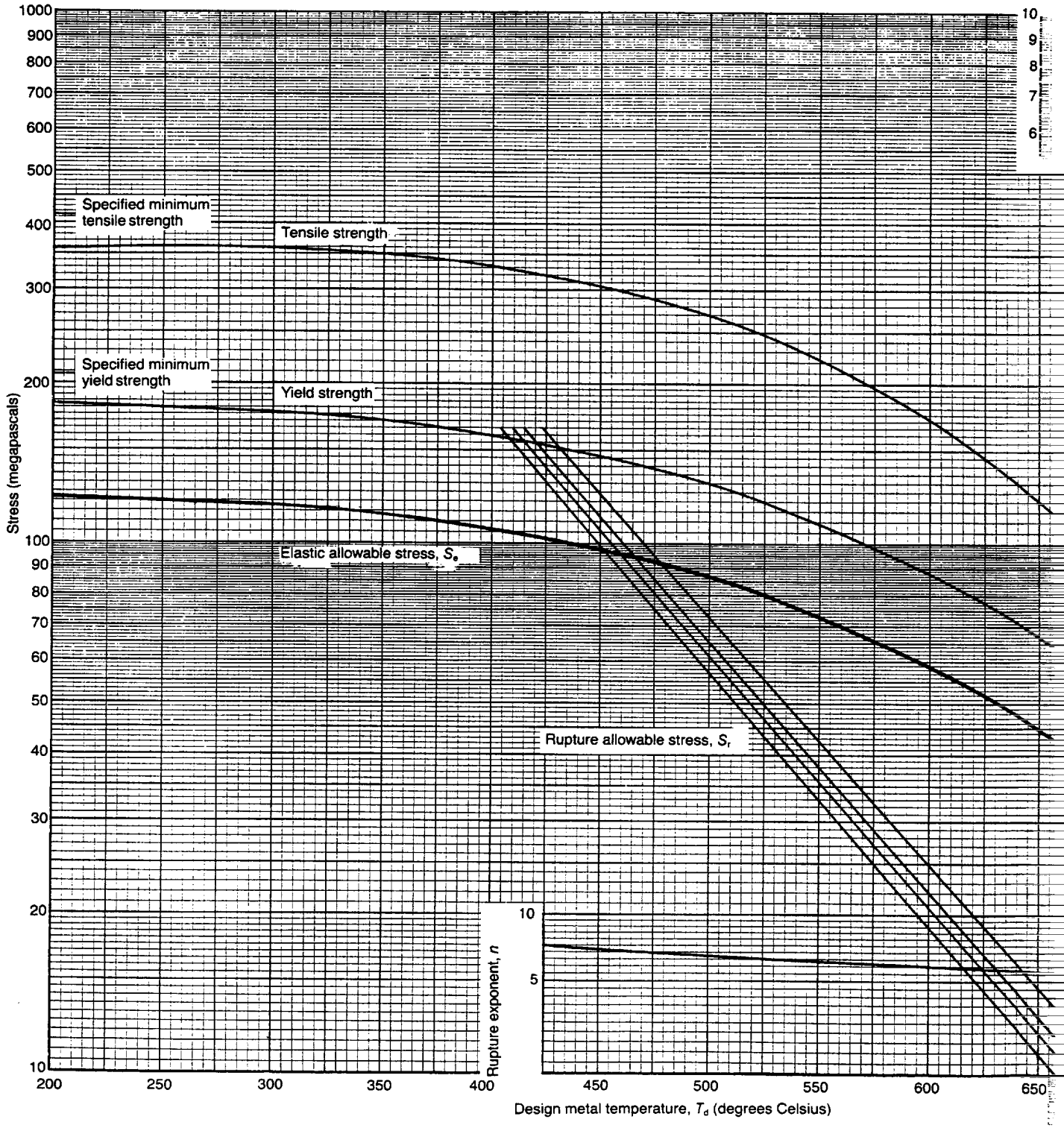


Figure 4H (SI)—5Cr–½Mo Si Steel, ASTM A 213 T5b,

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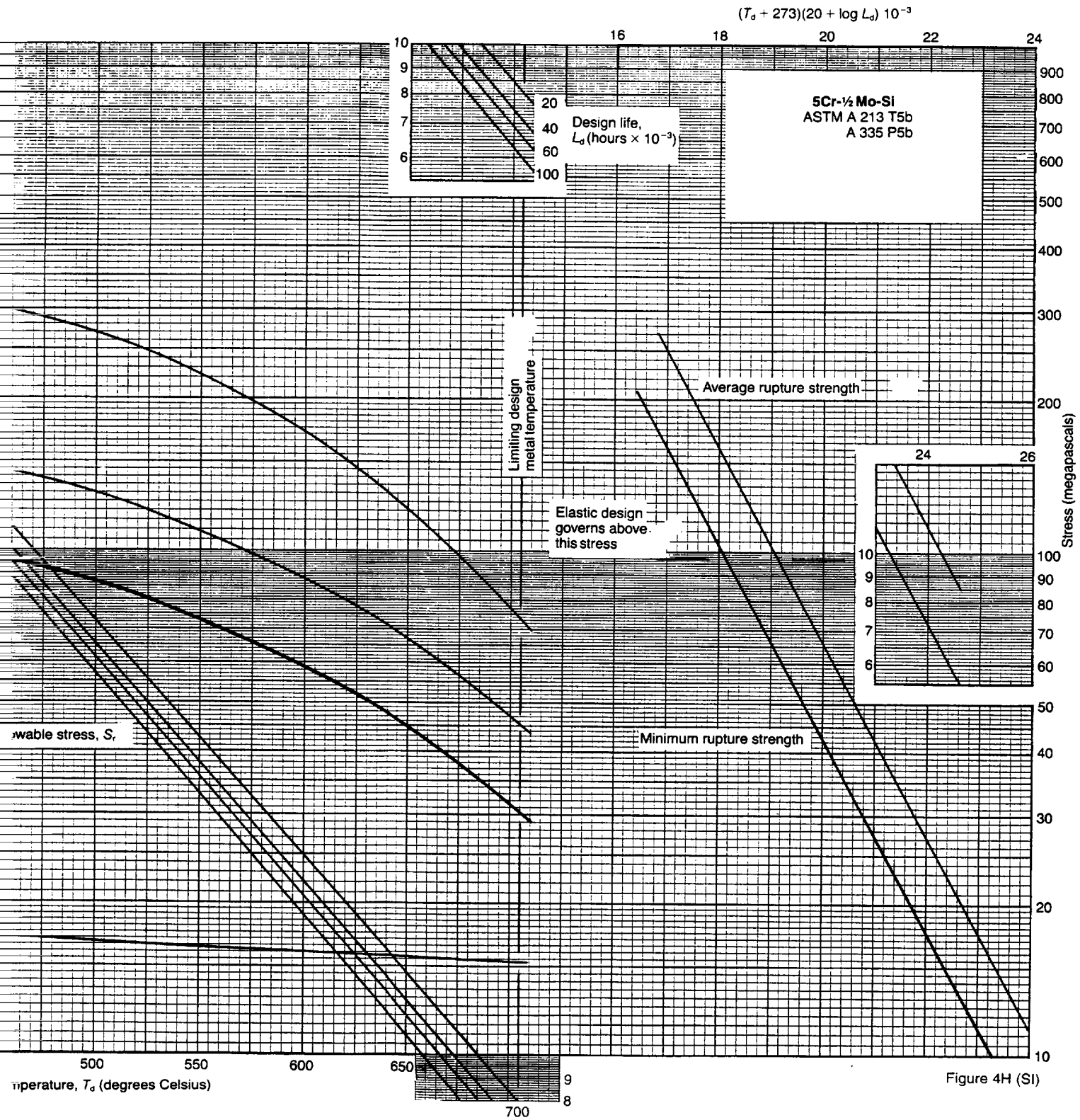


Figure 4H (SI)

re 4H (SI)—5Cr–½Mo Si Steel, ASTM A 213 T5b, A 335 P5b

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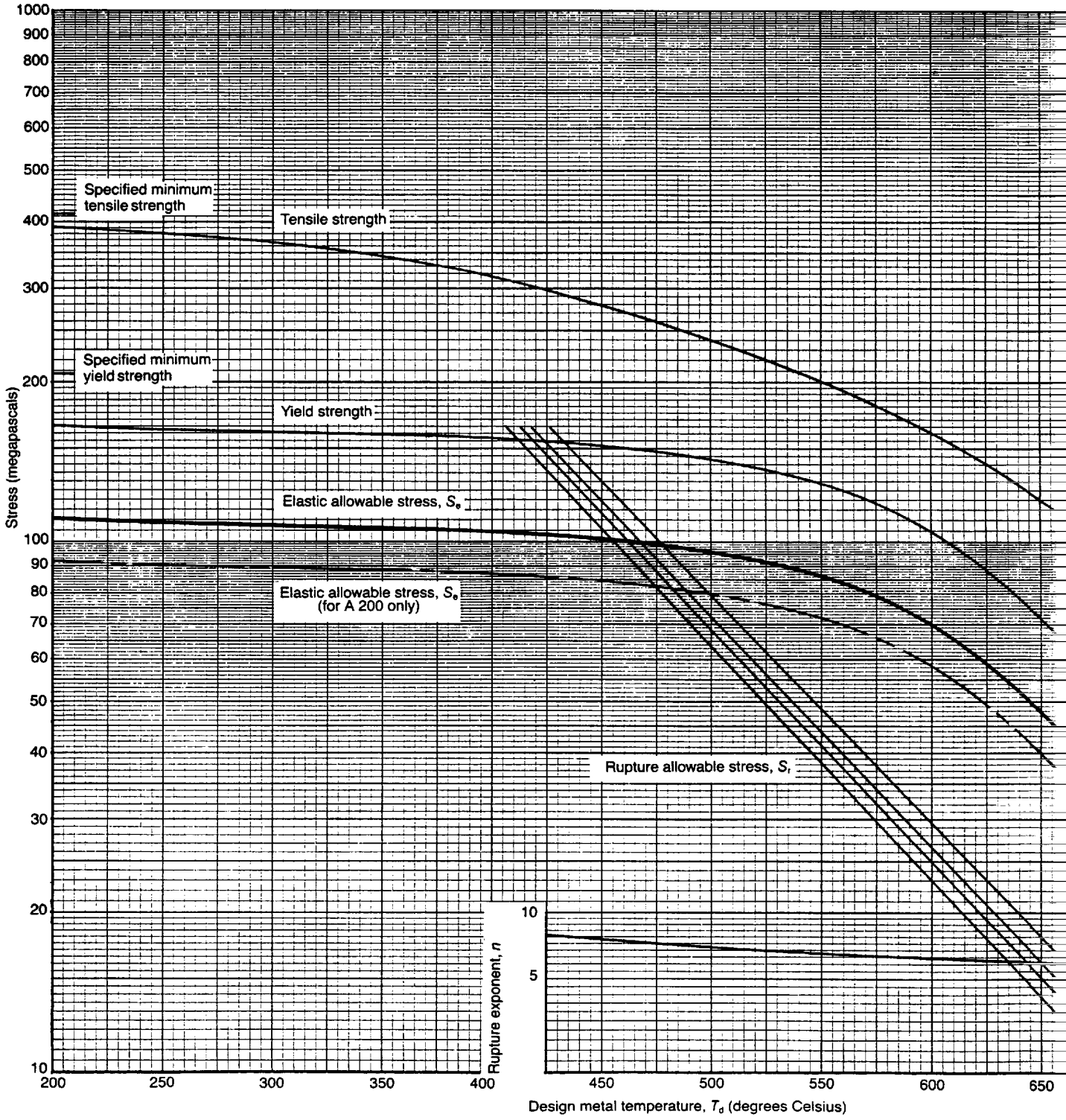


Figure 4I (SI)—7Cr–½Mo Steel, ASTM A 213 T7, A 33

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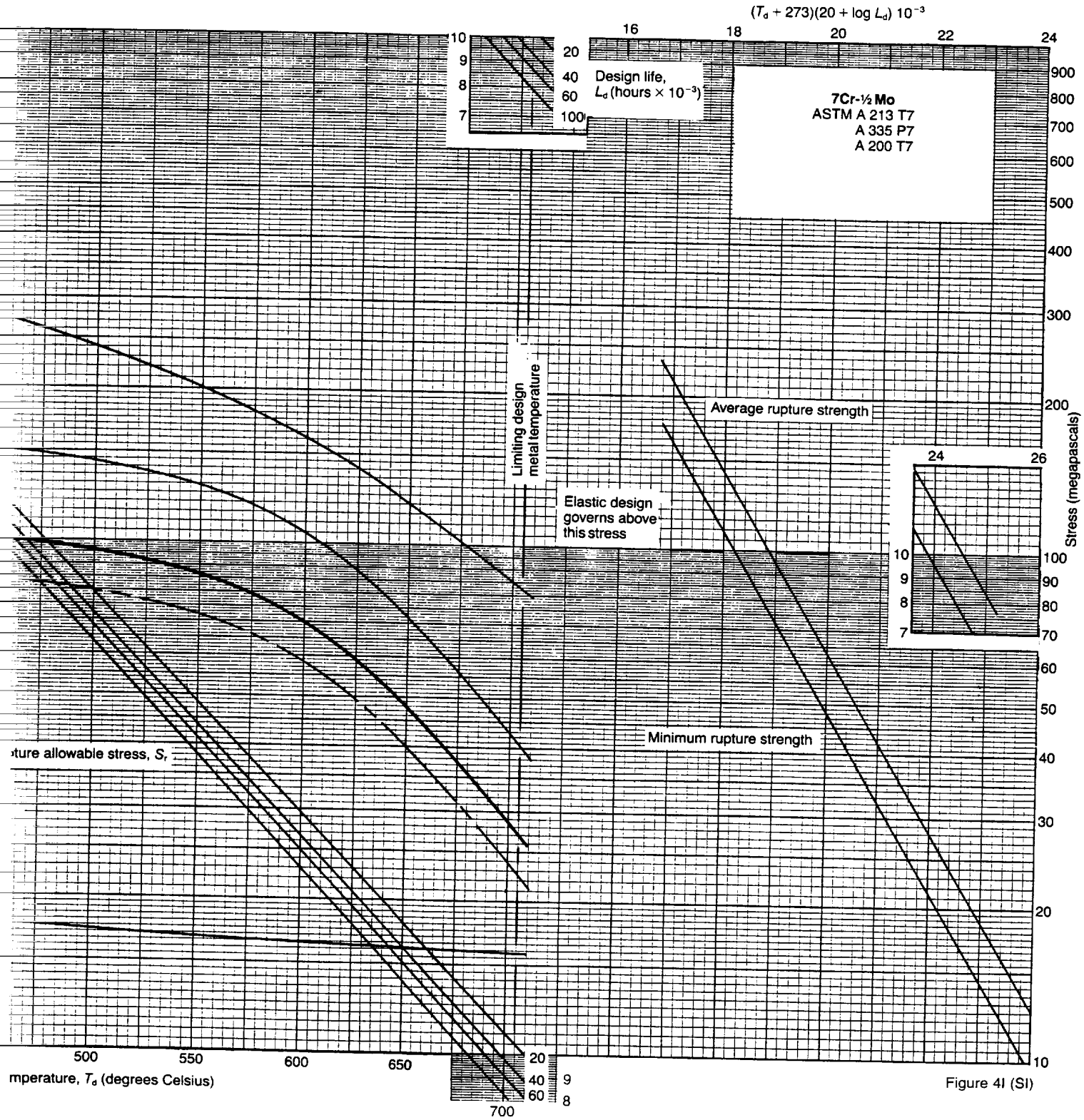


Figure 4I (SI)

4I (SI)—7Cr–½Mo Steel, ASTM A 213 T7, A 335 P7, A 200 T7

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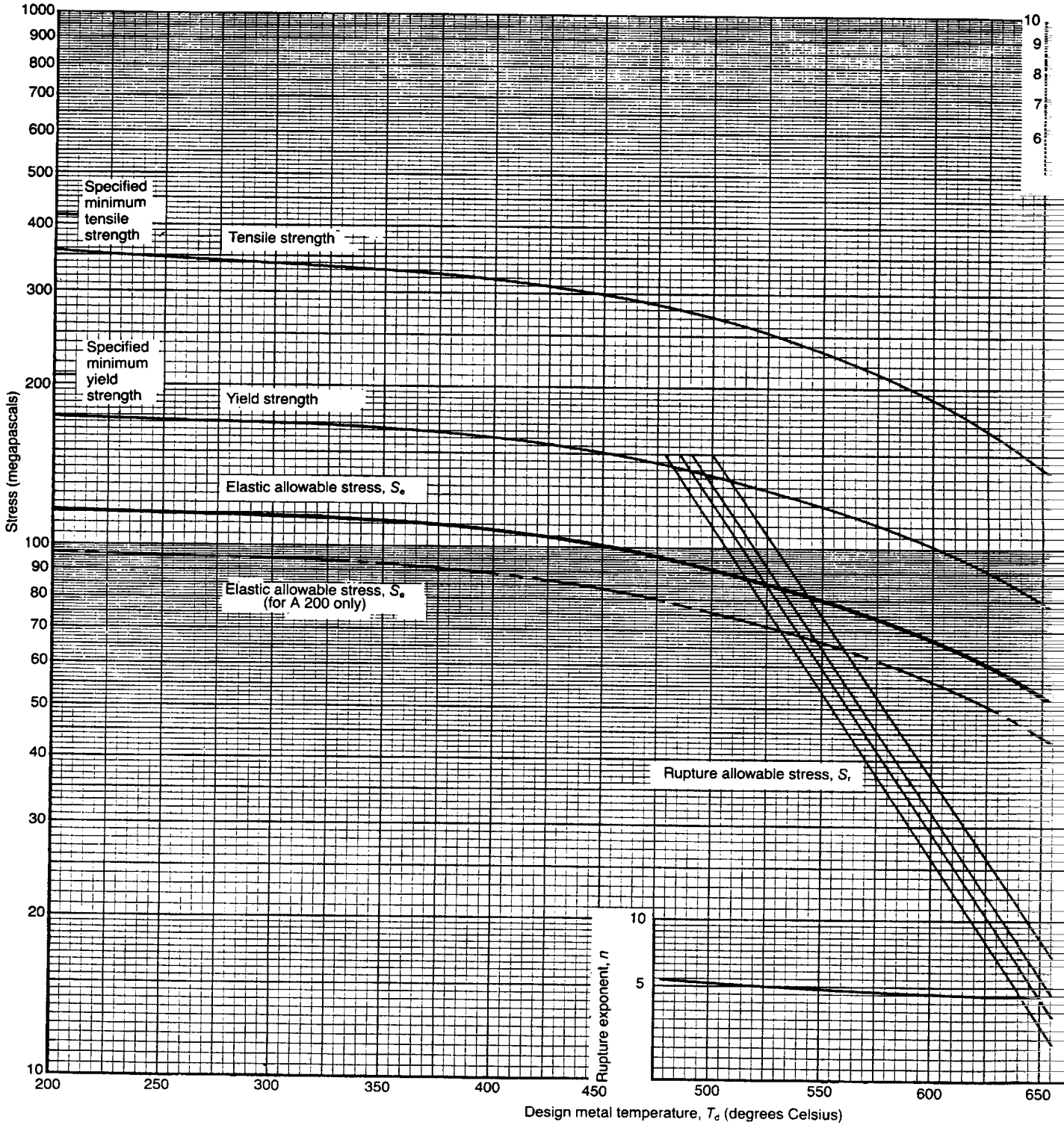


Figure 4J (SI)—9Cr–1Mo Steel, ASTM A 213 T9, A 335

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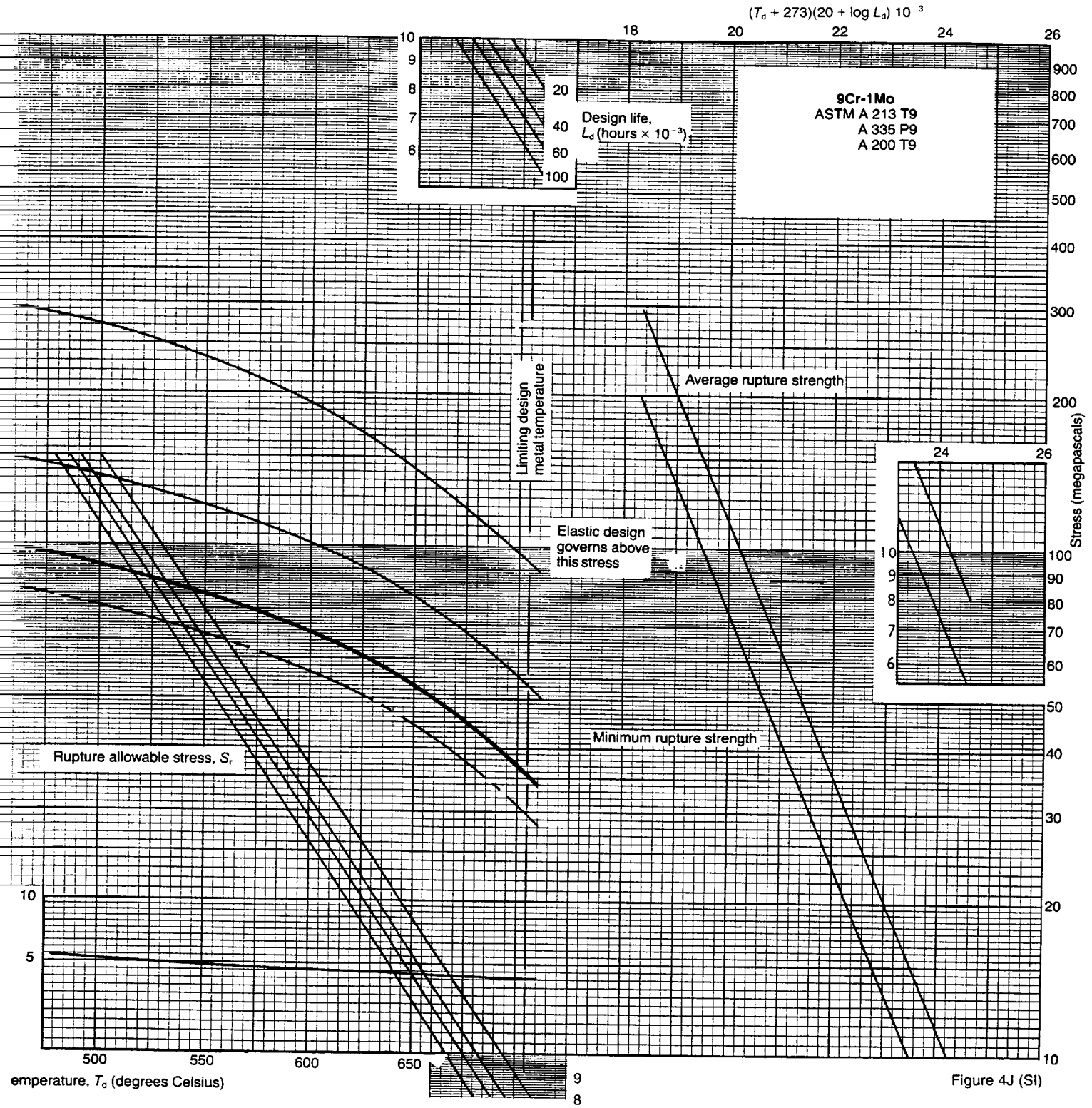


Figure 4J (SI)

: 4J (SI)—9Cr–1Mo Steel, ASTM A 213 T9, A 335 P9, A 200 T9

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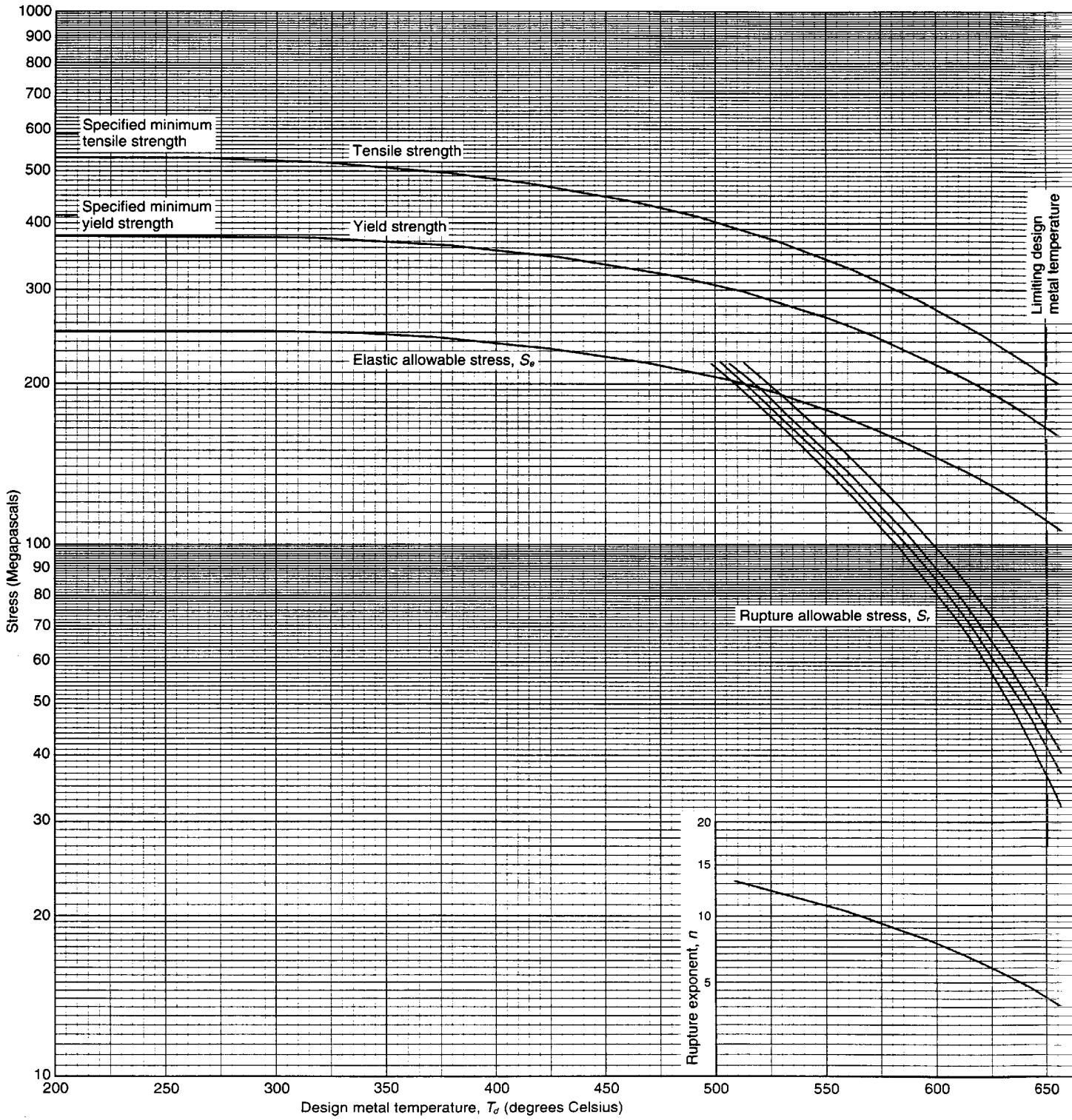


Figure 4K (SI)—9Cr–1Mo–Va Steel, ASTM A 213 T91, A 3

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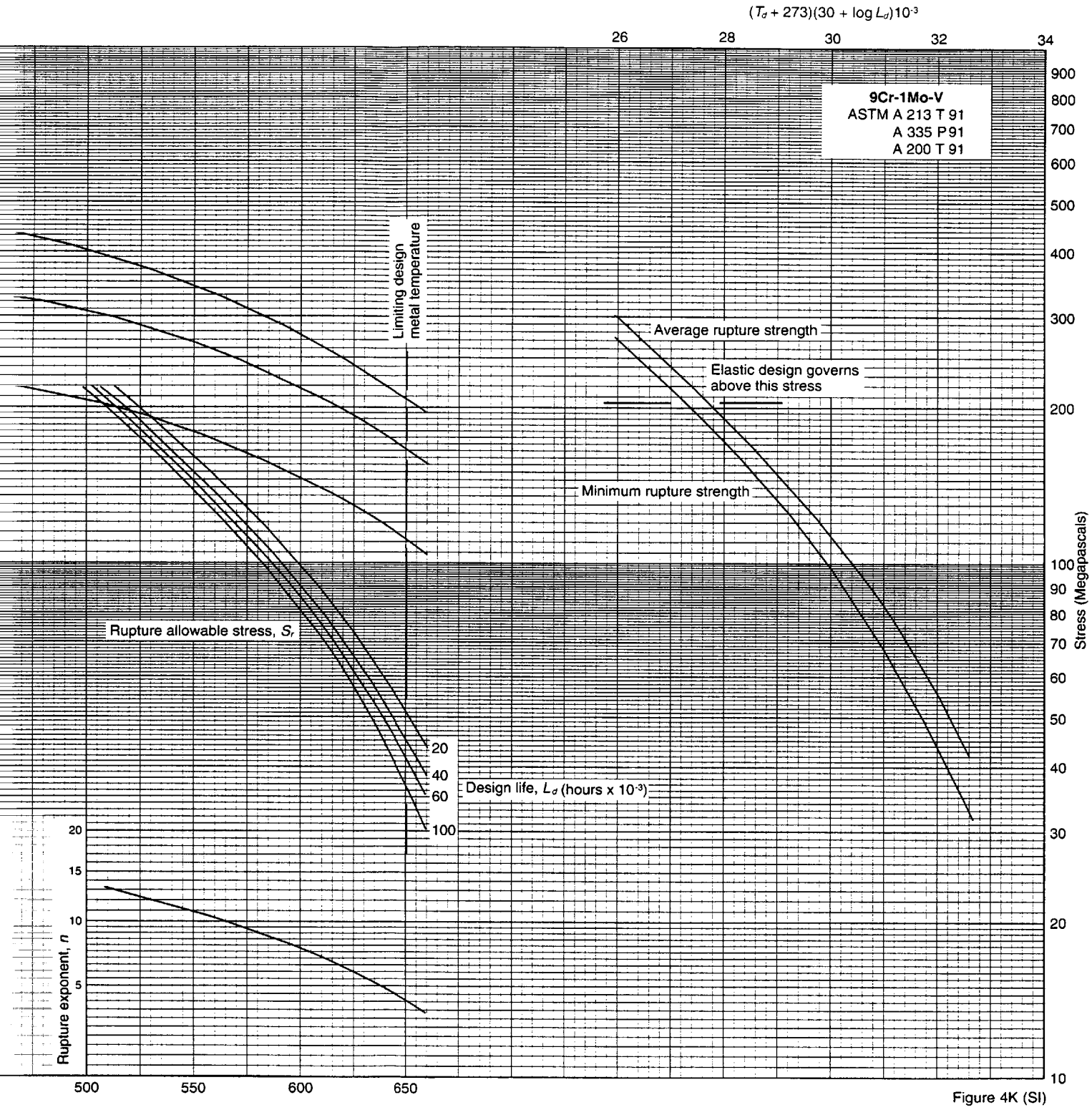


Figure 4K (SI)

(SI)—9Cr–1Mo–Va Steel, ASTM A 213 T91, A 335 P91, A 200 T91

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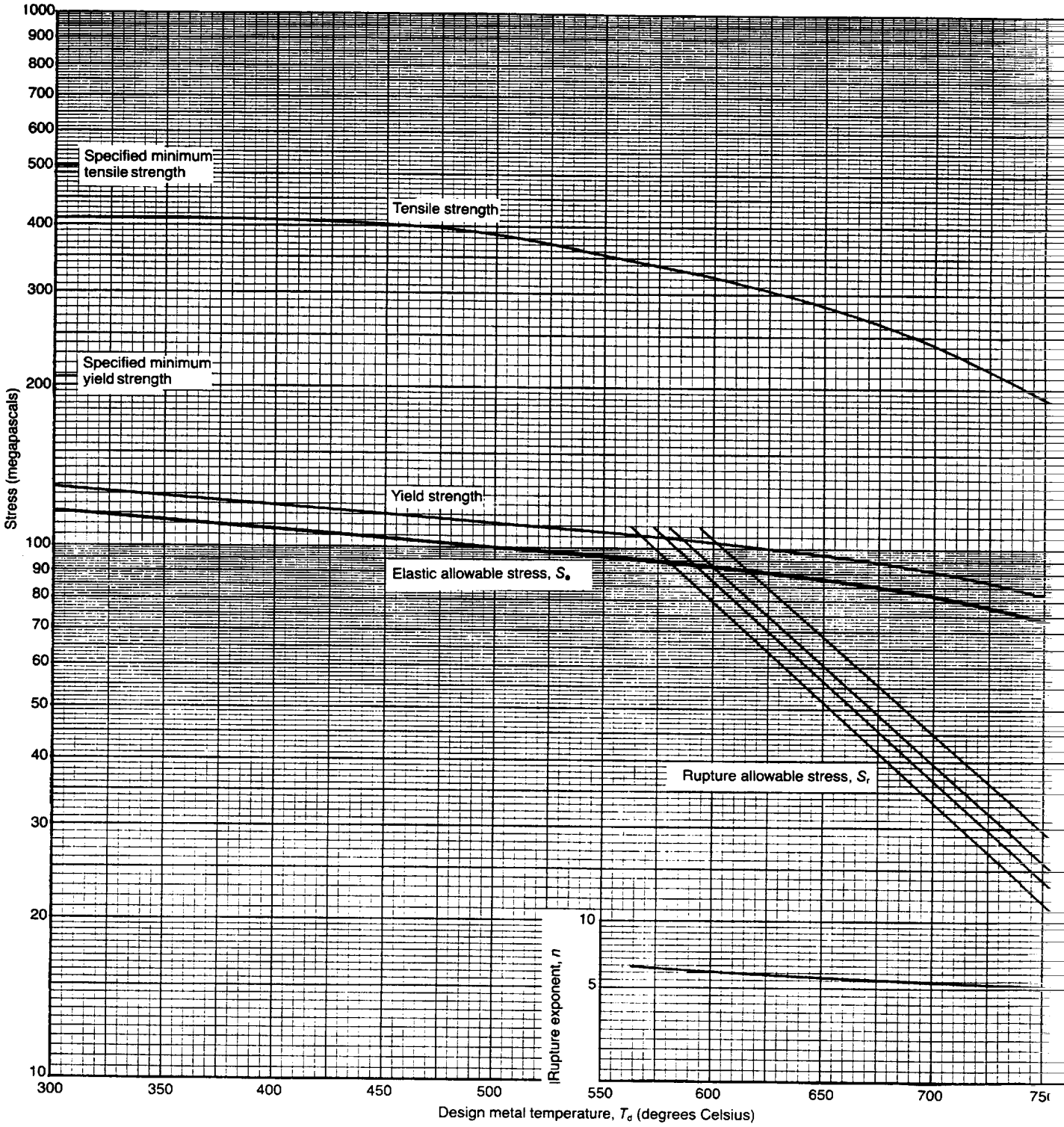


Figure 4L (SI)—Types 304 and 304H Stainless Steel, ASTM A 2

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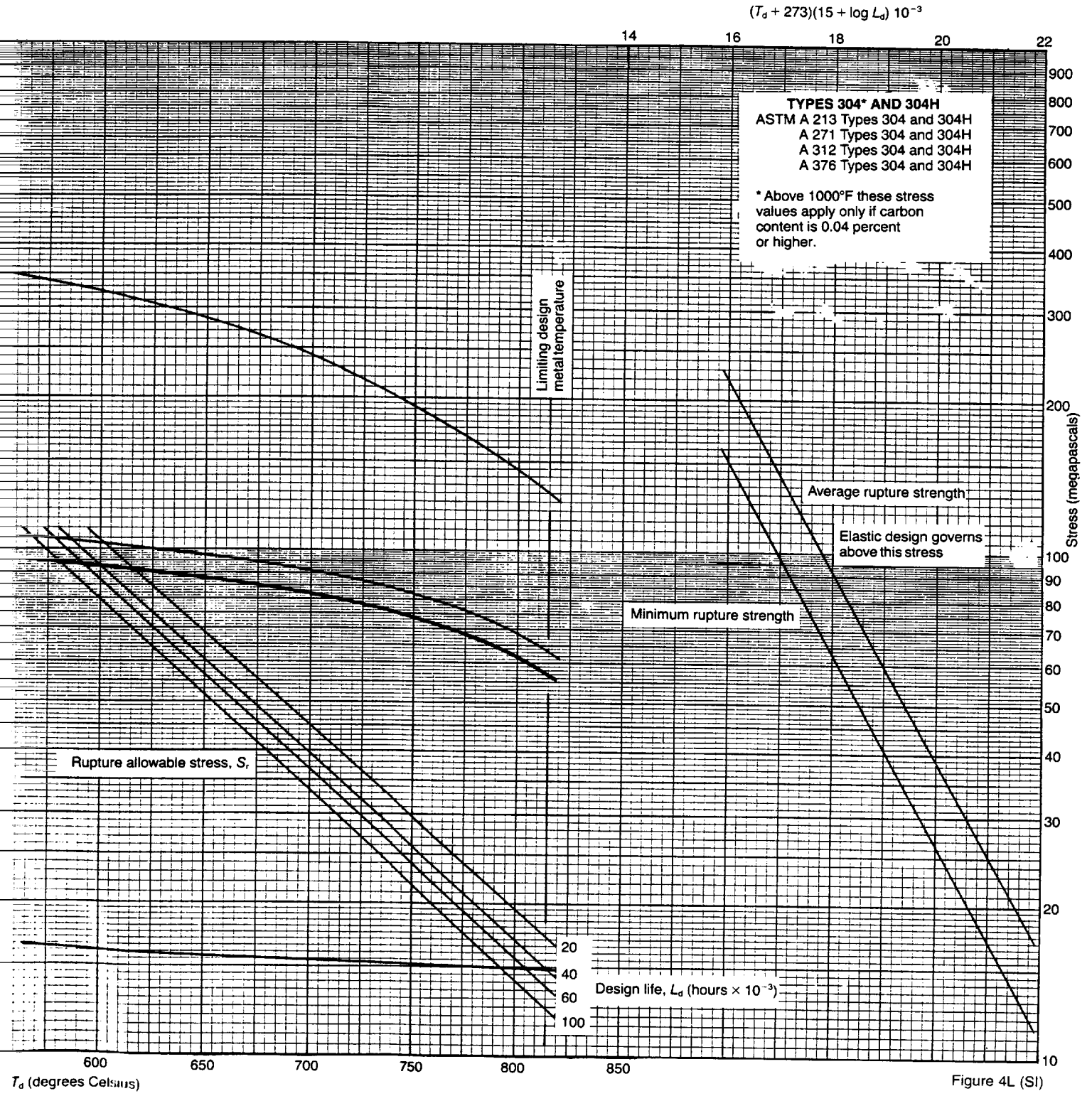


Figure 4L (SI)

Types 304 and 304H Stainless Steel, ASTM A 213, A 271, A 312, A 376

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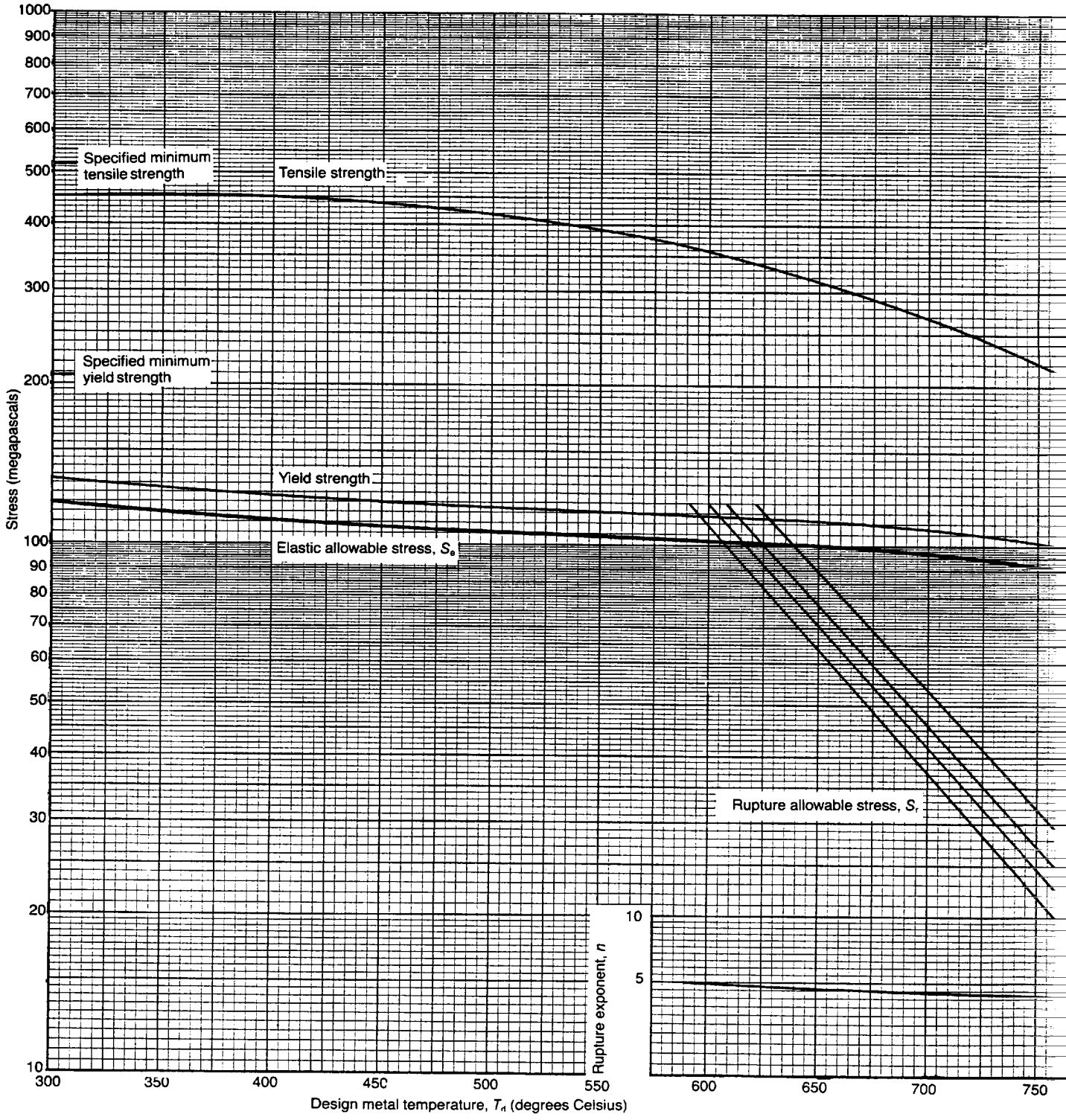


Figure 4M (SI)—Types 316 and 316H Stainless Steel, ASTM A 2

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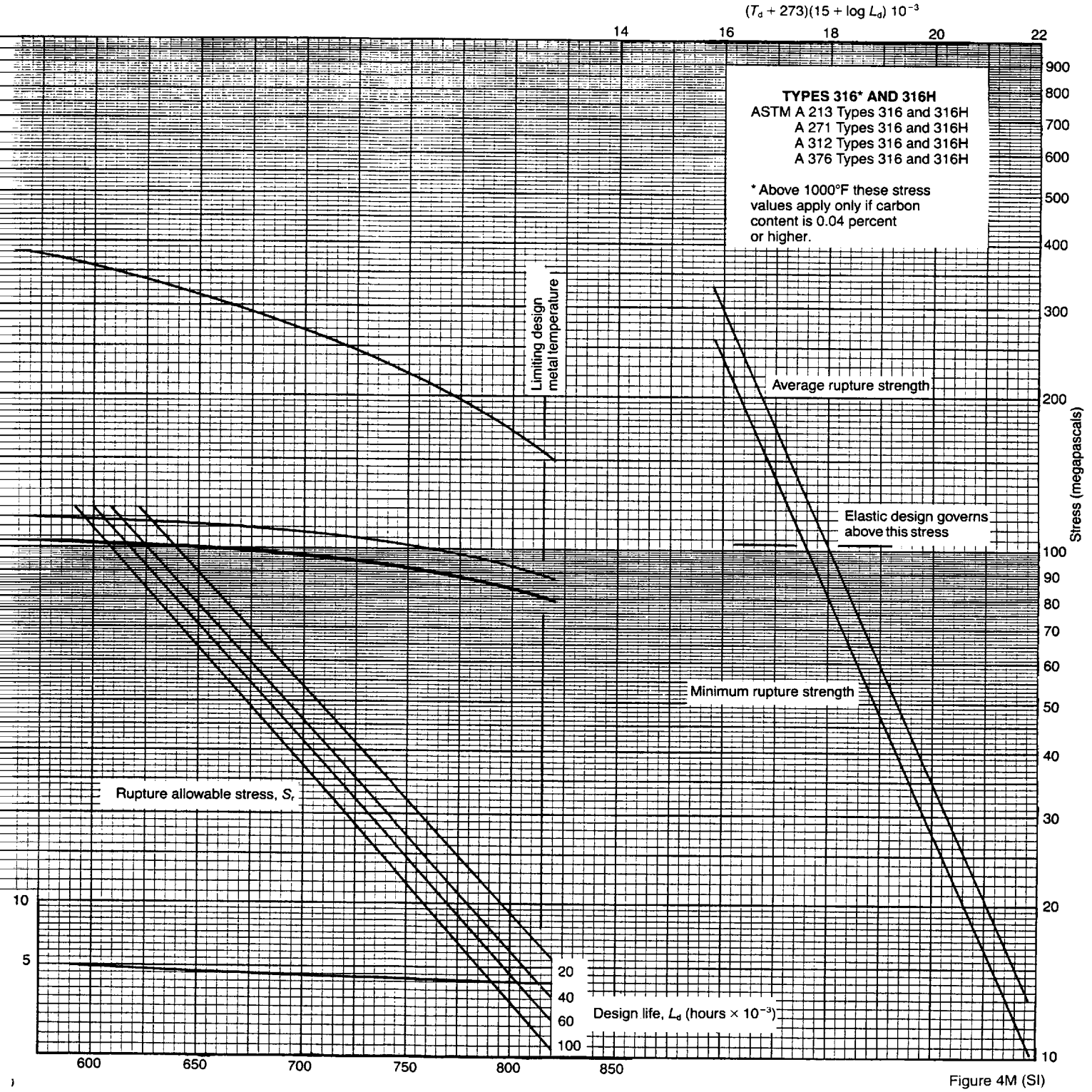


Figure 4M (SI)

Types 316 and 316H Stainless Steel, ASTM A 213, A 271, A 312, A 376

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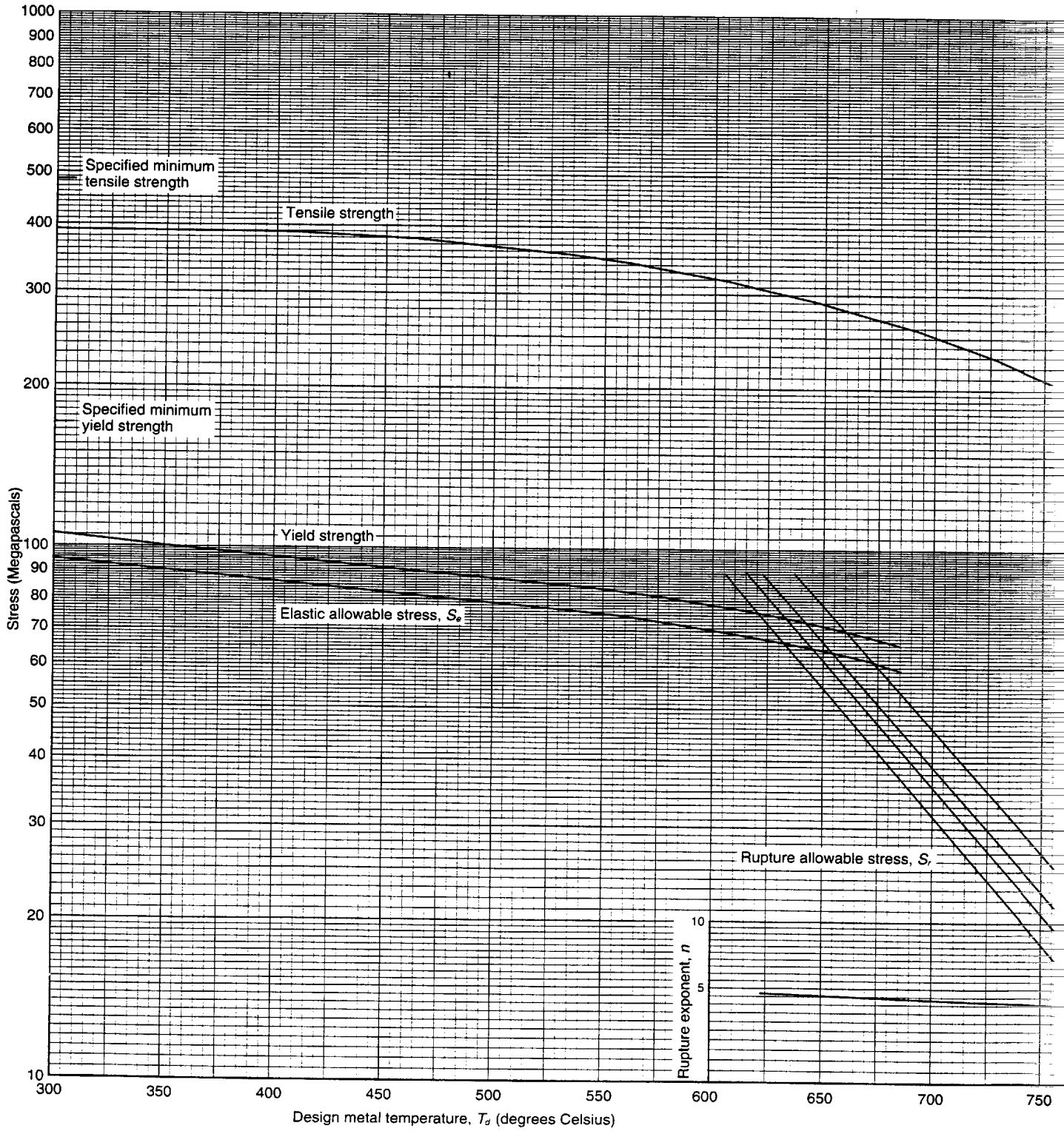


Figure 4N (SI)—Type 316L Stainless Steel

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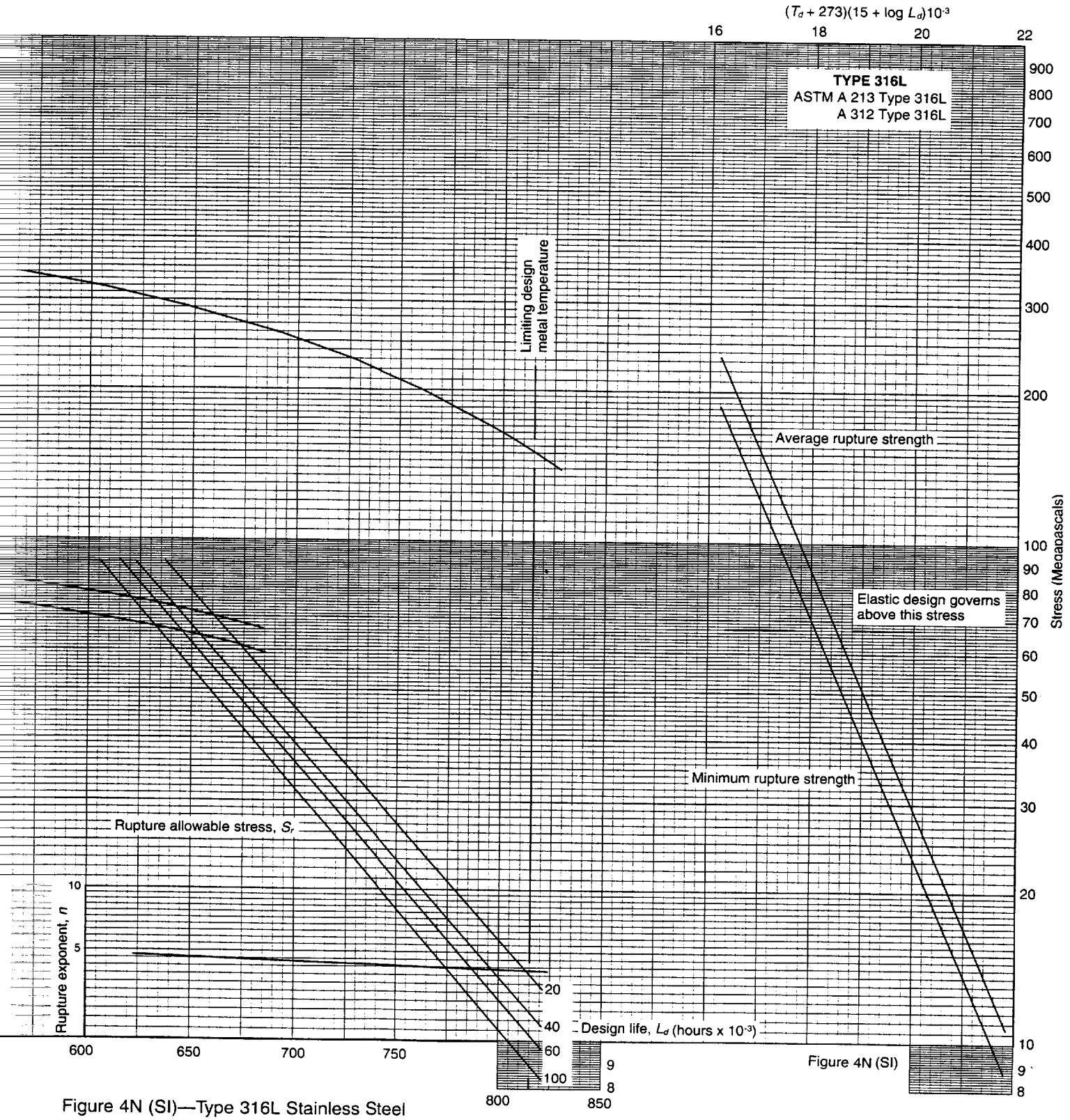


Figure 4N (SI)—Type 316L Stainless Steel

Figure 4N (SI)

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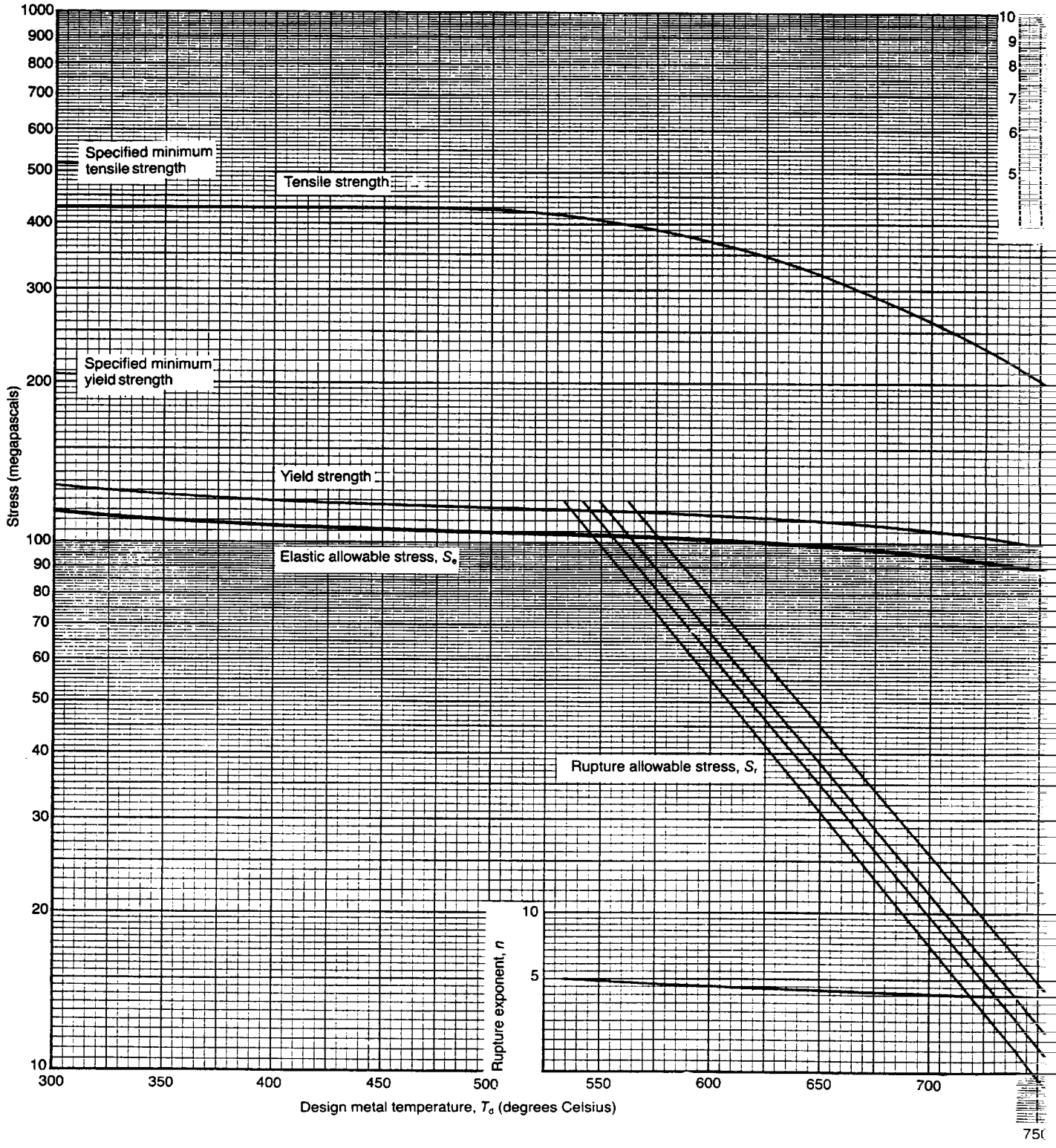


Figure 4O (SI)—Type 321 Stainless Steel, ASTM A 213, A

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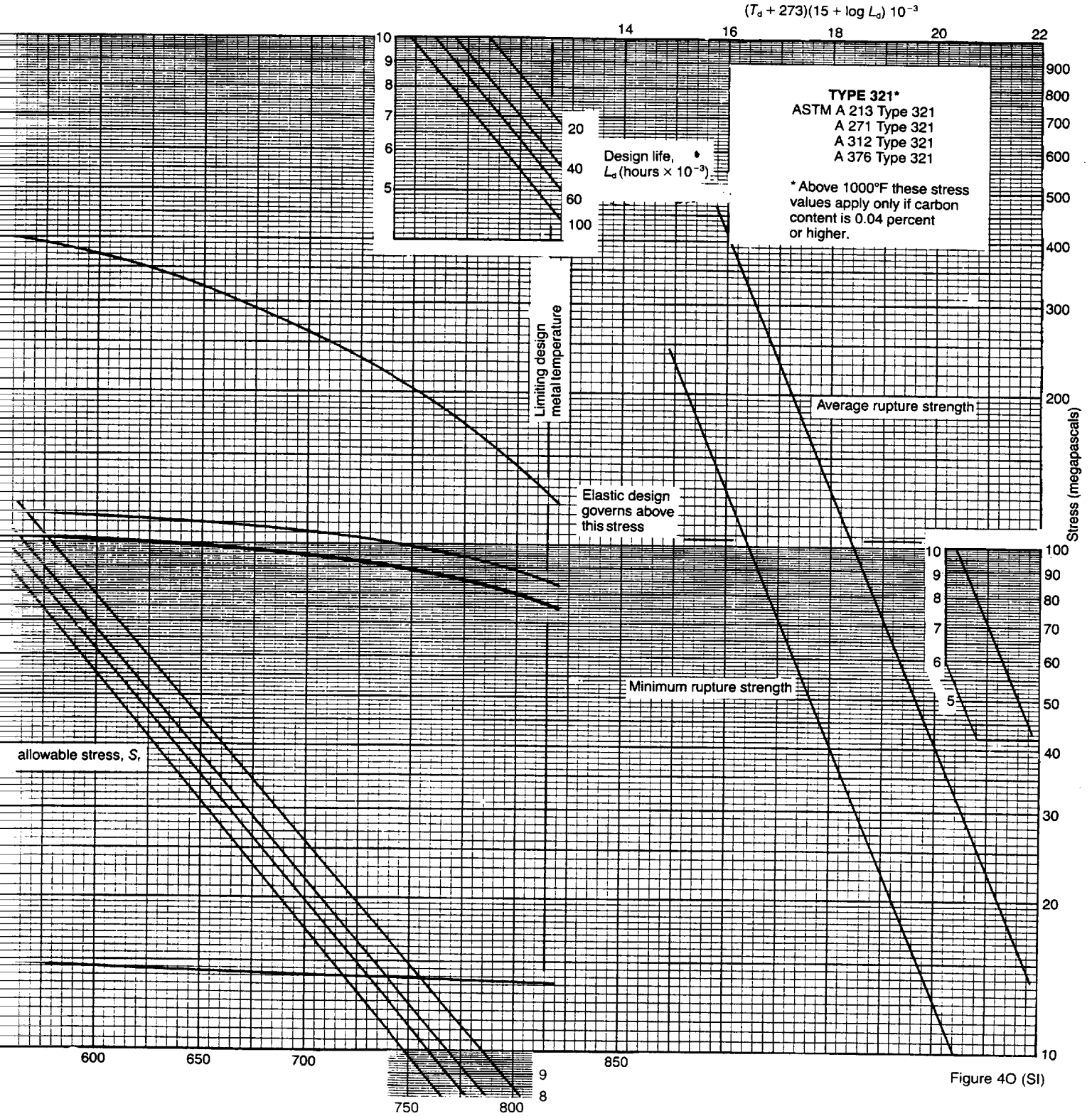


Figure 4O (SI)

(SI)—Type 321 Stainless Steel, ASTM A 213, A 271, A 312, A 376

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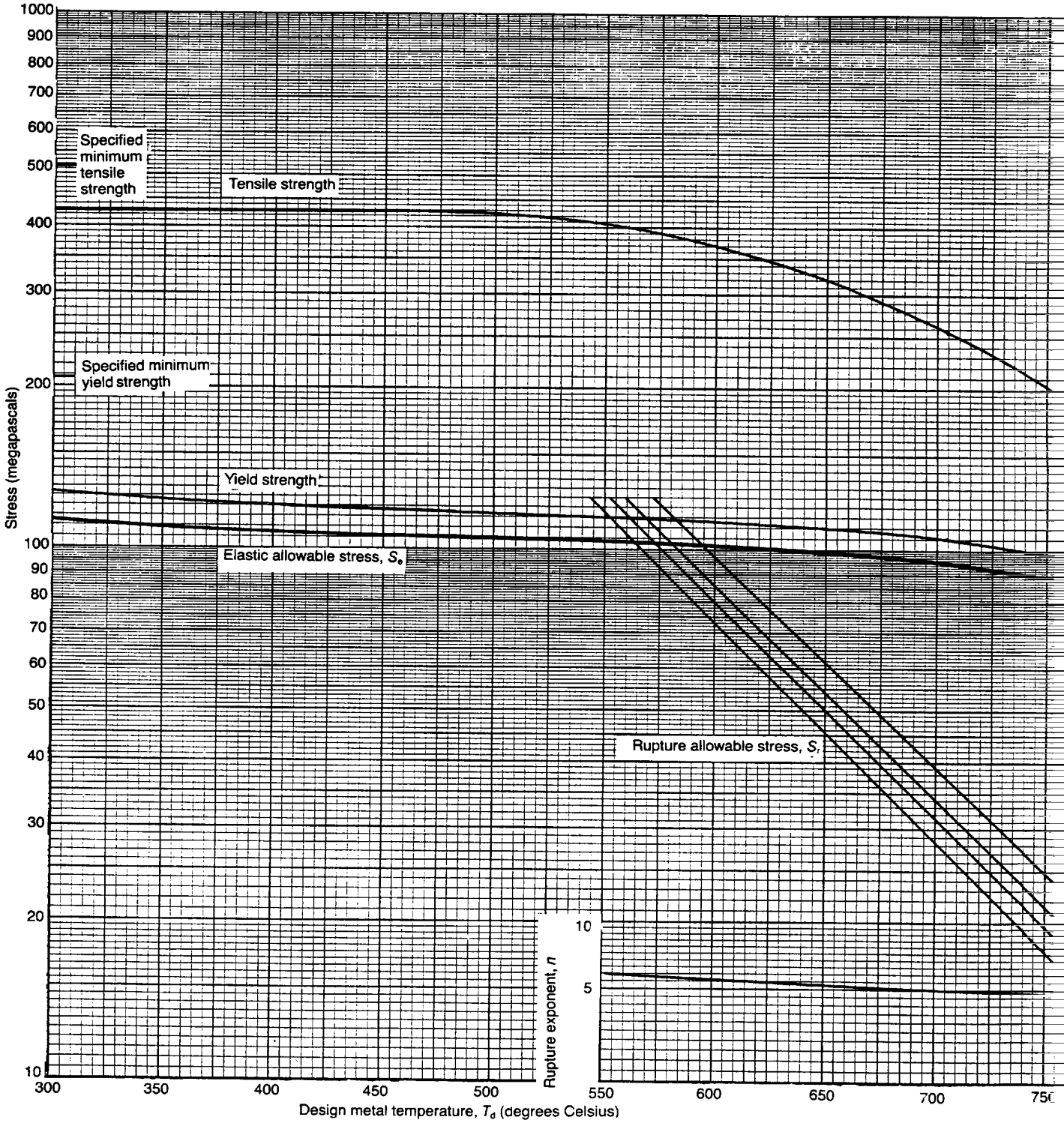


Figure 4P (SI)—Type 321H Stainless Steel, ASTM A 213, A 27

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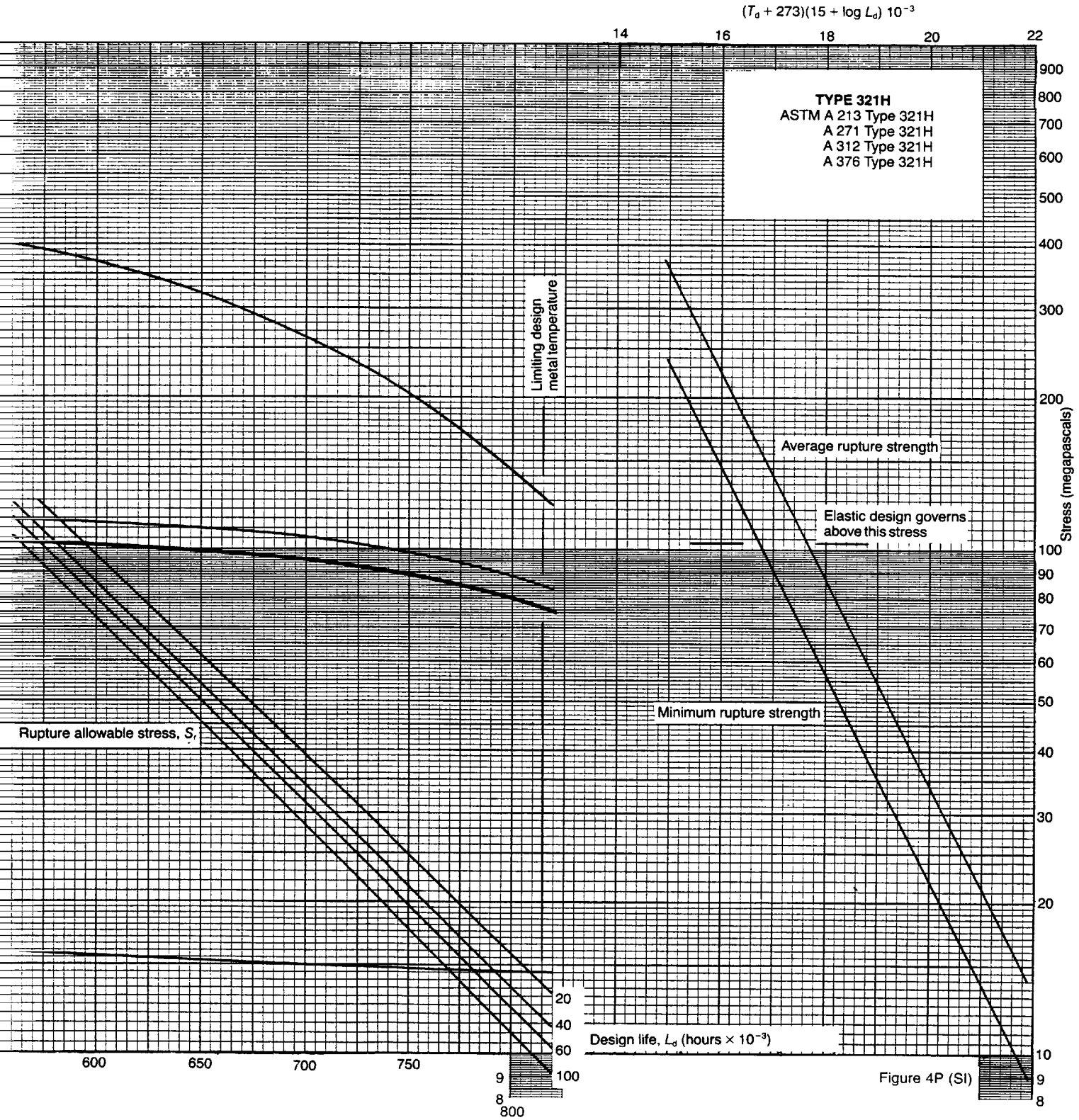


Figure 4P (SI)

)—Type 321H Stainless Steel, ASTM A 213, A 271, A 312, A 376

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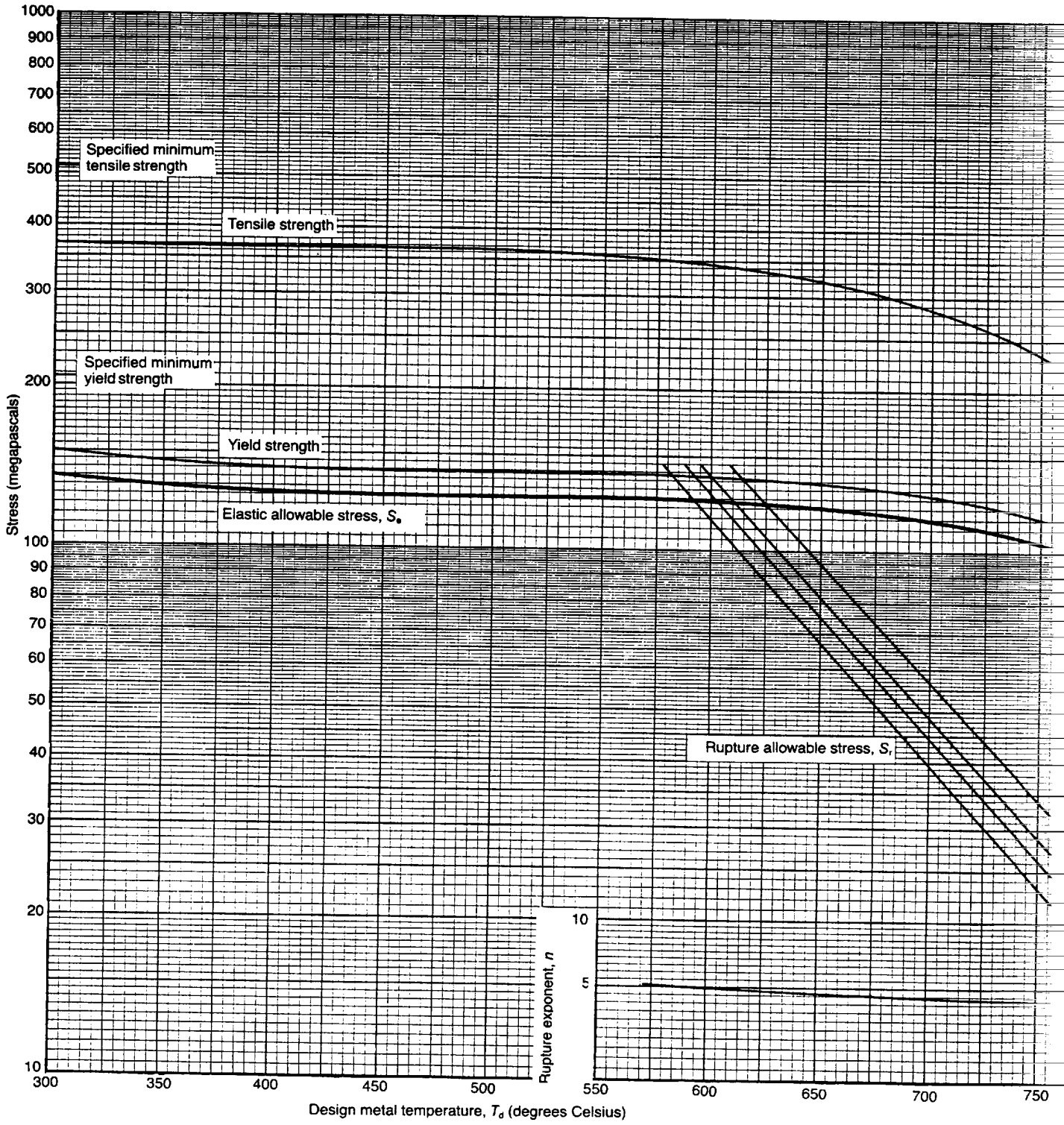


Figure 4Q (SI)—Type 347 and 347H Stainless Steel, ASTM A 21

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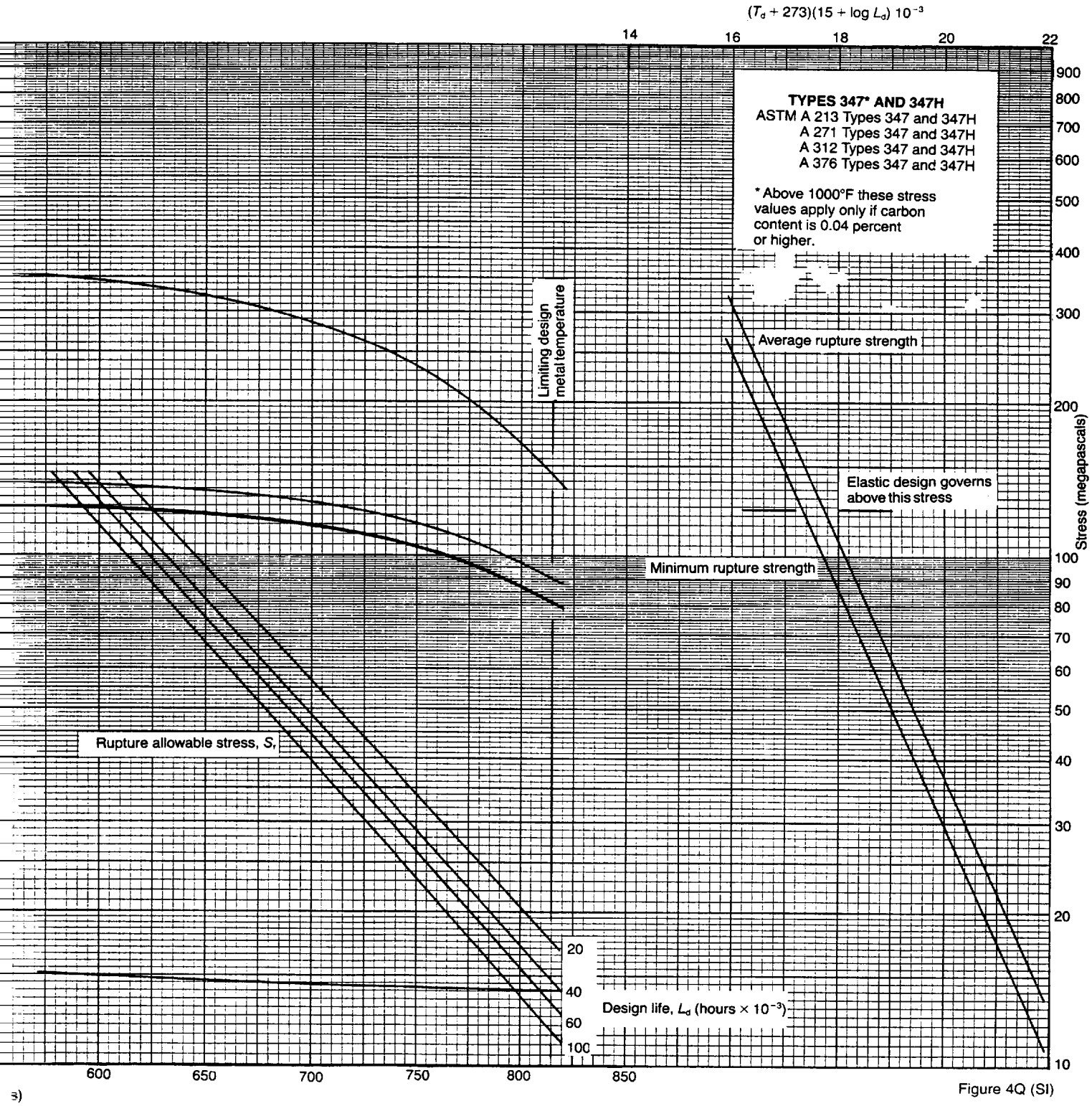


Figure 4Q (SI)

3)

—Type 347 and 347H Stainless Steel, ASTM A 213, A 271, A 312, A 376

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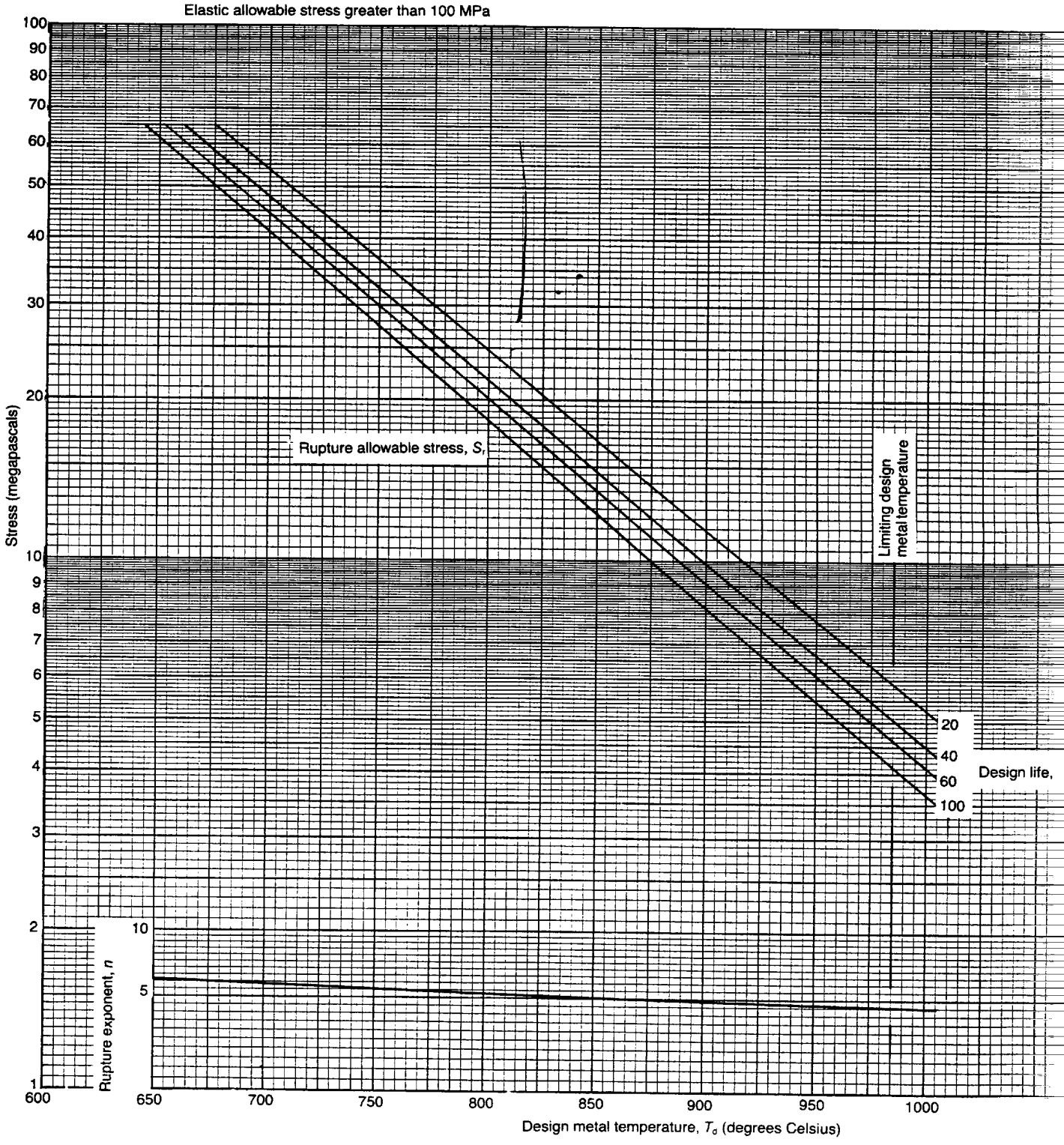


Figure 4R (SI)—Alloy 800H, ASTM B 407 UNS

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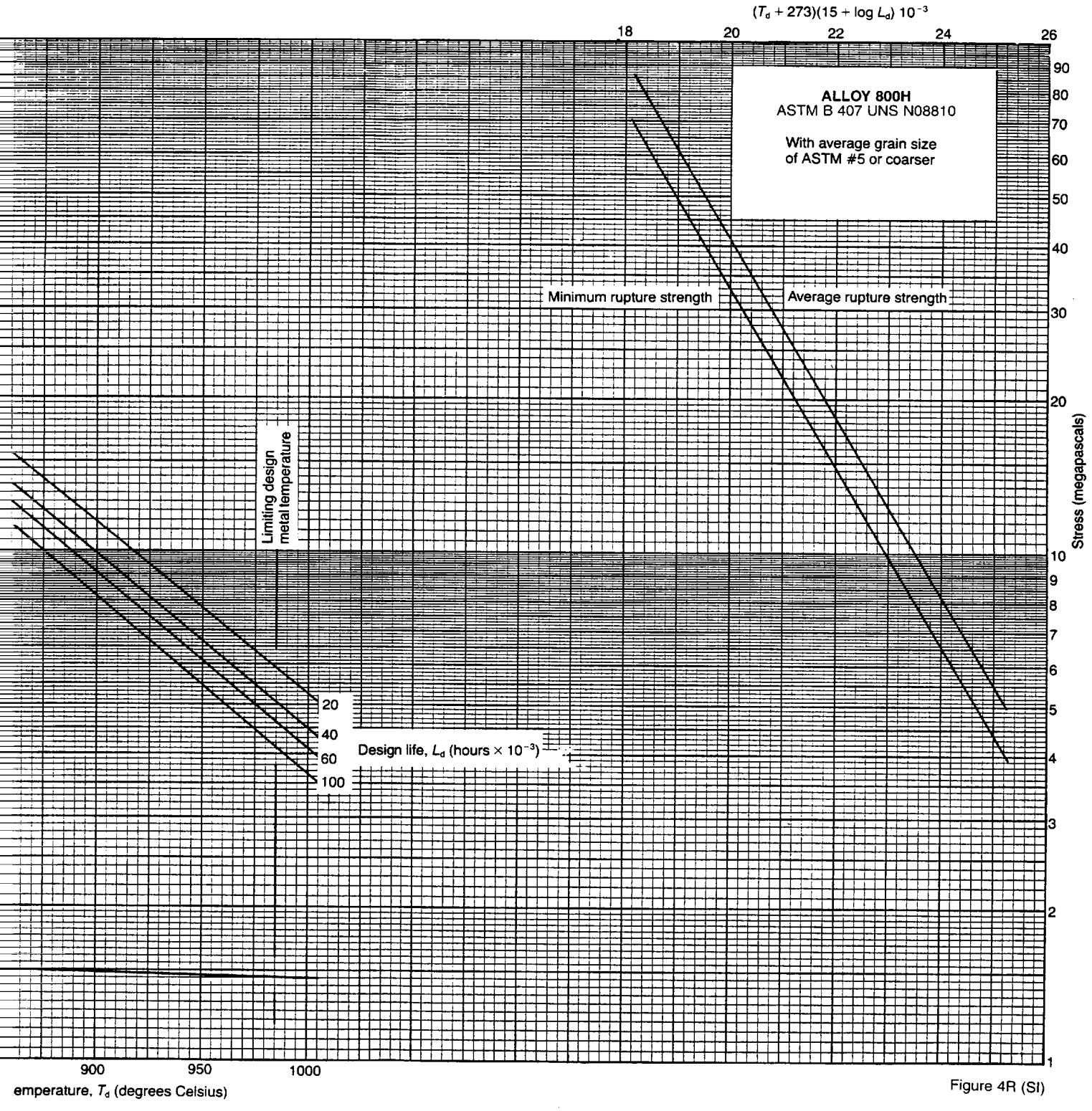


Figure 4R (SI)

Figure 4R (SI)—Alloy 800H, ASTM B 407 UNS N08810

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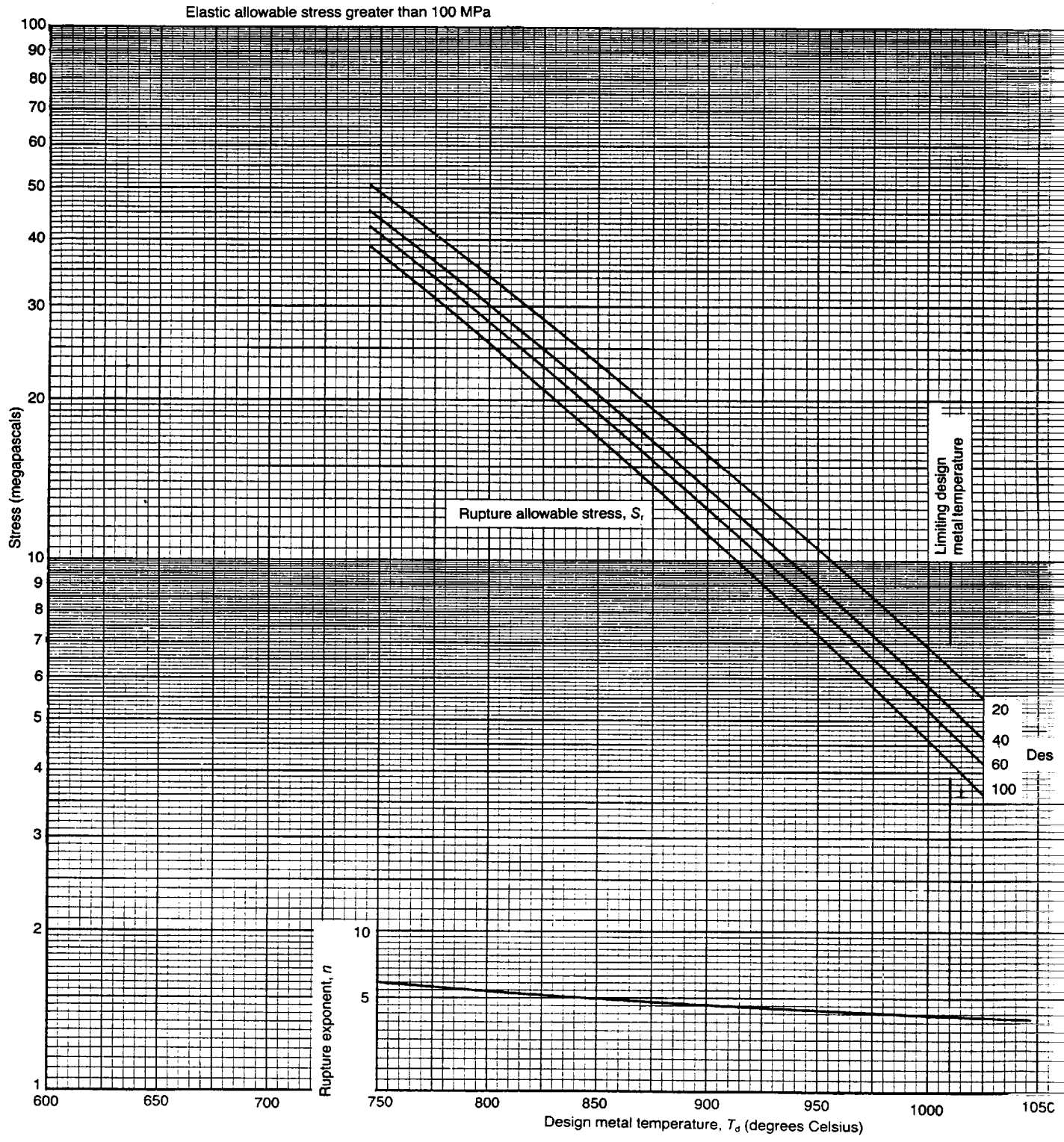


Figure 4S (SI)—HK-40, ASTM A 608 Grade HK

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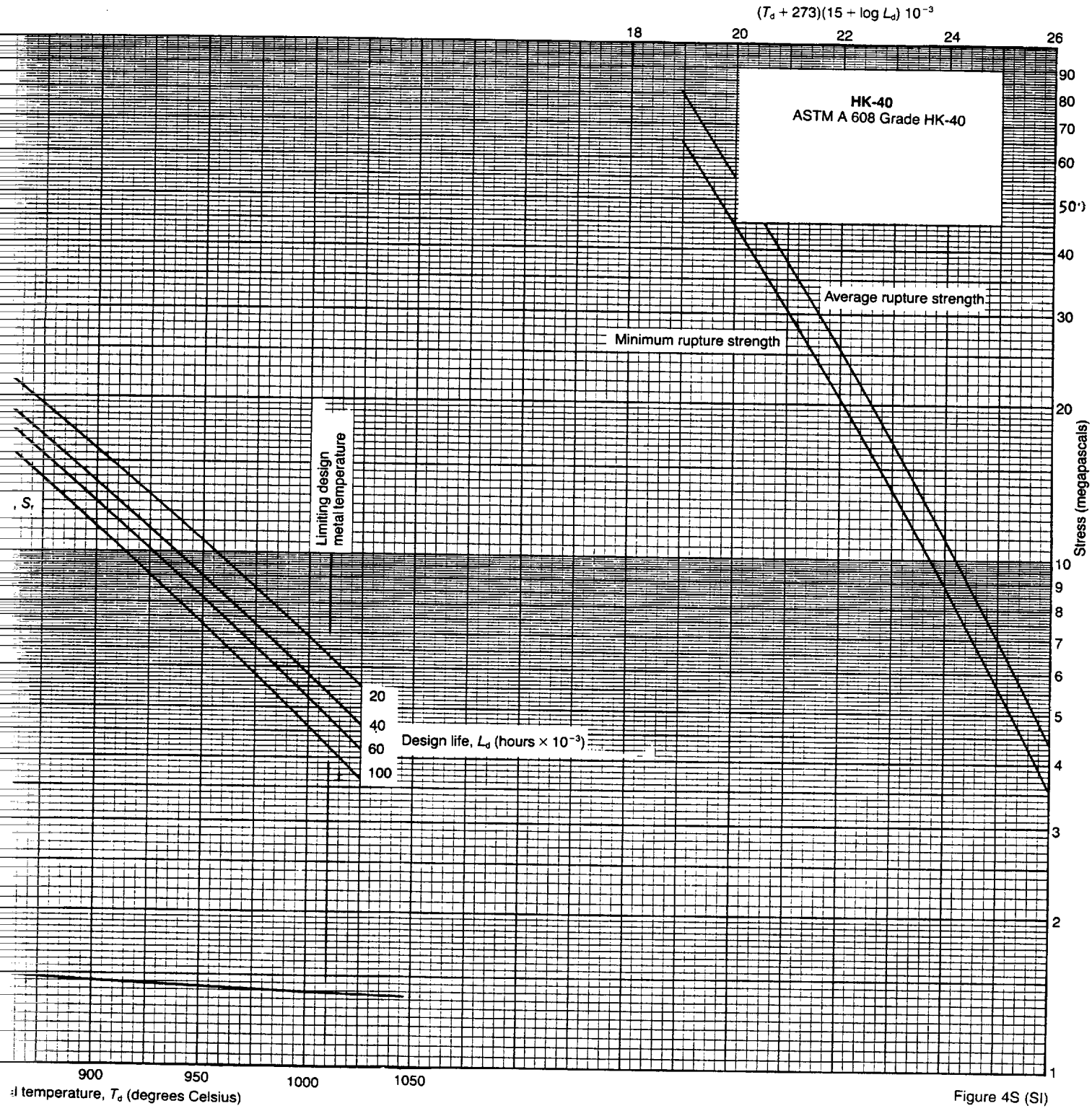


Figure 4S (SI)

Figure 4S (SI)—HK-40, ASTM A 608 Grade HK-40

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SECTION 4—SAMPLE CALCULATIONS

4.1 Elastic Design

Following is an example that illustrates how the design equations are used for the elastic range. Suppose the following information is given (the SI conversions in parentheses are approximate):

Material = 18Cr-10Ni-Cb. Type 347 stainless steel.
D_o = 6.625 inches (168.3 millimeters).
P_e = 900 pounds per square inch gauge (6.2 megapascals gauge).
T_d = 800°F (425°C).
CA = 0.125 inch (3.2 millimeters).

From Figure 4Q or Figure 4Q (SI).

S_e = 18,250 pounds per square inch (125 megapascals).
S_y = 20,200 pounds per square inch (140 megapascals).

Using Equations 2 and 3,

$$t_s = \frac{(900)(6.625)}{2(18,250) + 900} = 0.159 \text{ inch}$$

$$t_m = 0.159 + 0.125 = 0.284 \text{ inch}$$

In SI units,

$$t_s = \frac{(6.2)(168.3)}{2(125) + 6.2} = 4.0 \text{ millimeters}$$

$$t_m = 4.0 + 3.2 = 7.2 \text{ millimeters}$$

This design calculation is summarized on the calculation sheet in Figure 5.

4.2 Thermal-Stress Check (for Elastic Range Only)

The thermal stresses in the tube designed according to 4.1 shall be checked using the equations given in Appendix D as follows:

API STD 530
CALCULATION SHEET
Customary Units

Heater _____ Plant _____ Refinery _____
Coil _____ Material Type 347 ASTM Spec. A 213

CALCULATION OF MINIMUM THICKNESS	Elastic Design	Rupture Design
Outside diameter, inches	D_o = 6.625	D_o =
Design pressure, pounds per square inch gauge	P_e = 900	P_r =
Maximum or equivalent metal temperature, degrees Fahrenheit	T_m =	T_m =
Temperature allowance, degrees Fahrenheit	TA =	TA =
Design metal temperature, degrees Fahrenheit	T_d = 800	T_d =
Design life, hours	—	L_d =
Allowable stress at T_d, Figures 4A–4S, pounds per square inch	S_y = 18,250	S_r =
Stress thickness, Equation 2 or 4, inches	t_s = 0.159	t_s =
Corrosion allowance, inches	CA = 0.125	CA =
Corrosion fraction, Figure 1, n = B =	—	f =
Minimum thickness, Equation 3 or 5, inches	t_m = 0.284	t_m =

Figure 5—Sample Calculation, Elastic Design

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a = 10.05×10^{-6} °F^{-1} (1.81×10^{-5} °C^{-1}) (thermal expansion coefficient from Table C-3, ASME B31.3).
E = 24.1×10^{6} pounds per square inch (1.66×10^{5} megapascals) (modulus of elasticity from Table C-6, ASME B31.3).
v = 0.3 (value commonly used for steels).
q_o = 20,000 Btu/hr-ft^2 (63.1 kW/m^2) (assumed heat flux).
k = 11.9 Btu/hr-ft-°F (20.6 W/m-°C) (thermal conductivity from Table TCD in the ASME *Boiler and Pressure Vessel Code*, Section II Part D).

Using Equation D-2,

$$X = \left[\frac{(10.05)(24.1)}{4(1-0.3)}\right]\left[\frac{(20,000)(6.625)}{(11.9)(12)}\right]$$

= 8.026×10^4 pounds per square inch

In SI units,

$$X = \left[\frac{(1.81)(1.66)}{4(1-0.3)}\right]\left[\frac{(63.1)(168.3)}{(20.6)}\right]$$

= 553.2 megapascals

The thickness calculated in 4.1 is the minimum. The average thickness shall be used in the thermal-stress calculation. The average thickness (see 2.7) is calculated as follows:

(0.284)(1 + 0.14) = 0.324 inch

In SI units,

(7.2)(1 + 0.14) = 8.2 millimeters

The actual inside diameter is calculated as follows:

D_i = 6.625 - 2(0.324) = 5.977 inches
Y = 6.625/5.977 = 1.108

In SI units,

D_i = 168.3 - 2(8.2) = 151.9 millimeters
Y= 168.3/151.9 = 1.108

The term in brackets in Equation D-1 is calculated as follows:

$$\frac{2(1.108)^2}{(1.108)^2 - 1} \ln(1.108) - 1 = 0.106$$

Using Equation D-1, the maximum thermal stress is calculated as follows:

S_{th} = (8.026 x 10^4) (0.106)

= 8508 pounds per square inch

In SI units,

S_{th} = (553.2) (0.106)

= 58.6 megapascals

The limits for this stress for austenitic steels are given by Equations D-4 and D-6, in which the yield strength is 20,200 pounds per square inch (140 megapascals).

$S_{th.lim1}$ = [2.7 - 0.9(1.108)](20,200)

= 34,400 pounds per square inch

$S_{th.lim_2}$ = (1.8)(20, 000)

= 36,360 pounds per square inch

In SI units,

$S_{th.lim_1}$ = [2.7 – 0.9(1.108)](140)

= 238 megapascals

$S_{th.lim_2}$ = (1.8)(140)

= 252 megapascals

Since the maximum thermal stress is less than these limits, the design is acceptable.

If a thicker tube is specified arbitrarily (as Schedule 80S might be in this example), the actual average tube thickness shall be used in calculating the thermal stress and its limits as follows:

The inside diameter of a 6-inch Schedule 80S tube is as follows:

D_i = 5.761 inches

So,

Y = 6.625/5.761 = 1.150

In SI units,

D_i = 146.3 millimeters

Y = 168.3/146.3 = 1.150

The term in brackets in Equation D-1 is calculated as follows:

$$\frac{2(1.150)^2}{(1.150)^2 - 1} \ln(1.150) - 1 = 0.146$$

Using Equation D-1, the maximum thermal stress is calculated as follows:

S_{th} = (8.026×10^4)(0.146)

= 11,793 pounds per square inch

In SI units,

S_{th} = (553.2)(0.146)

= 80.9 megapascals

The average thickness of this tube is 0.432 inch (11.0 millimeters), so the minimum thickness is calculated follows:

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$$t_m = \frac{0.432}{1 + 0.14} = 0.379 \text{ inch}$$

In SI units,

$$t_m = \frac{11.0}{1 + 0.14} = 9.6 \text{ millimeters}$$

Using Equation D-7, the stress is calculated as follows:

$$S_{pm} = \frac{900}{2}\left(\frac{6.625}{0.379} - 1\right) = 7416 \text{ pounds per square inch}$$

In SI units,

$$S_{pm} = \frac{6.2}{2}\left(\frac{168.3}{9.6} - 1\right) = 51.2 \text{ megapascals}$$

The thermal-stress limit based on the primary plus secondary stress intensity is calculated using Equation D-9. Using the values above, this limit is calculated as follows:

$$S_{thlim_1} = 2.7(20, 200) - (1.15)(7416)$$
$$= 46{,}010 \text{ pounds per square inch}$$

In SI units,

$$S_{thlim_1} = 2.7(140) - (1.15)(51.2)$$
$$= 319.1 \text{ megapascals}$$

The thermal-stress ratchet limit is calculated using Equation D-12. In this case, the limit is as follows:

$$S_{thlim_2} = 4[(1.35)(20, 000) - 7416]$$
$$= 79{,}416 \text{ pounds per square inch}$$

In SI units,

$$S_{thlim_2} = 4[(1.35)(140) - 51.2]$$
$$= 551.2 \text{ megapascals}$$

The thermal stress in the thicker tube is well below these limits.

4.3 Rupture Design With Constant Temperature

A modification of the example in 4.1 illustrates how the design equations are used for the creep-rupture range. Suppose the tube described in 4.1 is to be designed for the following conditions:

T_d = 1300°F (705°C).
L_d = 100,000 hours.
P_r = 840 pounds per square inch gauge (5.8 megapascals gauge).

From Figure 4Q or Figure 4Q (SI),

S_r = 5450 pounds per square inch (37.3 megapascals)

Using Equation 4,

$$t_s = \frac{(840)(6.625)}{2(5450) + 840} = 0.474 \text{ inch}$$

In SI units,

$$t_s = \frac{(5.8)(168.3)}{2(37.3) + 5.8} = 12.1 \text{ millimeters}$$

From this,

$$B = \frac{0.125}{0.474} = 0.264$$

In SI units,

$$B = \frac{3.2}{12.1} = 0.264$$

From Figure 4Q or Figure 4Q (SI),

n = 4.4

Using these values for B and n, use Figure 1 to obtain the following corrosion fraction:

f = 0.558

Hence, using Equation 5,

$$t_m = 0.474 + (0.558)(0.125)$$
$$= 0.544 \text{ inch}$$

In SI units,

$$t_m = 12.1 + (0.558)(3.2)$$
$$= 13.9 \text{ millimeters}$$

To confirm that this is an appropriate design, the elastic design shall be checked using the elastic design pressure instead of the rupture design pressure. Using Equations 2 and 3 with the conditions given above,

S_e = 16,400 pounds per square inch

$$t_s = \frac{(900)(6.625)}{2(16, 400) + 900} = 0.177 \text{ inch}$$

$t_m = 0.177 + 0.125 = 0.302$ inch

In SI units,

S_e = 113 megapascals

$$t_s = \frac{(6.2)(168.3)}{2(113) + 6.2} = 4.5 \text{ millimeters}$$

$t_m = 4.5 + 3.2 = 7.7$ millimeters

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Since t_m based on rupture design is greater, it governs the design. This design calculation is summarized on the calculation sheet in Figure 6.

4.4 Rupture Design With Linearly Changing Temperature

Suppose the tube described in 4.3 will operate in a service for which the estimated tube metal temperature varies from 1175°F (635°C) at the start of run to 1275°F (690°C) at the end of run. Assume that the run lasts a year, during which the thickness will change about 0.013 inch (0.33 millimeter).

Assume that the initial minimum thickness is 0.315 inch (8.0 millimeters); therefore, using Equation 1, the initial stress will be as follows:

$$S_0 = \frac{840}{2}\left(\frac{6.625}{0.315} - 1\right) = 8413 \text{ pounds per square inch}$$

In SI units,

$$S_0 = \frac{5.8}{2}\left(\frac{168.3}{8.0} - 1\right) = 58.1 \text{ megapascals}$$

At the start-of-run temperature, $n_0 = 4.8$. From Table 3, A is 1.79×10^8 pounds per square inch (1.23×10^6 megapascals). The parameters for the temperature fraction are therefore as follows:

$$V = 4.8\left(\frac{100}{1635}\right) \ln\left(\frac{1.79 \times 10^8}{8413}\right) = 2.9$$

$$N = 4.8\left(\frac{0.013}{0.315}\right) = 0.2$$

In SI units,

$$V = 4.8\left(\frac{55}{908}\right) \ln\left(\frac{1.23 \times 10^6}{58.1}\right) = 2.9$$

$$N = 4.8\left(\frac{0.33}{8.0}\right) = 0.2$$

From Figure 2, $f_T = 0.62$, and the equivalent temperature is calculated using Equation 6 as follows:

$$T_e = 1175 + (0.62)(100) = 1237°F$$

In SI units,

$$T_e = 635 + (0.62)(55) = 669°C$$

API STD 530
CALCULATION SHEET
Customary Units

Heater _____ Plant _____ Refinery _____
Coil _____ Material Type 347 ASTM Spec. A 213

CALCULATION OF MINIMUM THICKNESS	Elastic Design	Rupture Design
Outside diameter, inches	D_o = 6.625	D_o = 6.625
Design pressure, pounds per square inch gauge	P_e = 900	P_r = 840
Maximum or equivalent metal temperature, degrees Fahrenheit	T_m =	T_m =
Temperature allowance, degrees Fahrenheit	TA =	TA =
Design metal temperature, degrees Fahrenheit	T_d = 1300	T_d = 1300
Design life, hours	—	L_d = 100,000
Allowable stress at T_d, Figures 4A–4S, pounds per square inch	S_e = 16,400	S_r = 5450
Stress thickness, Equation 2 or 4, inches	t_s = 0.177	t_s = 0.474
Corrosion allowance, inches	CA = 0.125	CA = 0.125
Corrosion fraction, Figure 1, n = 4.4 B = 0.264	—	f = 0.558
Minimum thickness, Equation 3 or 5, inches	t_m = 0.302	t_m = 0.544

Figure 6—Sample Calculation, Rupture Design (Constant Temperature)

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A temperature allowance of 25°F (15°C) is added to yield a design temperature of 1262°F, which is rounded up to 1265°F (685°C). Using this temperature to carry out the design procedure illustrated in 4.3 yields the following:

t_s = 0.388 inch

t_m = 0.388 + (0.572)(0.125)
= 0.460 inch

In SI units,

t_s = 9.9 millimeters

t_m = 9.9 + (0.572)(3.2)
= 11.7 millimeters

This thickness is different from the 0.315-inch (8.0-millimeter) thickness that was initially assumed. Using this thickness, the stress is calculated as follows:

$$S_0 = \frac{840}{2}\left(\frac{6.625}{0.460} - 1\right) = 5629 \text{ pounds per square inch}$$

In SI units,

$$S_0 = \frac{5.7}{2}\left(\frac{168.3}{11.7} - 1\right) = 38.8 \text{ megapascals}$$

With this stress, the temperature-fraction parameters V and N become the following:

$$V = 4.8\left(\frac{100}{1635}\right) \ln\left(\frac{1.79 \times 10^8}{5629}\right) = 3.0$$

$$N = 4.8\left(\frac{0.013}{0.460}\right) = 0.1$$

In SI units,

$$V = 4.8\left(\frac{55}{908}\right) \ln\left(\frac{1.23 \times 10^6}{38.8}\right) = 3.0$$

$$N = 4.8\left(\frac{0.33}{11.7}\right) = 0.1$$

Using these values in Figure 2, f_T = 0.62, the value that was determined in the first calculation. Since the temperature fraction did not change, further iteration is not necessary. This design calculation is summarized on the calculation sheet in Figure 7.

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CALCULATION SHEET
Customary Units

Heater _____ Plant _____ Refinery _____
Coil _____ Material Type 347 ASTM Spec. A 213

CALCULATION OF MINIMUM THICKNESS	Elastic Design	Rupture Design
Outside diameter, inches	D_o =	D_o = 6.625
Design pressure, pounds per square inch gauge	P_e =	P_r = 840
Maximum or equivalent metal temperature, degrees Fahrenheit	T_m =	T_m = 1237
Temperature allowance, degrees Fahrenheit	TA =	TA = 25
Design metal temperature, degrees Fahrenheit	T_d =	T_d = 1265
Design life, hours	—	L_d = 100,000
Allowable stress at T_d, Figures 4A–4S, pounds per square inch	S_e =	S_r = 6750
Stress thickness, Equation 2 or 4, inches	t_s =	t_s = 0.388
Corrosion allowance, inches	CA =	CA = 0.125
Corrosion fraction, Figure 1, n = 4.5 B = 0.322	—	f = 0.572
Minimum thickness, Equation 3 or 5, inches	t_m =	t_m = 0.460

CALCULATION OF EQUIVALENT TUBE METAL TEMPERATURE	
Duration of operating period, years	L_o = 1.0
Metal temperature, start of run, degrees Fahrenheit	T_{sor} = 1175
Metal temperature, end of run, degrees Fahrenheit	T_{eor} = 1275
Temperature change during operating period, degrees Fahrenheit	ΔT = 100
Metal temperature, start ot run, degrees Rankine	T_a = 1635
Thickness change during operating period, inches	Δt = 0.013
Assumed initial thickness, inches	t_0 = 0.315
Corresponding initial stress, Equation 1, pounds per square inch	S_0 = 8413
Material constant, Table 3, pounds per square inch	A = 1.79×10^8
Rupture exponent at T_{sor} Figures 4A–4S	n_0 = 4.8
Temperature fraction Figure 2, V = 2.9 N = 0.2	f_r = 0.62
Equivalent metal temperature, Equation 6, degrees Fahrenheit	T_e = 1237

Figure 7—Sample Calculation, Rupture Design (Changing Temperature)

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APPENDIX A—DATA SOURCES

A.1 General

Whenever possible, the yield-, tensile- and rupture-strength data displayed in Figures 4A–4S were taken from the ASTM Data Series Publications [References 8, 9, 10, 11, 12, 13] (see Table A-1). These publications contain discussions and detailed descriptions of the data that are not repeated in this appendix. The material that follows is limited to a discussion of deviations from published data and of data that have been used but are not generally available.

A.2 Minimum Rupture Strength

The ASTM Data Series Publications contain evaluations of various rupture-strength extrapolation techniques. From these evaluations, the most reliable extrapolation was selected. The average and minimum 100,000-hour rupture strengths, calculated by this method are used in this standard. The minimum rupture strength used is the lower 95-percent confidence limit; 95 percent of all samples should have rupture strengths greater than this value. This minimum rupture strength is obtained by using least-squares techniques to calculate a curve for the average rupture strength and subtracting 1.65 times the standard deviation of the data from this average. The specific figure number and Data Series reference for each alloy are listed in Table A-1.

A.3 Larson-Miller Parameter Curves

The Larson-Miller parameter combines design metal temperature, T_d, and design life, L_d, in hours, as follows: When T_d is expressed in degrees Fahrenheit,

$$(T_d + 460)(C + \log_{10} L_d) \times 10^{-3}$$

When T_d is expressed in degrees Celsius,

$$(T_d + 237)(C + \log_{10} L_d) \times 10^{-3}$$

The generally accepted empirical values of $C = 20$ and $C = 15$ are used for ferritic steels and austenitic steels, respectively. The value of $C = 30$ is used for T91 or P91, 9Cr-1Mo-Va steel. To calculate the rupture allowable stress for any given design metal temperature and design life, the appropriate value of C should be used to calculate the parameter, and one of the Larson-Miller parameter curves should then be used to find the corresponding rupture strength.

To the right in Figures 4A–4S are Larson-Miller parameter curves that permit tubes to be designed for lives other than 100,000 hours. These curves were developed from the average and minimum 100,000-hour rupture strengths. They can be used to estimate the rupture allowable stress (minimum rupture strength) for design lives from 20,000 hours to 200,000 hours. The resulting 20,000-, 40,000-, and 60,000-hour rupture allowable stresses are shown with the 100,000-hour rupture allowable stress to the left in Figures 4A–4S.

This is not the normal use of the Larson-Miller parameter. The Larson-Miller curve is traditionally developed from rupture-strength test data as one way to extrapolate long-term rupture strengths from short-term data. The resulting extrapolation is suitable for some alloys but not for all. Most of the ASTM Data Series Publications listed in Table A-1 examine the suitability of this Larson-Miller extrapolation.

Table A-1—Sources of Data for Yield, Tensile, and Rupture Strengths

Alloy	ASTM Strength	Yield Strength	Tensile Strength	Rupture Strength	Method Used	Comments
Carbon steels	DS 11S1	Figure 7c	Figure 7d	(See A.5.1)	LM	Fine-grained, tempered values used.
C-½Mo steel	DS 47	Figure 7a	Figure 7b	(See A.5.2)	LM	
1 ¼Cr-½Mo steel	DS 50	Figure 6c	Figure 6d	(See A.5.3)	IL	Nonplate values used.
2 ¼Cr-1Mo steel	DS 6S2	Figure 7a	Figure 7b	(See A.5.4)	MC	
3Cr-1Mo steel	DS 58	Figure 7a	Figure 7b	Figure 17c	IL	
5Cr-½Mo steel	DS 58	Figure 8a	Figure 8b	Figure 26c	IL	
5Cr-½Mo-Si steel	DS 58	Figure 9a	Figure 9b	Figure 33c	IL	
7Cr-½Mo steel	DS 58	Figure 11a	Figure 11b	Figure 47c	IL	
9Cr-1Mo steel	DS 58	Figure 12a	Figure 12b	Figure 54c	IL	
9Cr-1Mo-Va steel	MPC[c]				LM	
18Cr-8Ni steel	DS 5S2	Figure 14b	Figure 15b	Tables 7 and 10	IL	Adjusted values used. Figures 14a and 15a used above 1000°F.
16Cr-12Ni-2Mo steel	DS 5S2	Figure 14e	Figure 15e	Tables 7 and 10	IL	Adjusted values used.
16Cr-12Ni-2Mo (316L) steel	DS 5S2	Figure 14f	Figure 15f	Table 7	IL	Minimum is 80% of average.
18Cr-10Ni-Ti steel	DS 5S2	Figure 14g	Figure 15g	Tables 7 and 10	IL	Adjusted values used.
18Cr-10Ni-Cb steel	DS 5S2	Figure 14h	Figure 15h	Tables 7 and 10	IL	Adjusted values used.
Ni-Fe-Cr (Alloy 800H)				(See A.5.5)	LM	
25CR-20Ni (HK-40)				(See A.5.6)	LM	

[a]See references 8, 9, 10, 11, 12, and 13 in Appendix G for ASTM Data Series publications.
[b]LM = Larson-Miller, IL = Individual lots (see ASTM DS publication for definition), MC = Manson compromise.
[c]Data from Materials Properties Council, Inc.

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The Larson-Miller parameter curves used in this standard were developed from the extrapolated values of the 100,000-hour rupture strength. The values used are those listed in the various ASTM Data Series Publications. They have been estimated in the manner believed to be most reliable. For low- and medium-carbon steels, Alloy 800H, and HK 40, the 100,000-hour rupture strength has been estimated using a Larson-Miller extrapolation: other means have been used for the other alloys. Table A-1 lists the extrapolation method used for each alloy. Consequently, the Larson-Miller parameter curves in this standard are not the same as those shown in the various ASTM Data Series Publications. For those cases in which the 100,000-hour rupture strength was determined by other means, the Larson-Miller parameter curves in this standard may not give reliable estimates of the rupture strength for times less than 20,000 hours or more than 200,000 hours.

A.4 Rupture Exponent

Constant-temperature creep-rupture data can be conveniently plotted on a log-log graph, log (stress) versus log (rupture time). These stress-rupture curves can often be represented by a straight line or can be approximated by a straight line in limited regions. The straight line can be expressed as follows:

$$L_r = mS^{-n}$$

Where:

L_r = rupture time.

m and n = material parameters that are functions of temperature.

S = stress.

The parameter n is the rupture exponent; it is related to the slope of the stress-rupture curve.

The value of the rupture exponent can be calculated from two points on the curve. If the rupture time for a stress S_1 is L_1 and the rupture time for a stress S_2 is L_2, then

$$n = \frac{\log(L_1/L_2)}{\log(S_2/S_1)}$$

If the stress-rupture curve is a straight line, any two points on that line will give the same value of n. If the stress-rupture curve is not a straight line, the value of n will depend on which two points are chosen, since the slope of the straight-line approximation depends on which part of the curve is approximated.

The rupture exponents plotted in Figures 4A–4S were determined from the 60,000-hour and 100,000-hour minimum rupture strengths as estimated by the Larson-Miller parameter curves. These particular times were chosen to give a straight-line approximation over the range of the usual operating stress levels.

A.5 Modification of and Additions to Published Data

Whenever possible, the data used to generate Figures 4A–4S were taken from the ASTM Data Series Publications [References 8, 9, 10, 11, 12, 13]. Specific figure and table references for the yield, tensile, and rupture strengths are given in Table A-1. In some cases, the rupture-strength extrapolations were modified for this practice, or the data were used to develop new extrapolations. These modifications and additions are described in A.5.1 through A.5.4. Alloy 800H and HK-40 are not covered by recent ASTM publications. The data used to develop the figures for these alloys are described in A.5.5 and A.5.6.

A.5.1 CARBON STEELS

The determination of rupture strength in Data Series 11S1 makes no distinction between low-carbon steel (A 192) and medium-carbon steel (A 106 and A 210). Data from all three alloys were used to calculate the Larson-Miller curve in Data Series 11S1. For this standard, the distinction was made for Figures 4A and 4B by separating the data and calculating two Larson-Miller curves. The procedure for establishing the average and minimum rupture strengths was otherwise identical to that used in Data Series 11S1. Larson-Miller curves that represent the average strength were generated by the least-squares method; curves that represent minimum strength were generated by subtracting from the average-strength curves 1.65 times the standard deviation of the data.

A.5.2 C-1/2 Mo STEEL

The Larson-Miller curves in Figure 18a of Data Series 47 have an inflection point close to a parameter value of 37. The upturn to the right is considered questionable. For this standard, the parameter curves shown in Figure 4C were arbitrarily extended by straight lines above a parameter value of 37. These extensions are shown as dashed lines in Figure 4C.

A.5.3 1 1/4 Cr-1/2 Mo STEEL

The regression of the individual lot extrapolations in Figure 27c of Data Series 50 used a polynomial of third degree or higher. The resulting average and minimum rupture-strength curves show an upturn to the right. This upturn also results when the data points shown on Figure 27c are fitted with a quadratic curve. Since this upturn is considered questionable, the data points shown in Figure 27c were used to calculate a first-degree curve for this standard. The resulting curves for average and minimum rupture strengths are shown in Figure 4D.

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A.5.4 2¼ Cr-1Mo STEEL

The most reasonable extrapolation in Data Series 6S2 is provided by the strength-temperature regression curve shown in Figure 22 and again in Figure 26. As with $1^1/_4Cr$-$^1/_2Mo$ steel in Data Series 50, the regression used a polynomial of third degree or higher. The resulting curve is considered questionable. For this standard the Manson compromise curve in Figure 26 was used below 1100°F (593°C) and was extended downward to intersect the strength-temperature regression curve at 1200°F (649°C). The resulting curves for average and minimum 100,000-hour rupture strength shown in Figure 4E of this standard are generally equal to or below the strength-temperature regression curves of Data Series 6S2.

A.5.5 Ni-Fe-Cr (ALLOY 800H)

The Larson-Miller curves for Alloy 800H in Figure 4R were developed from 91 rupture-test data points from one source. These tests used samples from six heats of Alloy 800H (with appropriate chemistry and grain size) that were made in bar, plate, and tube product forms. All tests were run at temperatures of 1800°F (982°C) or lower, except for one that was run at 1900°F (1038°C). The linear curves for the average and minimum rupture strengths were calculated using least-squares techniques. Using a quadratic curve did not appreciably improve the fit of these data.

A.5.6 25Cr-20Ni (HK-40)

The Larson-Miller curves for HK-40 in Figure 4S were developed from 87 rupture-test data points. These tests came from four sources and involved seven heats of HK-40. The carbon content of these heats ranged from 0.35 to 0.45. No data from tests that were run at temperatures of 1900°F (1038°C) or higher were used in this evaluation, since significant metallurgical changes that affect the rupture strength occur above this temperature. The quadratic curves for the average and minimum rupture strengths were calculated using least squares techniques.

A.5.7 25Cr-35Ni-HP-MODIFIED

Stress curves for HP-modified cast tubing are not included. This material is proprietary to individual foundries. As such, it is not feasible to develop generic stress data which would apply to all manufacturers of this material.

A.5.8 9Cr-1Mo-Va STEEL

The maximum limit for this material has been restricted to 1200°F (650°C) due to the lack of stress data above this temperature (See Figure 4K).

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APPENDIX B—DERIVATION OF CORROSION FRACTION AND TEMPERATURE FRACTION

B.1 General

The 1958 version of this document contained a method for designing tubes in the creep-rupture range. The method took into consideration the effects of stress reductions produced by the corrosion allowance. In developing this design method, the following ideas were used.

At temperatures in the creep-rupture range, the life of a tube is limited. The rate of using up the life depends on temperature and stress. Under the assumption of constant temperature, the rate of using up the life increases as the stress increases. In other words, the tube will last longer if the stress is lower.

If the tube is corroding or oxidizing, the tube thickness will decrease in time; therefore, under the assumption of constant pressure, the stress in the tube will increase in time. As a result, the rate of using up the rupture life will also increase in time.

An integral of this effect over the life of the tube can be solved graphically in the 1958 version of API Recommended Practice 530 and developed using the linear damage rule (see B.2). The result is a nonlinear equation that provides the initial tube thickness for various combinations of design temperature and design life.

The concept of corrosion fraction used in 2.4 and derived in this appendix is developed from the same ideas and is a simplified method of achieving the same results.

Suppose a tube has an initial thickness, t_s, calculated using Equation 4. This is the minimum thickness required to achieve the design life without corrosion. If the tube does not corrode, the stress in the tube will always equal the minimum rupture strength for the design life, S_r. This tube should fail after the end of the design life.

If this tube were designed for use in a corrosive environment and had a corrosion allowance of CA, the minimum thickness could be set as follows:

$$t_m = t_s + CA$$

The stress would initially be less than S_r. After operating for its design life, the corrosion allowance would be used up, and the stress would only then equal S_r. Since the stress would always have been lower than S_r, the tube would still have some time to operate before it failed.

Suppose instead that the initial thickness was set as follows:

$$t_m = t_s + fCA$$

In this equation, f is a fraction less than unity. The stress would initially be less than S_r, and the rate of using up the rupture life would be low. At the end of the design life, the tube thickness would be as follows:

$$t_m - CA = t_s - (1-f)CA$$

This thickness is less than t_s; therefore, at the end of the design life, the stress would be greater than S_r, and the rate of using up the rupture life would be high. If the value of f is selected properly, the integrated effect of this changing rate of using up the rupture life would yield a rupture life equal to the design life. The corrosion fraction, f, given in Figure 1 is such a value.

The curves in Figure 1 were developed by solving the nonlinear equation that results from applying the linear damage rule. Figure 1 can be applied to any design life, provided only that the corrosion allowance, CA, and rupture allowable stress, S_r, are based on the same design life.

B.2 Linear Damage Rule

Consider a tube that is operated at a constant stress, S, and a constant temperature, θ, for a period of time, ΔL. Corresponding to this stress and temperature is the following rupture life:

$$L_r = L_r(S, \theta)$$

The fraction $\Delta L/L_r$, would then be the fraction of the rupture life used up during this operating period. After M operating periods, each with a corresponding fraction—

$$\left(\frac{\Delta L}{L_r}\right)_i, i = 1, 2, 3, \ldots, M$$

—the total fraction of the rupture life used up, F (also known as the life fraction), would be the sum of the fractions used in each period:

$$F(M) = \sum_{i=1}^{M} \left(\frac{\Delta L}{L_r}\right)_i \qquad \text{(B-1)}$$

In developing this equation, no restrictions were placed on the stress and temperature from period to period. It was assumed only that during any one period the stress and temperature were constant. The life fraction therefore provides a way of estimating the rupture life used up after periods of varying stress and temperature.

The linear damage rule asserts that creep rupture will occur when the life fraction totals unity, that is, when $F(M) = 1$.

The limitations of this rule are not well understood. Nevertheless, the engineering utility of the rule is widely accepted, and the rule is frequently used in both creep-rupture and fatigue analysis [References 14, 15, 16, 17].

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B.3 Derivation of Equation for Corrosion Fraction

With continually varying stress and temperature, the life fraction can be expressed as an integral:

$$F(T) = \int_0^T \frac{d\tau}{L_r} \tag{B-2}$$

Where:

T = operating life.
$L_r = L_r(S, \theta)$
= rupture life at stress, S, and temperature, θ.
τ = time.

In general, both the stress, S, and the temperature, θ, are functions of time.

The rupture life and the stress can be related as follows, at least over limited regions of stress or time (see A.4):

$$L_r = mS^{-n}$$

Where:

m and n = material parameters that are functions of temperature.
n = rupture exponent.

For a specified design life, L_d, and corresponding rupture strength, S_r

$$L_d = mS_r^{-n}$$

So,

$$m = L_d S_r^n$$

Hence,

$$L_r = L_d\left(\frac{S_r}{S_r}\right)^n \tag{B-3}$$

Using Equation B-3 in Equation B-2, the life fraction can be expressed as follows:

$$F(T) = \int_0^T \left[\frac{S(\tau)}{S_r}\right]^n \frac{dy}{L_d} \tag{B-4}$$

Where:

$S(\tau)$ = stress expressed as a function of time.

This integral can be calculated once the temperature and stress history are known, but in general this calculation is difficult to perform. For the purposes of this development for tube design, the temperature is assumed to be constant. (This assumption is relaxed in B.5.) The remaining variable is therefore the stress as a function of time. This is given by the mean-diameter equation for stress as follows:

$$S(\tau) = \frac{P_r}{2}\left(\frac{D_o}{t(\tau)} - 1\right) \tag{B-5}$$

Where:

P_r = rupture design pressure.
D_o = outside diameter.
$t(\tau)$ = thickness expressed as a function of time.

In general, the rupture design pressure (operating pressure) is also a function of time; however, like temperature, it is assumed to be constant for the purposes of tube design. The thickness is determined by the following equation:

$$t(\tau) = t_0 - R\tau \tag{B-6}$$

Where:

t_0 = initial thickness.
R = corrosion rate.

Calculating $F(T)$ is then simply a matter of substituting Equations B-5 and B-6 in Equation B-4 and integrating. This integration cannot be done in closed form; a simplifying assumption is needed.

Let t_s be the thickness calculated from S_r as follows:

$$t_s = \frac{P_r D_o}{2S_r + P_r}$$

To a first approximation,

$$S(\tau) \cong S_r\left[\frac{t_s}{t(\tau)}\right] \tag{B-7}$$

Substituting Equations B-5, B-6, and B-7 in Equation B-4 and integrating results in the following equation:

$$F(T) = \frac{t_s^n}{(n-1)RL_d}\left[\left(\frac{1}{t_0 - RT}\right)^{n-1} - \left(\frac{1}{t_0}\right)^{n-1}\right] \tag{B-8}$$

At $T = L_d$, $F(L_d)$ should equal unity; that is, the accumulated damage fraction should equal unity at the end of the design life. Using $F(T) = 1$ and $T = L_d$ in Equation B-8 results in the following equation:

$$1 = \frac{t_s^n}{(n-1)RL_d}\left[\left(\frac{1}{t_0 - RL_d}\right)^{n-1} - \left(\frac{1}{t_0}\right)^{n-1}\right] \tag{B-9}$$

Now let $t_0 = t_s + fCA$ and $B = CA/t_s$, where $CA = RL_d$; that is, the corrosion allowance is defined as being equal to the corrosion rate times the design life. With these changes, Equation B-9 becomes an equation for f as follows:

$$1 = \frac{1}{(n-1)B}\left[\left(\frac{1}{1 + fB - B}\right)^{n-1} - \left(\frac{1}{1 + fB}\right)^{n-1}\right] \tag{B-10}$$

For given values of B and n, Equation B-10 can be solved for the corrosion fraction, f. The solutions are shown in Figure 1.

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B.4 Limitations of the Corrosion Fraction

In addition to the limitations of the linear damage rule mentioned in B.2, the corrosion fraction has other limitations. For the derivation, the temperature, pressure, and corrosion rate were assumed to be constant throughout the operating life. In an operating heater, these factors are usually not constant; nevertheless, the assumptions of constant pressure, temperature, and corrosion rate are made for any tube design. The assumptions are therefore justified in this case, since the corrosion fraction is part of the rupture design procedure. (The assumption of constant temperature can be relaxed as shown in B.5.)

The derivation of the corrosion fraction also relies on the relationship between rupture life and stress expressed in Equation B-3. For those materials that show a straight-line Larson-Miller parameter curve in Figures 4A–4S this representation is exact. For those materials that show a curvilinear Larson-Miller parameter curve, using Equation B-3 is equivalent to making a straight-line approximation of the curve. To minimize the resulting error, the values of the rupture exponent shown in Figures 4A–4S were developed from the minimum 60,000- and 100,000-hour rupture strengths (see A.4). In effect, this applies the straight-line approximation to a shorter segment of the curved line and minimizes the error over the usual range of application.

Finally, the mathematical approximation of Equation B-7 was used. A more accurate approximation is available; however, when it is used, the resulting graphical solution for the corrosion fraction is more difficult to use. In addition, the resulting corrosion fraction differs from that given in Figure 1 by less than one half of 1 percent. This small error and the simplicity of using Figure 1 justify the approximation of Equation B-7.

B.5 Derivation of Equation for Temperature Fraction

Since tube design in the creep-rupture range is very sensitive to temperature, special consideration should be given to cases in which a big difference exists between start-of-run and end-of-run temperatures. In the derivation of the corrosion fraction B.3, the temperature was assumed to remain constant. The corrosion fraction can be applied to cases in which the temperature varies if an equivalent temperature can be calculated. The equivalent temperature should be such that a tube operating at this constant equivalent temperature would sustain the same creep damage as a tube operating at the changing temperature. Equation 6 can be used to calculate an equivalent temperature for a case in which the temperature changes linearly from start-of-run to end-of-run.

Equation B-3 was developed to relate the rupture life, L_r, to the applied stress, S. A comparable equation is needed to relate the rupture life to both stress and temperature. This equation can be derived by means of the Larson-Miller parameter plot. When this plot is a straight line (or when the curve can be approximated by a straight line), the stress and the Larson-Miller parameter, P, can be related as follows:

$$S = A10^{-bP} \tag{B-11}$$

Where:

A, b = curve-fit constants.
P = $\theta(C + \log L_r) \times 10^{-3}$.
θ = temperature, in absolute degrees.
C = Larson-Miller constant.
L_r = rupture time, in hours.

Solving Equation B-11 for L_r yields the following equation:

$$L_r = \frac{1}{10^c}\left(\frac{A}{S}\right)^{1000/b\theta} \tag{B-12}$$

Using Equation B-12, the life fraction given by Equation B-2 becomes the following:

$$F(T) = \int_0^T 10^c \left(\frac{S}{A}\right)^{1000/b\theta} d\tau \tag{B-13}$$

Where:

S = stress as a function of time.
θ = temperature as a function of time.

The thickness, which is also a function of time, can be expressed as follows:

$$t(\tau) = t_0 - \left(\frac{\Delta t}{T}\right)\tau = t_0\left[1 - \left(\frac{\Delta t}{t_0}\right)\left(\frac{\tau}{T}\right)\right]$$

Where:

t_0 = initial thickness.
Δt = thickness change in time T.
T = duration of operating period.

For this derivation, let

$$B = \frac{\Delta t}{t_0} \tag{B-14}$$

$$\rho = \frac{\tau}{T} \tag{B-15}$$

Therefore,

$$t(\tau) = t_0(1 - B\rho) \tag{B-16}$$

Using Equations B-5 and B-16 and the approximation given by Equation B-7, the stress can be expressed as follows:

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$$S(\tau) \cong S_0\left[\frac{t_0}{t(\tau)}\right] = \frac{S_0}{1 - B\rho} \qquad \text{(B-17)}$$

Where:

$$S_o = \frac{P_r}{2}\left(\frac{D_o}{t_0} - 1\right)$$

If a linear change in temperature occurs during the time T, then the temperature, θ, can be expressed as a function of time, τ, as follows:

$$\theta(\tau) = \theta_0 + \left(\frac{\Delta\theta}{T}\right)\tau = \theta_0\left[1 + \left(\frac{\Delta\theta}{\theta_0}\right)\left(\frac{\tau}{T}\right)\right]$$

Where:

θ_o = initial temperature, in absolute degrees.

$\Delta\theta$ = temperature change in time T

$= T_{eor} - T_{sor}$

Let

$$D = \frac{\Delta\theta}{\theta_0} \qquad \text{(B-18)}$$

Using Equations B-15 and B-18, the equation for temperature becomes the following:

$$\theta(\tau) = \theta_0(1 + D\rho) \qquad \text{(B-19)}$$

Using Equations B-17 and B-19, Equation B-13 can be written as follows:

$$F(T) = \int_0^1 10^c\left[\left(\frac{S_0}{A}\right)\left(\frac{1}{1 - B\rho}\right)\right]^{n_0/(1 + D\rho)} Td\rho \qquad \text{(B-20)}$$

Where:

$$n_0 = \frac{1000}{b\theta_0}$$

This is the rupture exponent at the initial temperature, θ_o.

The aim of this analysis is to find a constant equivalent temperature, θ_e, between θ_o and $\theta_o + \Delta\theta$ such that the life fraction at the end of the period T with the linearly changing temperature will be equal to the life fraction with the equivalent temperature. This equivalent temperature can be expressed as follows:

$$\theta_e = \theta_0(1 + D\lambda),\ 0 < \lambda < 1 \qquad \text{(B-21)}$$

From Equation B-20, the resulting life fraction is as follows:

$$F(T) = \int_0^1 10^c\left[\left(\frac{S_0}{A}\right)\left(\frac{1}{1 - B\rho}\right)\right]^{n_0/(1 + D\lambda)} Td\rho \qquad \text{(B-22)}$$

Equating Equations B-20 and B-22 and dividing out common terms yields an integral equation for the parameter λ:

$$\int_0^1\left[\left(\frac{S_0}{A}\right)\left(\frac{1}{1 - B\rho}\right)\right]^{n_0/(1 + D\rho)} d\rho$$

$$\int_0^1\left[\left(\frac{S_0}{A}\right)\left(\frac{1}{1 - B\rho}\right)\right]^{n_0/(1 + D\lambda)} d\rho \qquad \text{(B-23)}$$

For given values for S_0, A, n_0, B, and D, Equation B-23 can be solved numerically for λ. Using λ and Equations B-18 and B-21, the equivalent temperature is calculated as follows:

$$\theta_e = \theta_0\left(1 + \frac{\Delta\theta}{\theta_0}\lambda\right) = \theta_0 + \lambda\Delta\theta \qquad \text{(B-24)}$$

The parameter λ is the temperature fraction, f_T, in 2.8.

The solutions to Equation B-23 can be approximated by a graph if the given values are combined into two parameters as follows:

$$V = n_0 D \ln\left(\frac{A}{S_0}\right) = n_0\left(\frac{\Delta\theta}{\theta_0}\right) \ln\left(\frac{A}{S_0}\right)$$

$$N = n_0 B = n_0\left(\frac{\Delta t}{t_0}\right)$$

Using these two parameters, the solutions to Equation B-23 are shown in Figure 2.

The constant A in Table 3 is one of the least-squares curve-fit constants, A and b, in the equation $S = A10^{-bP}$, where P is the Larson-Miller parameter and S is the minimum rupture strength. For materials that have a straight Larson-Miller parameter curve, A can be calculated directly from any two points on the curve. For all other materials, a least-squares approximation of the minimum rupture strength was calculated in the stress region below the intersection of the rupture and elastic allowable stresses, since this is the region of most applications. For the purpose of calculating the temperature fraction, this accuracy is sufficient.

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APPENDIX C—CALCULATION OF MAXIMUM RADIANT SECTION TUBE SKIN TEMPERATURE

C.1 General

This appendix provides a procedure for calculating the maximum radiant section tube metal (skin) temperature. Correlations for estimating the fluid-film heat-transfer coefficient are given in C.2. A method for estimating the maximum local heat flux is given in C.3. The equations used to calculate the maximum tube skin temperature and the temperature distribution through the tube wall are described in C.4. The sample problem in C.5 demonstrates the use of these equations.

C.2 Heat-Transfer Coefficient

A value necessary for calculating the maximum tube metal temperature is the fluid heat-transfer coefficient at the inside wall of the tube. Although the following correlations are extensively used and accepted in heater design, they have inherent inaccuracies associated with all simplified correlations that are used to describe complex relationships.

For single-phase fluids, the heat-transfer coefficient is calculated by one of the two equations below, in which *Re* is the Reynolds number and *Pr* is the Prandtl number. No correlation is included for the heat-transfer coefficient in laminar flow, since this flow regime is rare in process heaters. There is inadequate information for reliably determining the inside coefficient in laminar flow for oil in tube sizes that are normally used in process heaters.

For liquid flow with $Re \geq 10{,}000$ [Reference 18],

$$h_l = 0.023\left(\frac{k}{D_i}\right) Re^{0.8} Pr^{0.33} \left(\frac{\mu_b}{\mu_w}\right)^{0.14} \qquad \text{(C-1)}$$

For vapor flow with $Re \geq 15{,}000$ [Reference 19],

$$h_v = 0.021\left(\frac{k}{D_i}\right) Re^{0.8} Pr^{0.4} \left(\frac{T_b}{T_w}\right)^{0.5} \qquad \text{(C-2)}$$

$$Re = \frac{D_i G}{\mu_b} \qquad \text{(C-3)}$$

$$Pr = \frac{C\mu_b}{k} \qquad \text{(C-4)}$$

Where:

h_l = heat-transfer coefficient, liquid phase, in British thermal units per hour per square foot per degree Fahrenheit (watts per square meter per degree Celsius).

k = thermal conductivity of fluid at bulk temperature, in British thermal units per hour per foot per degree Fahrenheit (watts per meter per degree Celsius).

D_i = inside diameter of tube, in feet (meters).

μ_b = absolute viscosity of fluid at bulk temperature, in pounds per foot per hour (pascal-seconds).

μ_w = absolute viscosity of fluid at wall temperature, in pounds per foot per hour (pascal-seconds).

h_v = heat-transfer coefficient, vapor phase, in British thermal units per hour per square foot per degree Fahrenheit (watts per square meter per degree Celsius).

T_b = absolute bulk temperature of vapor, in degrees Rankine (kelvins).

T_w = absolute wall temperature of vapor, in degrees Rankine (kelvins).

G = mass flux of fluid, in pounds per hour per square foot (kilograms per second per square meter).

C = heat capacity of fluid at bulk temperature, in British thermal units per pound per degree Fahrenheit (joules per kilogram per degree Celsius).

All of the material properties except μ_w are evaluated at the bulk fluid temperature. To convert absolute viscosity in centipoise to pounds per foot per hour, multiply by 2.42. To convert centipoise to pascal-seconds, divide by 1000.

For two-phase flows, the heat-transfer coefficient can be approximated using the following equation:

$$h_{tp} = h_l W_l + h_v W_v \qquad \text{(C-5)}$$

Where:

h_{tp} = heat-transfer coefficient, two phase, in British thermal units per hour per square foot per degree Fahrenheit (watts per square meter per degree Celsius).

W_l = weight fraction of liquid.

W_v = weight fraction of vapor.

The liquid and vapor heat-transfer coefficients, h_l and h_v should be calculated using the mixed-phase mass flux but using the liquid and vapor material properties, respectively.

Note: In two phase flow applications where dispersed flow or mist flow regimes occur due to entrainment of liquid droplets in the vapor (e.g. towards the outlet of vacuum heaters), the heat transfer coefficient may be calculated using the correlation for vapor phase per Equation C-2, based on the total flow rate, rather than approximated by Equation C-5.

C.3 Maximum Local Heat Flux

The average heat flux in the radiant section of a heater (or in a zone of the radiant section) is equal to the duty in the section or zone divided by the total outside surface area of the coil in the section or zone. The maximum local heat flux at any point in the coil can be estimated from the average

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heat flux. The maximum local heat flux is used with the equations in C.4 to calculate the maximum metal temperature.

Local heat fluxes vary considerably throughout a heater because of nonuniformities around and along each tube. Circumferential variations result from variations in the radiant heat flux produced by shadings of other tubes or from placement of the tubes next to a wall. Conduction around the tubes and convection flows of flue gases tend to reduce the circumferential variations in the heat flux. The longitudinal variations result from proximity to burners and variations in radiant firebox and bulk fluid temperatures. In addition to variations in the radiant section, the tubes in the shock section of a heater can have a high convective heat flux.

The maximum heat flux at any point in a coil can be estimated as follows:

$$q_m = F_c F_L F_T q_a + q_c \qquad \text{(C-6)}$$

Where:

q_m = maximum radiant heat flux, outside surface, in British thermal units per hour per square foot (watts per square meter).

F_C = factor accounting for circumferential heat-flux variations.

F_L = factor accounting for longitudinal heat-flux variations.

F_T = factor accounting for the effect of tube metal temperature on radiant heat flux.

q_a = average radiant heat flux, outside surface, in British thermal units per hour per square foot (watts per square meter).

q_c = average convective heat flux, outside surface, in British thermal units per hour per square foot (watts per square meter).

The circumferential variation factor, F_C, is given as a function of tube spacing and coil geometry in Figure C-1. The factor given by this figure is the ratio of the maximum local heat flux at the fully exposed face of a tube to the average heat flux around the tube. This figure was developed from considerations of radiant heat transfer only. As mentioned above, influences such as conduction around the tube and convective flows of flue gases act to reduce this factor. Since these influences are not included in this calculation, the calculated value will be somewhat higher than the actual maximum heat flux.

The longitudinal variation factor, F_L, is not easy to quantify. Values between 1.0 and 1.5 are most often used. In a firebox that has a very uniform distribution of heat flux, a value of 1.0 may be appropriate. Values greater than 1.5 may be appropriate in a firebox that has an extremely uneven distribution of heat flux (for example, a long or a tall, narrow firebox with burners in one end only).

The tube metal temperature factor, F_T, will be less than 1.0 near the coil outlet or in areas of maximum tube metal temperature. It will be greater than 1.0 in areas of lower tube metal temperatures. For most applications, the factor can be approximated as follows:

$$F_T = \left(\frac{T_g^4 - T_s^4}{T_g^4 - T_a^4}\right) \qquad \text{(C-7)}$$

Where:

T_g = average flue-gas temperature in the radiant section, in degrees Rankine (kelvin).

T_s = tube metal temperature at the point under consideration, in degrees Rankine (kelvin).

T_a = average tube metal temperature in the radiant section, in degrees Rankine (kelvin).

The convective heat flux in most parts of a radiant section is usually small compared with the radiant heat flux. In the shock section, however, the convective heat flux can be significant; it should therefore be added to the radiant heat flux when the maximum heat flux in the shock section is estimated.

C.4 Maximum Tube Metal Temperature

In addition to the heat-transfer coefficient and the maximum heat flux, the temperature profile of the fluid in the coil is necessary for calculating the maximum tube metal temperature in the radiant section of the heater. This profile, which is often calculated by the heater supplier, defines the variation of the bulk fluid temperature through the heater coil. For operation at or near design, the design profile can be used. For operation significantly different from design, a bulk temperature profile must be developed.

Once the bulk fluid temperature is known at any point in the coil, the maximum tube metal temperature can be calculated as follows:

$$T_m = T_b + \Delta T_f + \Delta T_c + \Delta T_w \qquad \text{(C-8)}$$

$$\Delta T_f = \frac{q_m}{h}\left(\frac{D_o}{D_i - 2t_c}\right) \qquad \text{(C-9)}$$

$$\Delta T_c = \frac{q_m t_c}{k_c}\left(\frac{D_o}{D_i - t_c}\right) \qquad \text{(C-10)}$$

$$\Delta T_w = \frac{q_m t_a}{k_w}\left(\frac{D_o}{D_o - t_a}\right) \qquad \text{(C-11)}$$

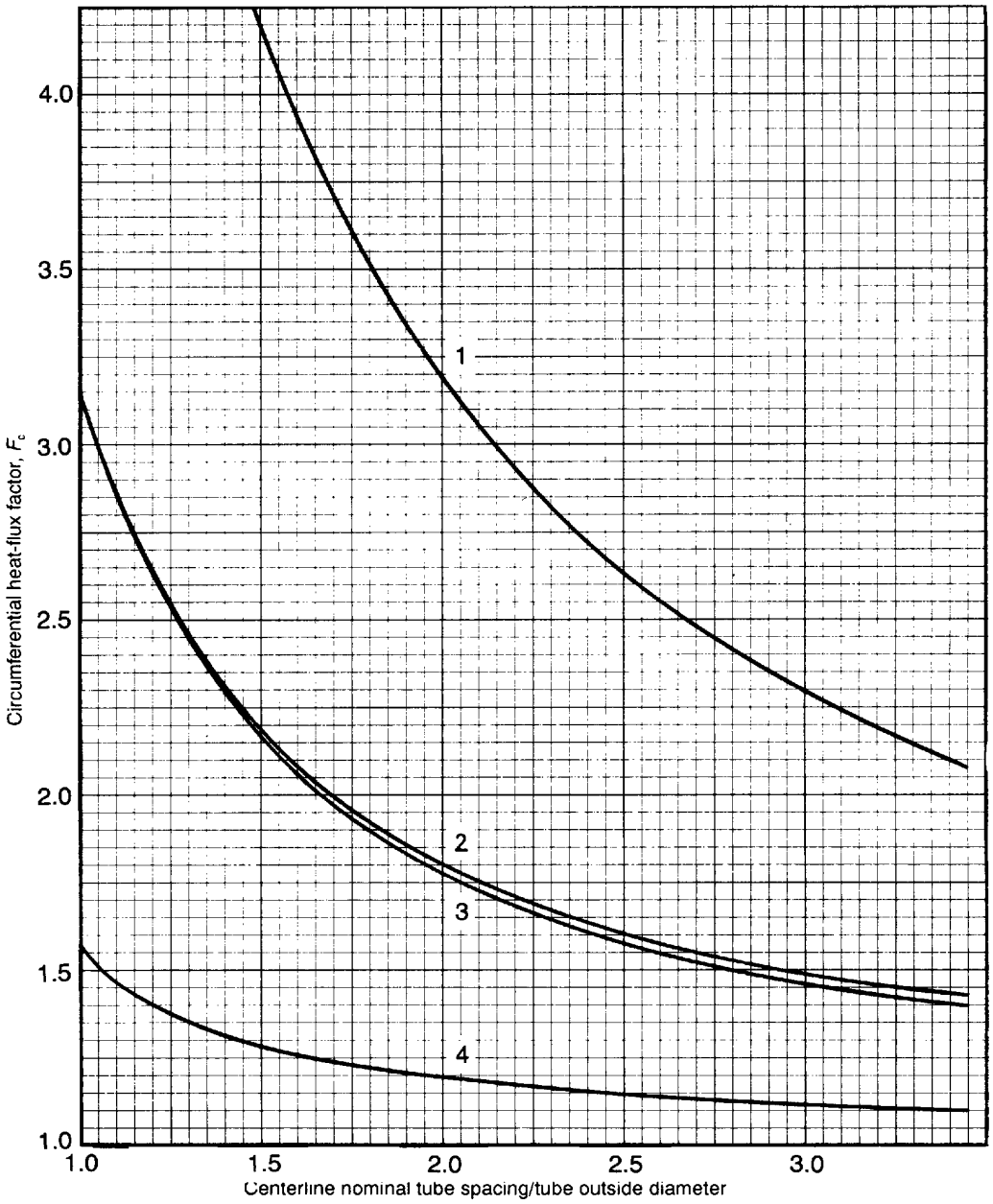
Where:

T_m = maximum tube metal temperature, in degrees Fahrenheit (degrees Celsius).

T_b = bulk fluid temperature, in degrees Fahrenheit (degrees Celsius).

ΔT_f = temperature difference across fluid film, in degrees

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Notes:
Note 1: Curve 1 = double row against wall, triangular spacing; Curve 2 = double row with equal radiation from both sides and two diameters between rows, equilateral spacing; Curve 3 = single row against wall; and Curve 4 = single row with equal radiation from both sides.
Note 2: These curves are valid when used with a tube-center-to-refractory-wall spacing of $1^1/_2$ times the nominal tube diameter. Any appreciable variation from this spacing must be given special consideration.
Note 3: These curves do not consider convection heat transfer to the tubes, circumferential heat transfer by conduction in the tube wall, or variations in heat flux in different zones of the radiant section.
Note 4: These curves are based on the work of H.C. Hottel, as reported on page 69 of Reference 18.

Figure C-1—Ratio of Maximum Local to Average Heat Flux

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Fahrenheit (degrees Celsius).

ΔT_c = temperature difference across coke or scale, in degrees Fahrenheit (degrees Celsius).

ΔT_w = temperature difference across tube wall, in degrees Fahrenheit (degrees Celsius).

q_m = maximum radiant heat flux, outside surface, in British thermal units per hour per square foot (watts per square meter).

h = fluid-film heat-transfer coefficient, in British thermal units per hour per square foot per degree Fahrenheit (watts per square meter per degree Celsius).

D_o = outside diameter of tube, in feet (meters).

D_i = inside diameter of tube, in feet (meters).

t_c = coke or scale thickness, in feet (meters).

k_c = thermal conductivity of coke or scale, in British thermal units per hour per foot per degree Fahrenheit (watts per meter per degree Celsius).

t_a = average tube thickness, in feet (meters).

k_w = thermal conductivity of tube metal, in British thermal units per hour per foot per degree Fahrenheit (watts per meter per degree Celsius).

In Equations C-10 and C-11, the denominators within the parentheses are the mean diameters of the coke layer and tube, respectively. The effect of coke or scale on the tube metal temperature can be estimated using Equation C-10.

The thermal conductivity of the tube material used in Equation C-11 should be evaluated at the average tube wall temperature. For cast tubes, the nominal as-cast thickness should be used for t_a in Equation C-11.

C.5 Sample Calculation

The following sample calculation demonstrates how to use the equations given in the previous paragraphs, (the SI conversions in parentheses are approximate).

In the heater under consideration, the medium carbon steel tubes are in a single row against the wall. Other aspects of the heater configuration are as follows:

Tube spacing = 8.0 inches = 0.667 foot (203.2 millimeters).

D_o = 4.5 inches = 0.375 foot (114.3 millimeters).

t_a = 0.25 inch = 0.0208 foot (6.4 millimeters).

D_i = 4.0 inches = 0.333 foot (101.5 millimeters).

t_c = 0 inches (0 millimeters).

k_w = 24.4 Btu/hr-ft-°F (42.2 W/m-°C) at an assumed tube metal temperature of 720°F (380°C) [from Table 1, Reference 7].

The flow in the tubes is two phase with 10 weight percent vapor. Other operating conditions are as follows:

Flow rate = 50,000 pounds per hour (6.3 kilograms per second) [total liquid plus vapor].

T_b = 520°F (270°C).

q_a = 10,000 Btu/hr-ft^2 (31,500 W/m^2).

The properties of the liquid at the bulk temperature are as follows:

μ = 2 centipoise = 4.84 lb/hr-ft (2.0×10^{-3} Pa-s).

k = 0.0672 Btu/hr-ft-°F (0.1163 W/m-°C).

C = 0.68 Btu/lb-°F (2850 J/kg-°C).

The properties of the vapor at the bulk temperature are as follows:

μ = 0.007 centipoise = 0.017 lb/hr-ft (7.0 × 10-6 Pa-s).

k = 0.020 Btu/hr-ft-°F (0.0346 W/m-°C).

C = 0.572 Btu/lb-°F (2395 J/kg-°C).

From the inside diameter, the flow area is equal to 0.0871 square foot (8.091×10^{-3} square meters). Using the total flow rate,

$G = (50{,}000/0.0871)$

$= 5.74 \times 10^5$ lb/hr-ft^2

In SI units,

$G = 6.3/(8.091 \times 10^{-3})$

$= 778.6$ kg/s-m^2

The Reynolds number (Equation C-3) is as follows:

For liquid,

$$Re = \frac{(0.333)(5.74 \times 10^5)}{4.84} = 3.95 \times 10^4$$

In SI units,

$$Re = \frac{(0.1015)(778.6)}{0.002} = 3.95 \times 10^4$$

For vapor,

$$Re = \frac{(0.333)(5.74 \times 10^5)}{0.017} = 1.13 \times 10^7$$

In SI units,

$$Re = \frac{(0.1015)(778.6)}{7.0 \times 10^{-6}} = 1.13 \times 10^7$$

The Prandtl number (Equation C-4) is as follows:

For liquid,

$$Pr = \frac{(0.68)(4.84)}{0.0672} = 49.0$$

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In SI units,

$$Pr = \frac{(2850)(0.002)}{0.1163} = 49.0$$

For vapor,

$$Pr = \frac{(0.572)(0.017)}{0.020} = 0.485$$

In SI units,

$$Pr = \frac{(2395)(7.0 \times 10^{-6})}{0.0346} = 0.485$$

Assume that for the liquid,

$$\left(\frac{\mu_b}{\mu_w}\right)^{0.14} = 1.1$$

Assume that for the vapor,

$$\left(\frac{T_b}{T_w}\right)^{0.5} = 0.91$$

These assumptions will be checked later. Using Equation C-1,

$$h_l = 0.023\left(\frac{k}{D_i}\right)(3.95 \times 10^4)^{0.8}(49.0)^{0.33}(1.1)$$

$$= 434.7\left(\frac{k}{D_i}\right)$$

Using Equation C-2,

$$h_v = 0.021\left(\frac{k}{D_i}\right)(1.13 \times 10^7)^{0.8}(0.485)^{0.4}(0.91)$$

$$= 6281\left(\frac{k}{D_i}\right)$$

Hence,

$$h_l = 434.7\left(\frac{0.0672}{0.333}\right) = 87.7 \text{ Btu/hr-ft}^2\text{-°F}$$

$$h_v = 6281\left(\frac{0.020}{0.333}\right) = 377 \text{ Btu/hr-ft}^2\text{-°F}$$

In SI units,

$$h_l = 434.7\left(\frac{0.1163}{0.1015}\right) = 498 \text{ W/m}^2\text{-°C}$$

$$h_v = 6281\left(\frac{0.0346}{0.1015}\right) = 2141 \text{ W/m}^2\text{-°C}$$

The two-phase heat-transfer coefficient can then be calculated using Equation C-5:

$$h_{tp} = (0.90)h_l + (0.10)h_v$$
$$= (0.90)(87.7) + (0.10)(377)$$
$$= 116.6 \text{ Btu/hr-ft}^2\text{-°F}$$

In SI units,

$$h_{tp} = (0.90)(498) + (0.10)(2141)$$
$$= 662.3 \text{ W/m}^2\text{-°C}$$

The ratio of tube spacing to tube diameter is as follows:

$$\frac{8.0}{4.5} = 1.78$$

In SI units,

$$\frac{203.2}{114.3} = 1.78$$

From Figure C-1, $F_C = 1.91$. Assume that for this heater, $F_L = 1.1$, $F_T = 1.0$, and $q_c = 0$ (that is, there is no convective heat flux at this point). Using Equation C-6,

$$q_m = (1.91)(1.1)(10{,}000)$$
$$= 21{,}000 \text{ Btu/hr-ft}^2$$

In SI units,

$$q_m = (1.91)(1.1)(31{,}500)$$
$$= 66{,}200 \text{ W/m}^2$$

The temperature difference through each part of the system can now be calculated. From Equation C-9 for the fluid film,

$$\Delta T_f = \left(\frac{21,000}{116.6}\right)\left(\frac{0.375}{0.333}\right) = 203°\text{F}$$

In SI units,

$$\Delta T_f = \left(\frac{66,200}{662.3}\right)\left(\frac{114.3}{101.5}\right) = 113°\text{C}$$

From Equation C-11 for the tube wall,

$$\Delta T_w = \left[\frac{(21,000)(0.0208)}{24.4}\right]\left[\frac{0.375}{0.375 - 0.0208}\right] = 19°\text{F}$$

In SI units,

$$\Delta T_w = \left[\frac{(66,200)(6.4)}{42.2}\right]\left[\frac{114.3}{114.3 - 6.4}\right] \times 10^{-3} = 11°\text{C}$$

Using Equation C-8, the maximum tube metal temperature is as follows:

$$T_m = 520 + 203 + 19 = 742°\text{F}$$

In SI units,

$$T_m = 270 + 113 + 11 = 394°\text{C}$$

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Checking the assumed viscosity ratio, at the oil-film temperature calculated above, 520 + 203 = 723°F (270 + 113 = 383°C), the viscosity is 1.1 centipoise = 2.66 pounds per hour-foot (1.1 x 10^{-3} pascal-seconds). So for the liquid,

$$\left(\frac{\mu_b}{\mu_w}\right)^{0.14} = \left(\frac{4.84}{2.66}\right)^{0.14} = (1.82)^{0.14} = 1.09$$

In SI units,

$$\left(\frac{\mu_b}{\mu_w}\right)^{0.14} = \left(\frac{0.0020}{0.0011}\right)^{0.14} = (1.82)^{0.14} = 1.09$$

For the vapor,

$$\left(\frac{T_b}{T_w}\right)^{0.5} = \left(\frac{520 + 460}{723 + 460}\right)^{0.5} = (0.83)^{0.5} = 0.91$$

In SI units,

$$\left(\frac{T_b}{T_w}\right)^{0.5} = \left(\frac{270 + 273}{383 + 273}\right)^{0.5} = (0.83)^{0.5} = 0.91$$

Both values are close to the values assumed for the calculation of h_l and h_v, so no additional work is needed.

The mean tube-wall temperature is as follows:

$$520 + 203 + \frac{19}{2} = 732°F$$

In SI units,

$$270 + 113 + \frac{11}{2} = 388°C$$

This is close to the temperature assumed for the tube conductivity, so no additional work is required.

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APPENDIX D—THERMAL-STRESS LIMITATIONS (ELASTIC RANGE)

D.1 General

In heater tubes, the thermal stress of greatest concern is the one developed by the radial distribution of temperature through the thickness. This stress can become particularly significant in thick stainless steel tubes exposed to high heat fluxes.

There are two limits for thermal stress; both are described in Reference 7, Paragraphs 4-134 and 5-130. These limits apply only in the elastic range; in the rupture range, an appropriate limit for thermal stress has not been established.

D.2 Equation for Thermal Stress

The following equation gives the maximum thermal stress in a tube:

$$S_{th} = X\left[\frac{2Y^2}{Y^2-1}\ln Y - 1\right] \qquad \text{(D-1)}$$

Where:

$$X = \left[\frac{aE}{2(1-\nu)}\right]\left[\frac{\Delta T}{\ln Y}\right]\left[\frac{aE}{4(1-\nu)}\right]\left[\frac{q_o D_o}{k}\right] \qquad \text{(D-2)}$$

a = coefficient of thermal expansion.
E = modulus of elasticity.
ν = Poisson's ratio.
ΔT = temperature difference across tube wall.
Y = D_o/D_i, ratio of outside diameter to actual inside diameter.
q_o = heat flux on outside surface of tube.
k = thermal conductivity of the steel.

The material properties a, E, ν, and k shall be evaluated at the mean temperature of the tube wall. The average wall thickness shall also be used in this equation (see 2.7).

D.3 Limits on Thermal Stress

The limitation on primary plus secondary stress intensity in Paragraph 4-134, Reference 7, can be approximated for thermal stress as follows (see D.4 for derivation):

For ferritic steels,

$$S_{th\,lim_1} \cong (2.0 - 0.67Y)S_y \qquad \text{(D-3)}$$

For austenitic steels,

$$S_{th\,lim_1} \cong (2.0 - 0.90Y)S_y \qquad \text{(D-4)}$$

Where:

S_y = yield strength.

The thermal-stress ratchet limit in Paragraph 5-130, Reference 7, can be approximated for thermal stress as follows (see D.5 for derivation):

For ferritic steels,

$$S_{th\,lim_2} \cong 1.33S_y \qquad \text{(D-5)}$$

For austenitic steels,

$$S_{th\,lim_2} \cong 1.8S_y \qquad \text{(D-6)}$$

Both the primary plus secondary stress limit $S_{th\,lim_1}$ and the thermal-stress ratchet limit $S_{th\,lim_2}$ shall be met if the tube is designed for the elastic range.

D.4 Derivation of Limits on Primary Plus Secondary Stress Intensity

The limit on primary plus secondary stress intensity can be expressed symbolically as $P_L + P_b + Q \le 3S_m$. For this application, Q is the maximum circumferential thermal stress, S_{th}, given by Equation D-1. For tubes with an internal pressure [Reference 7],

$$P_L + P_b = P_e\left(\frac{2Y^2}{Y^2-1}\right)$$

Where:

P_L = local primary membrane stress.
P_b = primary binding stress.
P_e = elastic design pressure.
Y = D_o/D_i, ratio of outside to actual inside diameter.

If S_{pm} is the primary membrane stress intensity given by Equation D-7 [Reference 7],

$$S_{pm} = \frac{P_e}{2}\left(\frac{D_o}{t} - 1\right) = \frac{P_e}{2}\left(\frac{Y+1}{Y-1}\right) \qquad \text{(D-7)}$$

It can then be easily shown that, to a first approximation and providing an upper bound,

$$P_L + P_b \cong YS_{pm}$$

In Reference 7, S_m is the allowable membrane stress intensity. For ferritic steels above about 650°F (340°C), S_m is equal to two-thirds the yield strength, S_y, so $3S_m = 2S_y$. For austenitic steels above about 500°F (260°C), S_m is 90 percent of S_y, so $3S_m = 2.7S_y$. Heater tubes usually operate above these temperatures.

Combining all of this, the primary plus secondary stress intensity limit on thermal stress can be expressed as follows:

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For ferritic steels,

$$S_{th \cdot lim_1} = 2S_y - YS_{pm} \quad \text{(D-8)}$$

For austenitic steels,

$$S_{th \cdot lim_1} = 2.7S_y - YS_{pm} \quad \text{(D-9)}$$

Where:

$S_{th \cdot lim_1}$ = maximum value permitted for the thermal stress S_{th}.

For ferritic steel heater tubes designed according to this standard,

$$S_{pm} \leq 0.67S_y \quad \text{(D-10)}$$

For austenitic steel tubes,

$$S_{pm} \leq 0.90S_y \quad \text{(D-11)}$$

The thermal-stress limit can therefore be approximated as follows:

For ferritic steels,

$$S_{th \cdot lim_1} \cong (2.0 - 0.67Y)S_y$$

For austenitic steels,

$$S_{th \cdot lim_1} \cong (2.0 - 0.90Y)S_y$$

The limits expressed by these equations are simple and appropriate. If the thermal stress is less than this limit, the design is approximate. If the thermal stress exceeds the limit given by these equations, then the more exact form of Equations D-8 or D-9 shall be used with the primary membrane stress intensity given by Equation D-7. Also, if the tube thickness is arbitrarily increased over the thickness calculated in 2.3, then the primary membrane stress intensity shall be calculated using the actual average thickness, and Equation D-8 or Equation D-9 shall be used to calculate the thermal-stress limit.

D.5 Derivation of Limits on Thermal-Stress Ratchet

The limit set to avoid thermal-stress ratchet can be expressed as follows [Reference 7]:

$$S_{th \cdot lim_2} = 4(S - S_{pm}) \quad \text{(D-12)}$$

For ferritic steels, $S = S_y$, For austenitic steels above about 500°F (260°C), $S = 1.5\ (0.9S_y) = 1.35S_y$. As before, S_{pm} is derived from Equation D-7. Using Equation D-10 or Equation D-11, this limit can be approximated as follows:

For ferritic steels,

$$S_{th \cdot lim_2} \cong 1.33S_y$$

For austenitic steels,

$$S_{th \cdot lim_2} \cong 1.8S_y$$

As with the limits developed in D.4, these limits are approximate. If the thermal stress exceeds this limit or if the tube thickness is arbitrarily increased, the exact limit expressed by Equation D-12 shall be used with the primary membrane stress intensity given by Equation D-7.

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APPENDIX E—ESTIMATION OF REMAINING TUBE LIFE

E.1 General

Figures 4A–4S and the ideas discussed in Appendix B have uses other than for the design of new tubes. They can also be used to help answer rerating and retirement questions about existing tubes that have operated in the creep-rupture range. This appendix describes how tube damage and remaining life can be estimated. Because of the uncertainties involved in these calculations, decisions about tube retirement should not be based solely on the results of these calculations. Other factors such as tube thickness or diameter-strain measurements should be primary considerations in decisions about tube retirement.

There are three primary areas of uncertainty in these calculations. First, to estimate the accumulated tube damage (the life fraction used up), the operating history must be approximated. This history must include the operating pressure, the tube metal temperature, and the corrosion rate. The uncertainties in these factors, particularly the temperature, can have a significant effect on the estimate. Second, knowledge of the actual rupture strength of a given tube is not precise. The example calculation in E.2 demonstrates the effects of this uncertainty. Finally, the tube damage calculation makes use of the linear damage rule described in B.2. As mentioned in B.2., the limitations of the hypothesis are not well understood. In spite of all these uncertainties, the estimation that is made using the procedure described in this appendix may provide information that will assist in making decisions about tube rerating and retirement.

The essence of this calculation procedure can be outlined as follows: The operating history is divided into periods of time in which the pressure, metal temperature, and corrosion rate are assumed constant. For each of these periods, the life fraction used up is calculated. The sum of these calculated life fractions is the total accumulated tube damage. The fraction remaining is calculated by subtracting this sum from unity. Finally, the remaining life fraction is transformed into an estimate of the expected life at specified operating conditions.

E.2 Estimation of Accumulated Tube Damage

Since the concepts required to estimate damage are developed elsewhere in this practice, they are not repeated here. The calculation procedure can best be explained by working through an example. For this example, the following conditions are assumed:

Material	= 8Cr-10Ni-Cb (Type 347) stainless steel
Outside diameter	= 6.625 inches (168.3 millimeters)
Initial minimum thickness	= 0.268 inch (6.8 millimeters)

It is also assumed that the operating history of the tube can be approximated as shown in Table E-1. (The SI conversions are approximate.)

The operating periods need not be of uniform length. In an actual heater, neither the operating pressure nor the metal temperature is uniform. Even so, for this calculation, they must be assumed to be uniform during each period. The values chosen for each period should represent typical values. The length of the operating period chosen depends on how the pressure and temperature vary.

The history of the tube thickness must be approximated. This history can usually be developed from thickness measurements made before the initial start-up and during routine heater-tube inspections. For all of these estimates, it is assumed that the outside diameter remains constant.

This information can be used to calculate the life fractions shown in Table E-2.

For a corroding tube, an equation similar to Equation B-8

Table E-1—Approximate Operating History

Operation Period	Duration (years)	Operating Pressure		Tube Metal Temperature		Minimum Thickness			
		Pounds per Square Inch Gauge	Megapascals (gauge)	Degrees Fahrenheit	Degrees Celsius	Beginning		End	
						Inches	Millimeters	Inches	Millimeters
1	1.3	575	3.96	1200	649	0.268	6.81	0.252	6.40
2	0.6	620	4.27	1230	665	0.252	6.40	0.244	6.20
3	2.1	590	4.07	1220	660	0.244	6.20	0.217	5.51
4	2.0	630	4.34	1230	665	0.217	5.51	0.190	4.83

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Table E-2—Life Fractions for Each Period

Operating Period	Average Stress		Larson-Miller Values				Rupture Time Based on Minimum Strength		Rupture Time Based on Average Strength	
			Minimum		Average					
	Pounds per Square Inch	Megapascals	Degrees Fahrenheit	Degrees Celsius	Degrees Fahrenheit	Degrees Celsius	Years	Life Fractions	Years	Life Fractions
1	7045	48.52	34.32	19.06	35.09	19.48	54.0	0.02	154.8	0.01
2	7973	54.91	33.90	18.83	34.66	19.25	13.1	0.05	35.8	0.02
3	8213	56.66	33.80	18.77	34.55	19.19	15.0	0.14	42.1	0.05
4	9985	68.78	33.15	18.41	33.90	18.83	4.7	0.43	13.1	0.15
							Accumulated damage = 0.64			0.23

can be developed for the life fraction; however, this is not necessary, since sufficient accuracy can be achieved for this calculation by using the average stress for each period (that is, the average of the stress at the beginning and at the end of the operating period).

The minimum and average Larson-Miller values in Table E-2 are determined from the average stress using the Larson-Miller parameter curves for minimum and average rupture strength in Figures 4A–4S. For this example, Figure 4Q was used.

With these Larson-Miller values and the metal temperature for each period, the expression for the Larson-Miller parameter was solved for the rupture time. This expression is at the top of Figures 4A–4S. Since this expression gives the rupture time in hours, the value must be converted to years. The resulting times based on the minimum rupture strength and the average rupture strength are shown in Table E-2.

The following example illustrates how to calculate the minimum-strength rupture time for the first operating period. The equation to be solved is as follows:

$$34.32 = (1200 + 460)(15 + \log L_r) \times 10^{-3}$$

In SI units,

$$19.06 = (649 + 273)(15 + \log L_r) \times 10^{-3}$$

Or,

$$\log L_r = 5.67$$
$$L_r = 4.73 \times 10^5 \text{ hours}$$
$$= 54.0 \text{ years}$$

The life fractions are simply the duration of the operating period divided by the rupture time that corresponds to that period. Using the minimum-strength rupture time calculated above, the fraction for the first line in Table E-2 is 1.3/54.0 = 0.02. The accumulated damage is the sum of the fractions.

The effect of the uncertainty about the rupture strength is evident in Table E-2. If the actual rupture strength of this tube is in the lower part of the scatter band (near the minimum rupture strength), then 64 percent of the tube life has been used. If the actual strength is in the middle of the scatter band (near the average rupture strength), then only 23 percent of the tube life has been used. If the actual rupture strength is higher, even less of the tube life has been used.

The effect of the uncertainty about the operating temperature can also be evaluated. Suppose the actual metal temperature of this tube were 9°F (5°C) higher than that shown in Table E-1. To estimate the effect of this difference, the life-fraction calculations in Table E-2 have been made with the slightly higher temperature. The corresponding accumulated damage fractions are 0.81 and 0.28, respectively. These should be compared with the values 0.64 and 0.23, which were calculated first.

E.3 Estimation of Remaining Tube Life

As in E.2, this calculation procedure is best explained by example. The example used is the one summarized in Tables E-1 and E-2. The life fraction remaining for this tube is as follows:

Minimum rupture strength: 1 – 0.64 = 0.36
Average rupture strength: 1 – 0.23 = 0.77

These fractions must be translated into expected life at the specified operating conditions.

The following related questions can be asked at this point:

a. What is the estimated life at a given operating pressure, metal temperature, and corrosion rate?

b. For a specified operating pressure and corrosion rate, what temperature limit must be imposed for the tube to last a minimum period of time?

c. How much should the operating pressure or metal temperature be reduced to extend the expected life by a given percentage?

Not all of these questions are answered in this appendix, but the method used to develop the answers should be clear from the following example.

For this example, the expected operating conditions are as follows:

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Operating pressure = 620 pounds per square inch gauge (4.27 megapascals gauge).
Metal temperature = 1220°F (660°C).
Corrosion rate = 0.013 inch per year (0.33 millimeter per year).

From these values a table of future-life fractions can be developed as shown in Table E-3 for the minimum rupture strength and in Table E-4 for the average rupture strength. As before, the average stress is the average of the stresses at the beginning and end of each operating period.

Since the tube in the example is corroding, the life estimation should be calculated in steps. For this example, a 1-year step was used. As can be seen from the two tables, the estimated life of this tube is between 1.5 and 4.5 years. If the rupture strength were in the upper part of the scatter band (above the average rupture strength), the estimated life would be even longer.

For tubes that are not corroding, estimating the life is easier. The rupture life is calculated as above from the anticipated stress and temperature. The estimated remaining life is simply the fraction remaining times the rupture life. In these cases, tables such as Tables E-3 and E-4 are not required.

The example given above describes a way to answer Question (a), posed at the beginning of this section: What is the estimated life for a specified set of operating conditions? Question (b), concerning the temperature limit that should be imposed for a specified pressure, corrosion rate, and minimum life, can be answered in the following way: The pressure and corrosion rate can be used to calculate an average stress from which a Larson-Miller value can be found using the curves in Figures 4A–4S With this value and a rupture life calculated by dividing the required life by the remaining life fraction, the Larson-Miller parameter equation can be solved for the maximum temperature. The other questions can be answered in similar ways.

Table E-3—Future Life Fractions, Minimum Strength

Time (years)	Minimum Thickness		Average Stress		Minimum Larson-Miller Value		Rupture Time (years)	Fraction	Remaining Fraction
	Inches	Millimeters	Pounds per Square Inch	Megapascals	Degrees Fahrenheit	Degrees Celsius			
0	0.190	4.83	—	—	—	—	—	—	0.36
1	0.177	4.50	10,896	74.99	32.85	18.25	4.1	0.24	0.12
1.5	0.171	4.34	11,497	79.19	32.66	18.14	3.1	0.16	-0.04

Table E-4—Future Life Fractions, Average Strength

Time (years)	Minimum Thickness		Average Stress		Minimum Larson-Miller Value		Rupture Time (years)	Fraction	Remaining Fraction
	Inches	Millimeters	Pounds per Square Inch	Megapascals	Degrees Fahrenheit	Degrees Celsius			
0	0.190	4.83	—	—	—	—	—	—	0.77
1	0.177	4.50	10,896	74.99	33.60	18.66	11.4	0.09	0.68
2	0.164	4.17	11,753	80.87	33.35	18.53	8.2	0.12	0.56
3	0.151	3.84	12,752	87.74	33.07	18.37	5.5	0.18	0.38
4	0.138	3.51	13,932	95.84	32.80	18.22	3.8	0.26	0.12
4.5	0.132	3.35	14,940	102.76	32.53	18.07	2.6	0.19	-0.07

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APPENDIX F—CALCULATION SHEETS

Appendix F contains calculation sheets that are useful in aiding and documenting the calculation of minimum thickness and equivalent tube metal temperature. Individual sheets are provided for calculations in customary or SI units. These calculation sheets may be reproduced.

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API STD 530
CALCULATION SHEET
Customary Units

Heater _____ Plant _____ Refinery _____
Coil _____ Material _____ ASTM Spec. _____

CALCULATION OF MINIMUM THICKNESS	Elastic Design	Rupture Design
Outside diameter, inches	D_o =	D_o =
Design pressure, pounds per square inch gauge	P_e =	P_r =
Maximum or equivalent metal temperature, degrees Fahrenheit	T_m =	T_m =
Temperature allowance, degrees Fahrenheit	TA =	TA =
Design metal temperature, degrees Fahrenheit	T_d =	T_d =
Design life, hours	—	L_d =
Allowable stress at T_d, Figures 4A-4S, pounds per square inch	S_e =	S_r =
Stress thickness, Equation 2 or 4, inches	t_s =	t_s =
Corrosion allowance, inches	CA =	CA =
Corrosion fraction, Figure 1, n = B =	—	f =
Minimum thickness, Equation 3 or 5, inches	t_m =	t_m =

CALCULATION OF EQUIVALENT TUBE METAL TEMPERATURE

Duration of operating period, years	L_o =
Metal temperature, start of run, degrees Fahrenheit	T_{sor} =
Metal temperature, end of run, degrees Fahrenheit	T_{eor} =
Temperature change during operating period, degrees Fahrenheit	DT =
Metal temperature, start ot run, degrees Rankine	T_a =
Thickness change during operating period, inches	Dt =
Assumed initial thickness, inches	t_0 =
Corresponding initial stress, Equation 1, pounds per square inch	S_0 =
Material constant, Table 3, pounds per square inch	A =
Rupture exponent at T_{sor1} Figures 4A-4S	n_0 =
Temperature fraction Figure 2, V = 2.9 N = 0.2	f_r =
Equivalent tube metal temperature, Equation 6, degrees Fahrenheit	T_e =

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API STD 530
CALCULATION SHEET
SI Units

Heater_____ Plant _____ Refinery _____
Coil _____ Material_____ ASTM Spec. _____

CALCULATION OF MINIMUM THICKNESS	Elastic Design	Rupture Design
Outside diameter, millimeters	D_o =	D_o =
Design pressure, megapascals gauge	P_e =	P_r =
Maximum or equivalent metal temperature, degrees Celsius	T_m =	T_m =
Temperature allowance, degrees Celsius	TA =	TA =
Design metal temperature, degrees Celsius	T_d =	T_d =
Design life, hours	—	L_d =
Allowable stress at T_d, Figures 4A-4S, megapascals	S_e =	S_r =
Stress thickness, Equation 2 or 4, millimeters	t_s =	t_s =
Corrosion allowance, millimeters	CA =	CA =
Corrosion fraction, Figure 1, n = B =	—	f =
Minimum thickness, Equation 3 or 5, millimeters	t_m =	t_m =

CALCULATION OF EQUIVALENT TUBE METAL TEMPERATURE

Duration of operating period, years	L_o =
Metal temperature, start of run, degrees Celsius	T_{sor} =
Metal temperature, end of run, degrees Celsius	T_{eor} =
Temperature change during operating period, degrees Celsius	DT =
Metal temperature, start ot run, degrees kelvin	T_a =
Thickness change during operating period, millimeters	Dt =
Assumed initial thickness, millimeters	t_0 =
Corresponding initial stress, Equation 1, megapascals	S_0 =
Material constant, Table 3, megapascals	A =
Rupture exponent at T_{sor1} Figures 4A-4S	n_0 =
Temperature fraction Figure 2, V = 2.9 N = 0.2	f_r =
Equivalent tube metal temperature, Equation 6, degrees Celsius	T_e =

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